LESSON 23

TOOLS FOR DATA COLLECTION

Broadly there are tools of data collection as part of communication surveys. These are:

1. Interview schedule
2. Questionnaire
3. Interview Guide

As discussed earlier interview schedule and questionnaires both are predesigned list of questions used for communication with the respondents. In the case of *interview schedule*, the list of questions remains in the hands of the interviewer who asks questions from the respondent, gets his/her response, and records the responses. *Questionnaire* is also a list of questions, which is handed over to the respondent, who reads the questions and records the answers himself. For purposes of convenience *questionnaire* will refer to both interview schedule as well as questionnaire.

Interview guide is list of topics that are to be covered during the course of interview. Interview guide is used for purposes of an in-depth interviewing. Questions on the topics are formulated on the spot. Most of the questions are open ended. The interviewer may not use the same wording for each respondent; the number of questions may be different; the sequence of questions may also be different.

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Guidelines for Questionnaire Design

A survey is only as good as the questions it asks. Questionnaire design is one of the most critical stages in the survey research process. While common sense and good grammar are important in question writing, more is required in the art of questionnaire design. To assume that people will understand the questions is common error. People may not simply know what is being asked. They may be unaware of topic of interest, they may confuse thee subject with something else, or the question may not mean the same thing to every respondent. Respondents may simply refuse to answer personal questions. Further, properly wording the questionnaire is crucial, as some problems may be minimized or avoided altogether if a skilled researcher composes the questions.

A good questionnaire forms an integrated whole. The researcher weaves questions together so they flow smoothly. He or she includes introductory remarks and instructions for clarification and measures each variable with one or more survey questions.

What should be asked?

The problem definition will indicate which type of information must be collected to answer the research question; different types of questions may be better at obtaining certain type of information than others.

1. Questionnaire Relevancy

A questionnaire is relevant if no unnecessary information is collected and if the information that is needed to solve the problem is obtained.

Asking the wrong or an irrelevant question is a pitfall to be avoided. If the task is to pinpoint compensation problems, for example, questions asking for general information about morale may be inappropriate. To ensure information relevancy, the researcher must be specific about data needs, and there should be a rationale for each item of information.

2. Questionnaire Accuracy

Once the researcher has decided what should be asked, the criterion of accuracy becomes of primary concern. Accuracy means that the information is reliable and valid. While experienced researchers believe that one should use simple, understandable, unbiased, unambiguous, and nonirritating words. Obtaining accurate answer from respondents is strongly influenced by the researcher's ability to design a questionnaire that facilitates recall and that will motivate the respondent to cooperate. Therefore avoid

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jargon, slang, and abbreviations. The respondents may not understand some basic terminology. Respondents can probably tell thee interviewer whether they are married, single, divorced, separated, or widowed, but providing their "marital status" may present a problem. Therefore, asking somebody about his/her *marital status* while the person may not understand the meaning of marital status is likely to mess up the information. Words used in the questionnaire should be readily understandable to all respondents.

3. Avoid Ambiguity, Confusion, and Vagueness.

Ambiguity and vagueness plague most question writers. A researcher might make implicit assumptions without thinking of respondents' perspectives. For example, the question, "what is your income?" could mean weekly, monthly, or annual: family or personal; before taxes or after taxes; for this year or last year; from salary or from all sources. The confusion causes inconsistencies in how different respondents assign meaning to and answer the question.

Another source of ambiguity is the use indefinite words or response categories. Consider the words such as *often, occasionally, usually, regularly, frequently, many, good, fair,* and *poor*. Each of these words has many meanings. For one person frequent reading of *Time* magazine may be reading six or seven issues a year; for another it may be two issues a year. The word *fair* has great variety of meanings; the same is true for many indefinite words.

4. Avoid Double-Barreled Questions

Make each question about one and only one. A double barreled question consists of two or more questions joined together. It makes the respondent's answer ambiguous. For example, if asked, "Does this company have pension and health insurance benefits?" a respondent at the company with health insurance benefits only might answer either yes or no. The response has an ambiguous meaning and the researcher cannot be certain of the respondent's intentions. When multiple questions are asked in one question, thee results may be exceedingly difficult to interpret.

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5. Avoid Leading Questions

Make respondents feel that all responses are legitimate. Do not let them aware of an answer that the researcher wants. A leading question is the one that leads the respondent to choose one response over another by its wording. For example, the question, "you don't smoke, do you?" leads respondents to state that they do not smoke. "Don't you think that women should be empowered?" In most the cases the respondent is likely to agree with the statement.

6. Avoid Loaded Questions

Loaded questions suggest a socially desirable answer or are emotionally charged. "Should the city government repair all the broken streets?" Most of the people are going to agree with this question simply because this is highly socially desirable. A question which may be challenging the traditionally set patterns of behavior may be considered as emotionally charged i.e. it is loaded with such material which may hit the emotions of the people. Look at some behaviors associated with masculinity in Pakistani society. Let us ask a husband "Have you ever been beaten up by your wife?" Straight away this question may be considered to be a challenge to the masculinity of the person. Hence it may be embarrassing for the person to admit such an experience. Therefore, even if the husband was beaten up by his wife, he might give a socially desirable answer.

7. Avoid Burdensome Questions that may Tax the Respondent's Memory

A simple fact of human life is that people forget. Researchers writing questions about past behavior or events should recognize that certain questions may make serious demand on the respondent's memory.

"How did you feel about your brother when you were 6 years old?" It may very difficult to recall something from the childhood.

8. Arrange Questions in a Proper Sequence

The order of question, or the question sequence, may serve several functions for the researcher. If the opening questions are interesting, simple to comprehend, and easy to answer, respondent's cooperation and involvement can be maintained throughout the questionnaire. If respondent's curiosity is not aroused at the outset, they can become disinterested and terminate the interview.

Sequencing specific questions before asking about broader issues is a common cause of question order bias. In some situations it may be advisable to ask general question before specific question to obtain the freest opinion of the respondent. This procedure, known as **funnel technique,** allows the researcher to understand the respondent's frame of reference before asking specific questions about thee level of respondent's information and intensity of his or her opinions.

9. Use Filter Question, if Needed

Asking a question that doesn't apply to the respondent or that the respondent is not qualified to answer may be irritating or may cause a biased response. Including filter question minimizes the chance of asking questions that are inapplicable. Filter question is that question which screens out respondents not qualified to answer a second question. For example the researcher wants to know about the bringing up of one's children. "How much time do you spend playing games with your oldest child?" What if the respondent is unmarried? Even if the respondent is married but does not have the child. In both these situations the question is inapplicable to him/her. Before this question the person may put a filter question whether or not the respondent is married.

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10. Layout of the questionnaire

There are two format or layout issues: the overall physical layout of the questionnaire and the format of questions and responses.
Good lay out and physical attractiveness is crucial in mail, Internet, and other self-administered questionnaires. For different reason it is also important to have a good layout in questionnaires designed for personal and telephone interviews.
Give each question a number and put identifying information on questionnaire. Never cramp questions together or create a confusing appearance.
Make a cover sheet or face sheet for each, for administrative use. Put the time and date of the interview, the interviewer, the respondent identification number, and interviewer's comments and observations on it. Give interviewers and respondents instructions on the questionnaire. Print instructions in a different style from question to distinguish them.
Lay out is important for mail questionnaires because there is no friendly interviewer to interact with thee respondent. Instead the questionnaire's appearance persuades the respondents. In mail surveys, include a polite, professional cover letter on letterhead stationery, identifying the researcher and offering a telephone number for any questions. Always end with "Thank you for your participation."

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LESSON 24

PILOT TESTING OF THE QUESTIONNAIRE

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire.

Conventional wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for extra training the research team. Ironically, professionals who have participated in scores of studies are more likely to pretest an instrument than is a beginning researcher hurrying to complete a project. Revising questions five or more times is not unusual. Yet inexperienced researchers often underestimate the need to follow the design-test-revise process.

It is important to pilot test the instrument to ensure that the questions are understood by the respondents and there are no problems with the wording or measurement. Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled. Making a mistake with 25 or so subjects can avert the disaster of administering an invalid questionnaire to several hundred individuals. Hence the main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

Administering a questionnaire exactly as planned in the actual study often is not possible. For example, mailing out a questionnaire might require several weeks. Pre-testing a questionnaire in this manner might provide important information on response rate, but it may not point out why questions were skipped or why respondents found certain questions ambiguous or confusing. The ability of personal interviewer to record requests for additional explanation and to register comments indicating respondent's difficulty with question sequence or other factors is the primary reason why interviewers are often used for pretest work.

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What aspects to be evaluated during pilot testing?

1. Reactions of Respondents:

The reactions of the respondents can be looked at from different angles. The researcher may be familiar with the local culture; still getting the first hand experience is always useful. Going to the field, contacting the people, and their reactions to the different aspects of research may be a learning experience.

- **Availability of study population timing.** In case we are doing interviewing then pre-testing might help to find out the most appropriate time when the respondent shall be available. The researcher can plan the interviewing accordingly.
- **Acceptability of the questions asked.** An important purpose of pre-testing is to discover participants' reaction to the questions. If the participants do not find the experience stimulating when an interviewer is physically present, how will they react on the phone, or in the self administered mode? Pre-testing should help to discover where repetitiveness or redundancy is bothersome or what topics were not covered that the participant expected. An alert interviewer will look for questions or even sections that the participant perceives to be sensitive or threatening or topics about which the participant knows nothing.

- **Pre-tes**ting will also provide the opportunity to see the acceptability of the wording of the questions in the local cultural context. Some of the issues may be discussed openly while for others people use a disguised language. If people consider the use of certain phrases as offensive, then it is high time to change the wording.

- **Willingness of the respondents to co-operate.** Field testing of the questionnaire will give the idea about the level of cooperation the research team is likely to get from the respondents, particularly if they have to interview them.

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2. Discovering errors in the instrument:

- **Do the tools provide you the information? Reliability. Suitability for analysis.** Tabulation of the results /of a pretest helps determine whether the questionnaire will meet the objectives of the research. A preliminary analysis often illustrates that although respondents can easily comprehend and answer a given question, it is an inappropriate question because it does not help solving the issue. The information may not be suitable for analysis.
- **Time taken/needed to interview/conduct the observation.** Pre-testing can indicate the time taken for interview or to conduct the observation. Too long questionnaires may not be recommended and, therefore, need modification. It can also help in estimating average time being taken to collect information form a respondent. Such an exercise can help in budget estimations.
- **If there is any need to revise the format of the tool.** Question arrangement can play a significant role in the success of the instrument. May be we should start with stimulating questions and place sensitive questions last. Such a situation might be handled through pre-testing. Therefore, pre-testing may help in putting questions in proper **sequence, using acceptable wording, doing appropriate translation, question spacing, structuring of answers, coding system, and needing instructions for interviewers (probing).**

3. Sampling procedure can be checked:

- **The extent to which instructions given are followed.** Field functionaries are given the instructions for following a sampling procedure. Depending upon the type of sampling to be followed, the field worker must follow the guidelines otherwise the quality of the study will be hampered. During the pre-testing one could see not only the extent to which the instructions are being followed but also locate the problems in carrying out those instructions. Also what could be the solutions to those problems?
- **How much time is needed to locate the respondents?** By following the instructions how easy it is to locate the respondents, and how much time is needed to do that activity. It could help in calculating the overall time for data collection, having relevancy for budgeting thee resources.

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4. Staffing and activities of research team can be checked:

- **How successful the training has been?** Pre-testing can be seen as a period of extra training. The pre-testing exercise can provide a good opportunity to make an evaluation of the achievement of the objectives of training. For any deficiencies additional training may be provided.
- **What is the work output of each member?** The researcher can calculate the average output of each fieldworker and accordingly calculate the number of workers needed to finish the work on time. It can also help in making the budget estimates.
- **How well the research team works together?** It is a good opportunity to observe the kind of coordination the research team has. The integrated work is likely affect the efficiency of the team. Any shortcomings could be looked after.
- **Is the logistical support adequate?** Of course we are leaving the field functionaries in isolation. They shall be in need of other logistical support like the transportation, boarding, lodging, guidance and supervision. Some of these aspects could also be appraised during the pre-testing

5. Procedure for data processing and analysis can be evaluated:

- **Make dummy tables.** See how can we tabulate the data and use the appropriate statistics for purposes of interpretations

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LESSON 26

SAMPLE AND SAMPLING TERMINOLOGY

A **sample** is a subset, or some part, of a larger whole. A larger whole could be anything out which sample is taken. That 'whole' could be a bucket of water, a bag of sugar, a group of organizations, a group of students, a group of customers, or a group mid-level managers in an organization. A complete group of entities sharing some common set of characteristics is population. In other words, the totality out of which sample is drawn is referred to as **population.**

Why sample?

1. Saves Cost, Labor, and Time

Applied research projects usually have budget and time constraints. Since sample can save financial cost as well as time, therefore, to go for sample study is pragmatic.
Of course, a researcher investigating a population with an extremely small number of population elements may elect to conduct a study on the total population rather than a sample because cost, labor, and time constraints are relatively insignificant.
Although sample study cuts costs, reduces labor requirements, and gathers vital information quickly, yet there could be other reasons.

2. Quality Management/supervision

Professional fieldworkers are a scarce commodity. In a large study rather than employing less qualified staff it may be advisable to do a sample study and employ highly qualified professional fieldworkers. It can certainly affect the quality of the study. At the same time it may be easier to manage a small group and produce quality information. Supervision, record keeping, training, and so forth would all be more difficult in a very large study.

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3. Accurate and Reliable Results

Another major reason for sampling is that samples, if properly selected, are sufficiently accurate in most of the cases. If the elements of a population are quite similar, only a small sample is necessary to accurately portray the characteristics of interest. Most of us have had blood samples taken from the finger, the arm, or another part of body. The assumption is that blood is sufficiently similar through out the body, the characteristics of the blood can be determined on the basis of sample.
When the elements of population are highly homogenous, samples are highly representative of the population. Under these circumstances almost any sample is as good as another.
Samples may be more accurate than census. In a census study of large population there is a greater likelihood of non-sampling errors. In a survey mistakes may occur that are unrelated to the selection of people in the study. For example, a response may be coded incorrectly, or the keyboard operator might make data entry error. Interviewer mistakes, tabulation errors, and other non-sampling errors may increase during census because of the increased volume of work. In sample increased accuracy is possible because the fieldwork and tabulation of the data can be closely supervised than would be possible in a census. In field survey, a small, well trained, closely supervised group may do a more careful and accurate job of collecting information than a large group of nonprofessional interviewers trying to contact everyone.

4. Sampling may be the Only Way

Many research projects, especially those in quality control testing, require the destruction of the items being tested. If the manufacturer of firecrackers wished to find out whether each product met a specific production standard, there would be no product left after testing. Similarly, consider the case of electric

bulbs. In testing the life of bulbs, if we were to burn every bulb produced, there would be none left to sell. This is destructive sampling.

5. Determine the Period of Study

Interviewing every element of a large population without sampling requires lot of time, may be a year or more. In such a long period study, even the seasonal variation may influence the response pattern of the respondents. For example, if the study was aimed at measuring the level of unemployment in a given large city, the unemployment rate produced by the survey data would not refer to the city as of the beginning of interviewing or as of the end. Researcher may be forced to attribute the unemployment to some hypothetical date, representing to the midpoint of the study period. Hence it will be difficult to determine the exact timing to which the data of the study pertains.

Sampling Terminology

There are a number of technical terms used in books on research and statistics which need explanation. Some of the important terms are:

Element

An element is that unit about which information is collected and which provides the basis of analysis. Typically, in survey research, elements are people or certain types of people. It is that unit about which information is collected and that provides the basis of analysis. It can be a person, groups, families, organizations, corporations, communities, and so forth.

Population

A population is the theoretically specified aggregation of study elements. It is translating the abstract concept into workable concept. For example, let us look at the study of "college students." Theoretically who are the college students? They might include students registered in government colleges and/or private colleges, students of intermediate classes and/or graduate classes, students of professional colleges and/or non-professional colleges, and many other variations. In this way the pool of all available elements is population.

Target Population

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Out of the conceptual variations what exactly the researcher wants to focus on. This may also be called a target population. Target population is the complete group of specific population elements relevant to the research project. Target population may also be called *survey population* i.e. that aggregation of elements from which the survey sample is actually selected.

At the outset of the sampling process, it is vitally important to carefully define the target population so the proper source from which the data are to collected can be identified. In our example of 'college students" finally we may decide to study the college students from government institutions located in Lahore, who are studying social sciences, who are aged 19 years of age, and hailing from rural areas.

Sampling

The process of using a small number of items or parts of a larger population to make conclusions about the whole population. It enables the researchers to estimate unknown characteristics of the population.

Sampling Frame

In actual practice the sample will be drawn from a list of population elements that is often different from the target population that has been defined. A sampling frame is the list of elements from which the

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sample may be drawn. A simple example could be listing of all college students meeting the criteria of target population and who are enrolled on the specified date.

A sampling frame is also called the *working population* because it provides the list that can be worked with operationally. In our example, such a list could be prepared with help of the staff of the selected colleges.

Sampling Frame Error

A sampling frame error occurs when certain sample elements are excluded or when the entire population is not accurately represented in the sampling frame. The error that occurs when certain sample elements are not listed or available and are not represented in the sampling frame.

Sampling Unit

A sampling unit is that element or set of elements considered for selection in some stage of sampling. Sampling may be done in single stage or in multiple stages. In a simple, single-stage sample, the sampling units are the same as the elements. In more complex samples, however, different levels of sampling units may be employed. For example, a researcher may select a sample of *Mohallahs* in a city, and then select a sample of households from the selected *Mohallahs*, and finally may select a sample of adults from the selected households. The sampling units of these three stages of sampling are respectively *Mohallah*, households, and adults, of which thee last of these are the elements. More specifically, the terms "primary sampling units," "secondary sampling units," and "final sampling units" would be used to designate the successive stages.

Observation Unit

An observation unit, or unit of data collection, is an element or aggregation of elements from which the information is collected. Often the unit of analysis and unit of observation are the same – the individual person – but this need not be the case. Thus the researcher may interview heads of household (the observation units) to collect information about every member of the household (the unit of analysis).

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Parameter

A parameter is the summary description of a given variable in a population. The mean income of all families in a city and thee age distribution of the city's population are parameters. An important part portion of survey research involves the estimation of population parameters on the basis of sample observation.

Statistic

A statistic is the summary description of a given variable in a survey sample. Thus the mean income computed from the survey sample and the age distribution of that sample are statistics. Sample statistics are used to make estimates of the population parameters.

Sampling Error

Probability sampling methods seldom, if ever, provide statistics exactly equal to the parameters that they are used to estimate. Probability theory, however, permits us to estimate the error to be expected for a given sample (more information to be sought from professional in Statistics).

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LESSON 27

PROBABILITY AND NON-PROBABILITY SAMPLING

There are several alternative ways of taking a sample. The major alternative sampling plans may be grouped into probability techniques and non-probability techniques. In **probability sampling** every element in the population has a *known nonzero probability* of selection. The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalisability. When time or other factors, rather than generalisability, become critical, non-probability sampling is generally used.

In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate. Nevertheless, there are occasions when non-probability samples are best suited for the researcher's purpose.

Types of non-probability sampling:

In non-probability sampling designs, the elements in the population do not have any probabilities attached to their being chosen as sample subjects. This means that the findings from the study of the sample cannot be confidently generalized to the population. However the researchers may at times be less concerned about generalisability than obtaining some preliminary information in a quick and inexpensive way. Sometimes non-probability could be thee only way to collect the data.

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Convenience Sampling

Convenience sampling (also called *haphazard or accidental sampling*) refers to sampling by obtaining units or people who are most conveniently available. For example, it may be convenient and economical to sample employees in companies in a nearby area, sample from a pool of friends and neighbors. The person-on-the street interview conducted by TV programs is another example. TV interviewers go on the street with camera and microphone to talk to few people who are convenient to interview. The people walking past a TV studio in thee middle of the day do not represent everyone (homemakers, people in the rural areas). Likewise, TV interviewers select people who look "normal" to them and avoid people who are unattractive, poor, very old, or inarticulate.

Another example of haphazard sample is that of a newspaper that asks the readers to clip a questionnaire from the paper and mail it in. Not everyone reads thee newspaper, has an interest in the topic, or will take the time to cut out the questionnaire, and mail it. Some will , and the number who do so may seem large, but the sample cannot be used to generalize accurately to the population.

Convenience samples are least reliable but normally the cheapest and easiest to conduct. Convenience sampling is most often used during the exploratory phase of a research project and is perhaps the best way of getting some basic information quickly and efficiently. Often such sample is taken to test ideas or even to gain ideas about a subject of interest.

Purposive Sampling

Depending upon the type of topic, the researcher lays down the criteria for the subjects to be included in the sample. Whoever meets that criteria could be selected in the sample. The researcher might select such cases or might provide the criteria to somebody else and leave it to his/her judgment for the actual selection of the subjects. That is why such a sample is also called as **judgmental or expert opinion sample.** For example a researcher is interested in studying students who are enrolled in a course on

research methods, are highly regular, are frequent participants in the class discussions, and often come with new ideas. The criteria has been laid down, the researcher may do this job himself/herself, or may ask the teacher of this class to select the students by using the said criteria. In the latter situation we are leaving it to the judgment of the teacher to select the subjects. Similarly we can give some criteria to the fieldworkers and leave it to their judgment to select the subjects accordingly. In a study of working women the researcher may lay down the criteria like: the lady is married, has two children, one of her child is school going age, and is living in nuclear family.

Quota Sampling

A sampling procedure that ensures that certain characteristics of a population sample will be represented to the exact extent that the researcher desires. In this case the researcher first identifies relevant categories of people (e.g. male and female; or under age 30, ages 30 to 60, over 60, etc) then decides how many to get in each category. Thus the number of people in various categories of sample is fixed. For example the researcher decides to select 5 males and 5 females under age 30, 10 males and 10 females aged 30 to 60, and 5 males and 5 females over age 60 for a 40 person sample. This is quota sampling.

Once the quota has been fixed then the researcher may use convenience sampling. The convenience sampling may introduce **bias**. For example, the field worker might select the individual according to his/her liking, who can easily be contacted, willing to be interviewed, and belong to middle class.
Quota sampling can be considered as a form of proportionate stratified sampling, in which a predetermined proportion of people are sampled from different groups, but on a convenience basis.
Speed of data collection, lower costs, and convenience are the major advantages of quota sampling compared to probability sampling. Quota sampling becomes necessary when a subset of a population is underrepresented, and may not get any representation if equal opportunity is provided to each. Although there are many problems with quota sampling, careful supervision of the data collection may provide a representative sample of the various subgroups within the population.

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Snowball Sampling

Snowball sampling (also called *network, chain referral, or reputational sampling*) is a method for identifying and sampling (or selecting) cases in the network. It is based on an analogy to a snowball, which begins small but becomes larger as it is rolled on wet snow and picks up additional snow. It begins with one or a few people or cases and spreads out on the basis of links to thee initial cases.
This design has been found quite useful where respondents are difficult to identify and are best located through referral networks. In the initial stage of snowball sampling, individuals are discovered and may or may not be selected through probability methods. This group is then used to locate others who possess similar characteristics and who, in turn, identify others. The "snowball" gather subjects as it rolls along.

For example, a researcher examines friendship networks among teenagers in a community. He or she begins with three teenagers who do not know each other. Each teen names four close friends. The researcher then goes to the four friends and asks each to name four close friends, then goes to those four and does the same thing again, and so forth. Before long, a large number of people are involved. Each person in the sample is directly or indirectly tied to the original teenagers, and several people may have named the same person. The researcher eventually stops, either because no new names are given, indicating a closed network, or because the network is so large that it is at thee limit of what he or she can study.

Sequential Sampling

Sequential sampling is similar to purposive sampling with one difference. In purposive sampling, the researcher tries to find as many relevant cases as possible, until time, financial resources, or his or her

energy is exhausted. The principle is to get every possible case. In sequential sampling, a researcher continues to gather cases until the amount of new information or diversity is filled. The principle is to gather cases until a saturation point is reached. In economic terms, information is gathered, or the incremental benefit for additional cases, levels off or drops significantly. It requires that the researcher continuously evaluates all the collected cases. For example, a researcher locates and plans in-depth interviews with 60 widows over 70 years old who have been living without a spouse for 10 or more years. Depending on the researcher's purposes, getting an additional 20 widows whose life experiences, social background, and worldview differ little from the first 60 may be unnecessary.

Theoretical Sampling

In theoretical sampling, what the researcher is sampling (e.g. people, situation, events, time periods, etc.) is carefully selected, as the researcher develops grounded theory. A growing theoretical interest guides the selection of sample cases. The researcher selects cases based on new insights they may provide. For example, a field researcher may be observing a site and a group of people during week days. Theoretically, the researcher may question whether the people act the same at other times or when other aspects of site change. He or she could then sample other time periods (e.g. nights and weekends) to get more full picture and learn whether important conditions are the same.

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LESSON 28

TYPES OF PROBABILITY SAMPLING

Probability samples that rely on random processes require more work than nonrandom ones. A researcher must identify specific sampling elements (e.g. persons) to include in the sample. For example, if conducting a telephone survey, the researcher needs to try to reach the specific sampled person, by calling back several times, to get an accurate sample.

Random samples are most likely to yield a sample that truly represents the population. In addition, random sampling lets a researcher statistically calculate the relationship between the sample and the population – that is the size of *sampling error.* A non-statistical definition of the sampling error is the deviation between sample result and a population parameter due to random process.

Simple Random Sample

The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

After numbering all elements in a sampling frame, the researcher uses a list of random numbers to decide which elements to select. He or she needs as many random numbers as there are elements to be sampled: for example, for a sample of 100, 100 random numbers are needed. The researcher can get random numbers from a random number table, a table of numbers chosen in a mathematically random way. Random-number tables are available in most statistics and research methods books. The numbers are generated by a pure random process so that any number has an equal probability of appearing in any position. Computer programs can also produce lists of random number.

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A random starting point should be selected at the outset.

Random sampling does not guarantee that every random sample perfectly represents the population. Instead, it means that most random samples will be close to the population most of the time, and that one can calculate the probability of a particular sample being inaccurate. A researcher estimates the chance that a particular sample is off or unrepresentative by using information from the sample to estimate the sampling distribution. The sampling distribution is the key idea that lets a researcher calculate sampling error and confidence interval.

Systematic Random Sample

Systematic random sampling is simple random sampling with a short cut for random selection. Again, the first step is to number each element in the sampling frame. Instead of using a list of random numbers, researcher calculates a *sampling interval*, and the interval becomes his or her own quasi random selection method. The sampling interval (i.e. 1 in *K where K is some number)* tells the researcher how to select elements from a sampling frame by skipping elements in the frame before one for the sample.

Sampling intervals are easy to compute. We need the sample size and the population size. You can think of the sample interval as the inverse of the sampling ratio. The sampling ratio for 300 names out of 900 will be 300/900 = .333 = 33.3 percent. The sampling interval is 900/300 = 3

Begin with a random start. The easiest way to do this is to point blindly at a number from those from the beginning that are likely to be part of the sampling interval.
When the elements are organized in some kind of cycle or pattern, the systematic sampling will not give a representative sample.

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Stratified Random Sample

When the population is heterogeneous, the use of simple random sample may not produce representative sample. Some of the bigger strata may get over representation while some of the small ones may entirely be eliminated. Look at the variables that are likely to affect the results, and stratify the population in such a way that each stratum becomes homogeneous group within itself. Then draw the required sample by using the table of random numbers. Hence in stratified random sampling a sub-sample is drawn utilizing simple random sampling within each stratum. (Randomization is not done for quota sampling).

There are three reasons why a researcher chooses a stratified random sample: (1) to increase a sample's statistical efficiency, (2) to provide adequate data for analyzing the various subpopulations, and (3) to enable different research methods and procedures to be used in different strata.

1. Stratification is usually more efficient statistically than simple random sampling and at worst it is equal to it. With the ideal stratification, each stratum is homogeneous internally and heterogeneous with other strata. This might occur in a sample that includes members of several distinct ethnic groups. In this instance, stratification makes a pronounced improvement in statistical efficiency.

   Stratified random sampling provides the assurance that the sample will accurately reflect the population on the basis of criterion or criteria used for stratification. This is a concern because occasionally simple random sampling yields a disproportionate number of one group or another, and the sample ends up being less representative than it could be.
   Random sampling error will be reduced with the use of stratified random sampling
   Because each group is internally homogeneous but there are comparative differences
   Between groups. More technically, a smaller standard error may result from stratified
   Sampling because the groups are adequately represented when strata are combined.

2. It is possible when the researcher wants to study the characteristics of a certain population subgroups. Thus if one wishes to draw some conclusions about activities in different classes of student body, stratified sampling would be used.
3. Stratified sampling is also called for when different methods of data collection are applied in different parts of the population. This might occur when we survey company employees at the home office with one method but mist use a different approach with employees scattered over the country.

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Stratification Process

The ideal stratification would be based on the primary variable (the dependent variable) under study. The criterion is identified as an efficient basis for stratification. The criterion for stratification is that it is a characteristic of the population elements known to be related to the dependent variable or other variables of interest. The variable chosen should increase homogeneity within each stratum and increase heterogeneity between strata.
Next, for each separate subgroup or stratum, a list of population elements must be obtained. Serially number the elements within each stratum. Using a table of random numbers or some other device, a separate simple random sample is taken within each stratum. Of course the researcher must determine how large a sample must be drawn from each stratum

Proportionate versus Disproportionate

If the number of sampling units drawn from each stratum is in proportion to the relative population size of the stratum, the sample is proportionate stratified sampling. Sometime, however, a disproportionate stratified sample will be selected to ensure an adequate number of sampling units in every stratum

In a disproportionate, sample size for each stratum is not allocated in proportion to the population size, but is dictated by analytical considerations.

Cluster Sampling

The purpose of cluster sampling is to sample economically while retaining the characteristics of a probability sample. Groups or chunks of elements that, ideally, would have heterogeneity among the members within each group are chosen for study in cluster sampling. This is in contrast to choosing some elements from the population as in simple random sampling, or stratifying and then choosing members from the strata, or choosing every nth case in the population in systematic sampling. When several groups with intra-group heterogeneity and inter-group homogeneity are found, then a random sampling of the clusters or groups can ideally be done and information gathered from each of the members in the randomly chosen clusters.

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Cluster samples offer more heterogeneity within groups and more homogeneity among and homogeneity within each group and heterogeneity across groups.

Cluster sampling addresses two problems: researchers lack a good sampling frame for a dispersed population and the cost to reach a sampled element is very high. A cluster is unit that contains final sampling elements but can be treated temporarily as a sampling element itself. Researcher first samples clusters, each of which contains elements, then draws a second a second sample from within the clusters selected in the first stage of sampling. In other words, the researcher randomly samples clusters, and then randomly samples elements from within the selected clusters. He or she can create a good sampling frame of clusters, even if it is impossible to create one for sampling elements. Once the researcher gets a sample of clusters, creating a sampling frame for elements within each cluster becomes more manageable. A second advantage for geographically dispersed populations is that elements within each cluster are physically closer to each other. This may produce a savings in locating or reaching each element.

A researcher draws several samples in stages in cluster sampling. In a three-stage sample, stage 1 is random sampling of big clusters; stage 2 is random sampling of small clusters within each selected big cluster; and the last stage is sampling of elements from within the sampled within the sampled small clusters. First, one randomly samples the city blocks, then households within blocks, then individuals within households. This can also be an example of **multistage area sampling.**
The unit costs of cluster sampling are much lower than those of other probability sampling designs. However, cluster sampling exposes itself to greater biases at each stage of sampling.

Double Sampling

This plan is adopted when further information is needed from a subset of the group from which some information has already been collected for the same study. A sampling design where initially a sample is used in a study to collect some preliminary information of interest, and later a sub-sample of this primary sample is used to examine the matter in more detail, is called double sampling.

What is the Appropriate Sample Design?

A researcher who must make a decision concerning the most appropriate sample design for a specific project will identify a number of sampling criteria and evaluate the relative importance of each criterion before selecting a sample design. The most common criteria

Degree of Accuracy

Selecting a representative sample is, of course, important to all researchers. However, the error may vary from project to project, especially when cost saving or another benefit may be a trade-off for reduction in accuracy.

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Resources

The costs associated with the different sampling techniques vary tremendously. If the researcher's financial and human resources are restricted, this limitation of resources will eliminate certain methods. For a graduate student working on a master's thesis, conducting a national survey is almost always out of the question because of limited resources. Managers usually weigh the cost of research versus the value of information often will opt to save money by using non-probability sampling design rather than make the decision to conduct no research at all.

Advance Knowledge of the Population

Advance knowledge of population characteristics, such as the availability of lists of population members, is an important criterion. A lack of adequate list may automatically rule out any type of probability sampling..

National versus Local Project

Geographic proximity of population elements will influence sample design. When population elements are unequally distributed geographically, a cluster sampling may become more attractive.

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Need for Statistical Analysis

The need for statistical projections based on the sample is often a criterion. Non-probability sampling techniques do not allow researcher to use statistical analysis to project the data beyond the sample.

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LESSON 31

DATA PRESENTATION

Tables and graphs (pictorial presentation of data) may simplify and clarify the research data. Tabular and graphic representation of data may take a number of forms, ranging from computer printouts to elaborate pictographs. The purpose of each table or graph, however, is to facilitate the summarization and communication of the meaning of the data.

Although there are a number of standardized forms for presenting data in table or graphs, the creative researcher can increase the effectiveness of particular presentation. Bar charts, pie charts, curve diagrams, pictograms, and other graphic forms of presentation create a strong visual impression.

The proliferation of computer technology in business and universities has greatly facilitated tabulation and statistical analysis. Commercial packages eliminate the need to write a new program every time you want to tabulate and analyze data with a computer. SAS, Statistical Package for the Social Sciences (SPSS), SYSTAT, Epi. Info. And MINITAB is commonly used statistical packages. These user friendly packages emphasize statistical calculations and hypothesis testing for varied types of data. They also provide programs for entering and editing data. Most of these packages contain sizeable arrays of programs for descriptive analysis and univariate, bivariate, and multivariate statistical analysis.

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Results with one variable

Frequency Distribution

Several useful techniques for displaying data are in use. The easiest way to describe the numerical data of one variable is with a frequency distribution. It can be used with nominal-, ordinal-, interval-, or ratio-level data and takes many forms. For example we have data of 400 students. We can summarize the data on the gender of the students at a glance with raw count or a *frequency distribution*

Table 1: Frequency distribution of students

Gender	Frequency	Percent
Male	300	75
Female	100	25
Total	**400**	**100**

We can present the same information in a graphic form. Some common types of graphic presentations are the *histograms, bar chart, and pie chart.* Bar charts or graphs are used for discrete variables. They can have vertical or horizontal orientation with small space between the bars. The terminology is not exact, but histograms are usually upright bar graphs for interval or ratio data.

Presentation of data in these forms lays emphasis on visual representation and graphical techniques over summary statistics. Summary statistics may obscure, conceal, or even misrepresent the underlying structure of the data. Therefore it is suggested that data analysis should begin with visual inspection.

The presented data has to be interpreted. The purpose of interpretation is to explain the meanings of the data so that we can make inferences and formulate conclusions. Therefore, **interpretation** refers to making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions. In order for interpretation, the data have to be meaningfully analyzed. For purposes of analysis the researchers use statistics.

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The word *statistics* has several meanings. It can mean a set of collected numbers (e.g. numbers telling how many people living in a city) as well as a branch of applied mathematics used to manipulate and summarize the features of numbers. Social researchers use both types of statistics. Here, we focus on the second type – ways to manipulate and summarize numbers that represent data from research project.

Descriptive statistics describe numerical data. They can be categorized by the number of variables involved: univariate, bivariate, or multivariate (for one, two, and three or more variables). Univariate statistics describe one variable.

Researchers often want to summarize the information about one variable into a single number. They use three measures of central tendency, or measures of the center of the frequency distribution: mean, median and mode, which are often called averages (a less precise and less clear way to say the same thing). The *mode* is simply the most common or frequently occurring number. The *median* is the middle point. The *mean* also called the arithmetic average, is the most widely used measure of central tendency. A particular central tendency is used depending upon the nature of the data.

Bivariate Tables

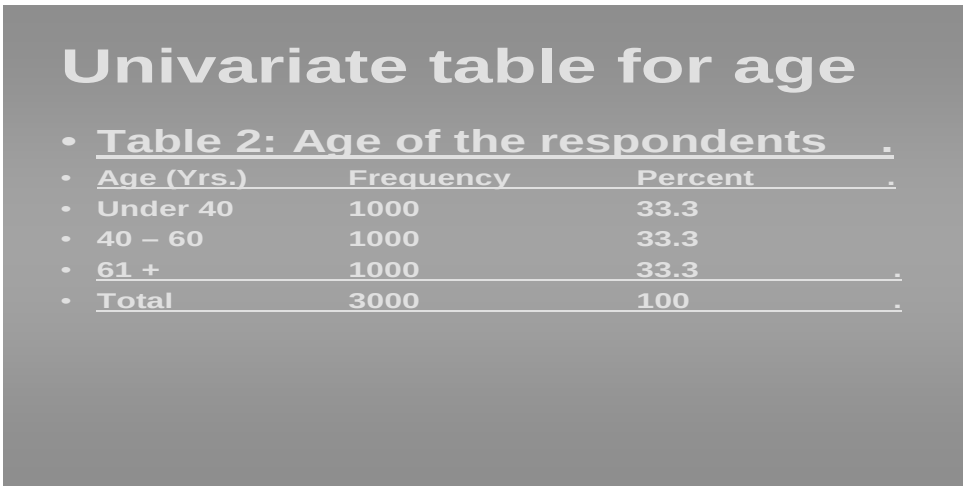
The bivariate contingency table is widely used. The table is based on cross-tabulation (cross-classification); that is the cases are organized in the table on the basis of two variables at the same time.

A contingency table is formed by cross-tabulating the two or more variables. It is contingent because the cases in each category of a variable get distributed into each category of a second variable. The table distributes cases into categories of multiple variables at the same time and shows how the cases, by the category of one variable, are "contingent upon" the categories of the other variables.

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Constructing Percentage Tables

It is to construct a percentage table, but there are ways to make it look professional. Let us take two variables like the age of the respondents and their attitude towards "women empowerment." Assuming that age affects the attitude towards women empowerment let us hypothesize: the lower the age, the higher the favorable attitude towards "women empowerment." The age range of the respondents is 25 to 70, and the attitude index has three categories of "highly favorable," "medium favorable," and "low favorable." The age variable has so many categories that making a table with that number becomes unwieldy and meaningless. Therefore, we regroup (recode) the age categories into three i.e. under 40 years, 40 – 60 years, and 61 + years.



Univariate table for attitude

- **Table 3: Attitude towards women empowerment**

Attitude	Frequency	Percent
Hi Favorable	1100	37
Med Favorable	1050	35
Lo Favorable	850	28
Total	3000	100

Bivariate table

- Table 4: Age by attitude towards women empowerment

	Age (in years)							
Level of attitude	under 40		40 –60		61 +		Total	
	F.	%	F.	%	F	%	F	%
Hi Favorable	600	60	300	30	200	20	1100	37
Med. Favorable	300	30	500	50	250	25	1050	35
Lo Favorable	100	10	200	20	500	50	850	28
Total	1000	100	1000	100	1000	100	3000	100

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LESSON 32

THE PARTS OF THE TABLE

1. Give each table a number.
2. Give each table a title, which names variables and provides background information
3. Label the row and columns variables and give name to each of the variable categories.
4. Include the totals of the columns and rows. These are called *marginals.*They equal the univariate frequency distribution for the variable.
5. Each number or place that corresponds to the intersection of a category for each variable is a *cell of a table.*
6. The numbers with the labeled variable categories and the totals are called the *body of the table.*
7. If there is missing information, report the number of missing cases near the table to account for all original cases.

Researchers convert raw count tables into percentages to see bi-variate relationship. There are three ways to percentage a table: by row, by column, and for the total. The first two are often used and show relationship.

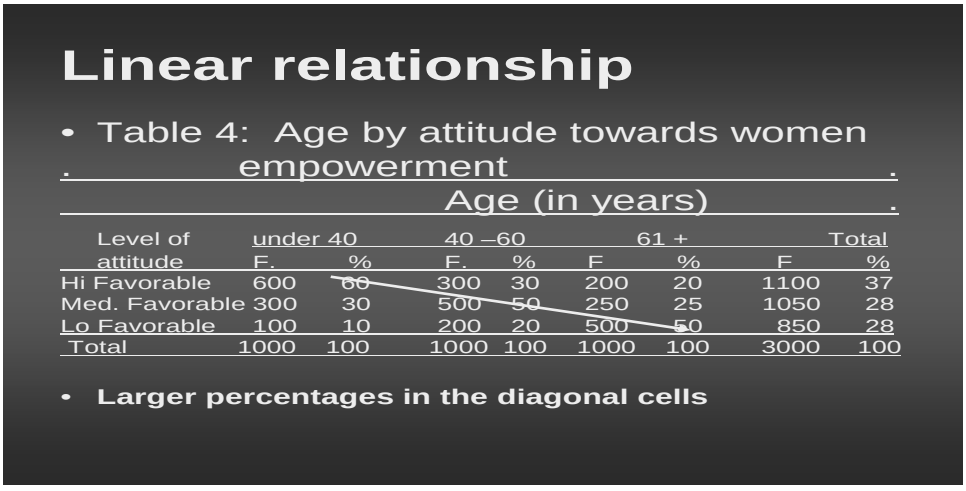
Is it best to percentage by row or column? Either could be appropriate. A researcher's hypothesis may imply looking at row percentages or the column percentages. Here, the hypothesis is that age affects attitude, so column percentages are most helpful. Whenever one factor in a cross-tabulation can be considered the cause of the other, percentage will be most illuminating if they are computed in the direction of the causal factor.

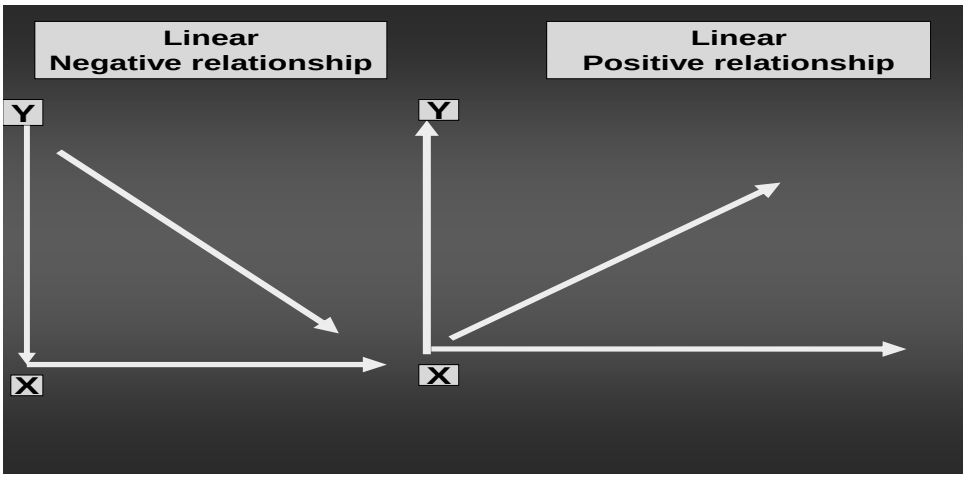
Reading a percentage Table: Once we understand how table is made, reading it and figuring out what it says are much easier. To read a table, first look at the title, the variable labels, and any background information. Next, look at the direction in which percentages have been computed – in rows or columns.

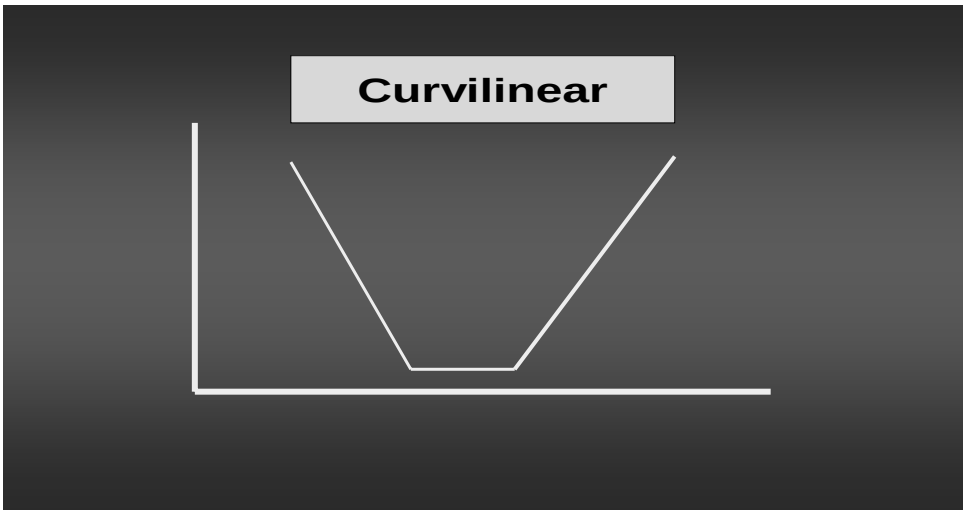
Researchers read percentaged tables to make comparisons. Comparisons are made in the opposite direction from that in which percentages are computed. A rule of thumb is to compare across rows if the table is percentaged down (i.e. by column) and to compare up and down in columns if the table is percentaged across (i.e. by row).

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It takes practice to see a relationship in a perentaged table. If there is no relationship in a table, the cell percentages look approximately equal across rows or columns. A linear relationship looks like larger percentages in the diagonal cells. If there is curvilinear relationship, the largest percentages form a pattern across cells. For example, the largest cells might be the upper right, the bottom middle, and the upper left. It is easiest to see a relationship in a moderate-sized table (9 to 16 cells) where most cells have some cases (at least five cases are recommended) and the relationship is strong and precise.







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A simple way to see strong relationships is to circle the largest percentage in each row (in row percentaged tables) or columns (for column-percentaged tables) and see if a line appears.

A simple way to see strong relationship is to circle the largest percentage in applicable row or column and see if a line appears

- Table 4: Age by attitude towards women empowerment

	Age (in years)							
Level of attitude	under 40		40 –60		61 +		Total	
	F.	%	F.	%	F	%	F	%
Hi Favorable	600	(60)	300	30	200	20	1100	37
Med. Favorable	300	30	500	(50)	250	25	1050	35
Lo Favorable	100	10	200	20	500	(50)	850	28
Total	1000	100	1000	100	1000	100	3000	100

The circle-the-largest-cell rule works – with one important caveat. The categories in the percentages table must be ordinal or interval. The lowest variable categories begin at the bottom left. If the categories in a table are not ordered the same way, the rule does not work.

Statistical Control

Showing an association or relationship between two variables is not sufficient to say that an independent variable *causes* a dependent variable. In addition to temporal order and association, a researcher must eliminate alternative explanations – explanations that can make the hypothetical relationship spurious. Experimental researchers do this by choosing a research design that physically controls potential alternative explanations for results (i.e. that threaten internal validity).

In non-experimental research, a researcher controls for alternative explanations with statistics. He or she measures possible alternative explanations with *control variables*, and then examines the control variables with multivariate tables and statistics that help him or her to decide whether a bivariate relationship is spurious. They also show the relative size of the effect of multiple independent variables on dependent variable.

A researcher controls for alternative explanation in multivariate (more than two variables) analysis by introducing a third (sometimes fourth, or fifth) variable. For example, a bivariate table shows that young people show more favorable attitude towards women empowerment. But the relationship between age and attitude towards women empowerment may be spurious because men and women may have different attitudes. To test whether the relationship is actually due to gender, a researcher must control for gender; in other words, effects of gender are statistically removed. Once this is done, a researcher can see whether the bivariate relationship between age and attitude towards women empowerment remains.

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A researcher controls for a third variable by seeing whether the bivariate relationship persists within categories of the control variable. For example controls for gender, and the relationship between age and attitude persists. This means that both male and females show negative association between age and attitude toward women empowerment. In other words, the control variable has no effect. When this is so, the bivariate relationship is not spurious.
If the bivariate relationship weakens or disappears after the control variable is considered, it means that the age is not real factor that makes the difference in attitude towards women empowerment, rather it is the gender of the respondents.

Statistical control is a key idea in advanced statistical techniques. A measure of association like the correlation co-efficient only suggests a relationship. Until a researcher considers control variables, the bivariate relationship could be spurious. Researchers are cautious in interpreting bivariate relationships until they have considered control variables.

After they introduce control variables, researchers talk about the *net effect* of an independent variable – the effect of independent variable "net of," or in spite of, the control variable. There are two ways to introduce control variables: trivariate percentaged tables and multiple regression analysis.

Constructing Trivariate Tables

In order to meet all the conditions needed for causality, researchers want to "control for" or see whether an alternative explanation explains away a causal relationship. If an alternative explanation explains a relationship, then bivariate relationship is spurious. Alternative explanations are operationalize as a third variable, which are called *control variables* because they control for alternative explanation.

One way to take such third variables into consideration and see whether they influence the bivariate relationship is to statistically introduce control variables using trivariate or three variable tables. Trivariate tables differ slightly from bivariate tables; they consist of multiple bivariate tables.

A trivariate table has a bivariate table of the independent and dependent variable for each category of the control variable. These new tables are called *partials.* The number of partials depends on the number of categories in control variable. Partial tables look like bivariate tables, but they use a subset of the cases. Only cases with a specific value on the control variable are in the partial. Thus it is possible to break apart a bivariate table to form partials, or combine the partials to restore the initial bivariate table.

Trivariate tables have three limitations. First, they are difficult to interpret if a control variable has more that four categories. Second, control variables can be at any level of measurement, but interval or ratio control variables must be grouped (i.e. converted to an ordinal level), and how cases are grouped can affect the interpretation of effects. Finally, the total number of cases is a limiting factor because the cases are divided among cells in partials. The number of cells in the partials equals the number of cells in the bivariate relationship multiplied by the number of categories in the control variables. For example if the control variable has three categories, and a bivariate table has 12 cells, the partials have 3 X 12 = 36 cells. An average of five cases per cell is recommended, so the researcher will need 5 X 36 = 180 cases at minimum.

Like a bivariate table construction, a trivariate table begins with a compound frequency distribution (CFD), but it is a three-way instead of two-way CFD. An example of a trivariate table with "gender" as control variable for the bivariate table is shown here:

Partial table for males

	Age (in years)							
Level of	Under 40		40—60		61+		Total	
Attitude	F	%	F	%	F.	%	F.	%.
High	300	60	200	33	30	6	530	33
Medium	140	28	270	45	120	24	530	33
Low	60	12	130	22	350	70	540	34
Total	500	100	600	100	500	100	1600	100

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Partial table for females

	Age (in years)							
Level of	Under 40		40—60		61+		Total	
Attitude	F	%	F	%	F.	%	F.	%
High	350	70	200	50	20	4	570	41
Medium	150	30	150	38	220	44	520	37
Low	-	-	50	12	260	52	310	22
Total	500	100	400	100	500	100	1400	100

The replication pattern is the easiest to understand. It is when the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable. It means that the control variable has no effect.

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The specification pattern is the next easiest pattern. It occurs when one partial replicate the initial bivariate relationship but other partials do not. For example, we find a strong (negative) bivariate relationship between age of the respondents and attitude towards women empowerment. We control for gender and discover the relationship holds only for males (i.e. the strong negative relationship was in the partial for males, but not for females). This is specification because a researcher can specify the category of the control variable in which the initial relationship persists.

The interpretation pattern describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

The suppressor variable pattern occurs when the bivariate tables suggest independence but relationship appears in one or both of the partials. For example, the age of the respondents and their attitudes towards women empowerment are independent in a bivariate table. Once the control variable "gender" is introduced, the relationship between the two variables appears in the partial tables. The control variable is suppressor variable because it suppressed the true relationship; the true relationship appears in partials.

Multiple Regression Analysis

Multiple regression controls for many alternative explanations of variables simultaneously (it is rarely possible to use more than one control variable using percentaged tables). Multiple regression is a technique whose calculation you may have learnt in the course on statistics.

Note

In the preceding discussion you have been exposed to the descriptive analysis of the data. Certainly there are statistical tests which can be applied to test the hypothesis, which you may have learnt in your course on statistics.

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LESSON 33

EXPERIMENTAL RESEARCH

Experimental research builds on the principles of positivist approach more directly than do the other research techniques. Researchers in the natural sciences (e.g. chemistry and physics), related applied fields (e.g. engineering, agriculture, and medicines) and the social sciences conduct experiments. The logic that guides an experiment on plant growth in biology or testing a metal in engineering is applied in experiments on human social behavior. Although it is most widely used in psychology, the experiment is found in education, criminal justice, journalism, marketing, nursing, political science, social work, and sociology.

The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated. The experimenter, therefore, manipulates a single variable in an investigation and holds constant all other, extraneous variables. (Events may be controlled in an experiment in a way that is not possible in a survey.) The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.
Experiment is a research design in which conditions are controlled so that one or more variables can be manipulated in order to test a hypothesis. Experimentation is a research design that allows evaluation of causal relationship among variables.

Experiments differ from other research methods in terms of degree of control over the research situation. In a typical experiment one variable (the *independent variable*) is manipulated and its effect on another variable (the *dependent variable*) is measured, while all other variables that may confound such relationship are eliminated or controlled. The experimenter either creates an artificial situation or deliberately manipulates a situation.

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Once the experimenter manipulates the independent variable, changes in the dependent variable are measured. The essence of a behavioral experiment is to do something to an individual and observe his or her reaction under conditions where this reaction can be measured against a known baseline.
To establish that variable X cause's variable Y, *all three* of the following conditions should be met:

1. Both X and Y should co-vary (i.e. when one goes up, the other should also simultaneously go up (or go down).
2. X (the presumed causal factor) should precede Y. In other words, there must be a time sequence in which the two occur.
3. No other factor should possibly cause the change in the dependent variable Y.

It may thus be seen that to establish causal relationships between two variables in an organizational setting, several variables that might co-vary with the dependent variable have to be controlled. This would then allow us to say that variable X and variable X alone causes the dependent variable Y. Useful as it is to know the cause-and-effect relationships, establishing them is not so easy, because several other variables that co-vary with the dependent variable have to be controlled. It is not always possible to control all the co-variates while manipulating the causal factor (the independent variable that is causing the dependent variable) in organizational settings, where events flow or occur naturally and normally. It is, however, possible to first isolate thee effects of a variable in a tightly controlled artificial setting (the lab setting), and after testing and establishing the cause-and-effect relationship under these tightly controlled conditions, see how generalizable such relationships are to the field setting.

The Language of Experiments

Experimental research has its own language or set of terms and concepts. One important term frequently used is *subjects* or *test units*. In experimental research, the cases or people used in research projects and on whom variables are measured are called thee *subjects* or *test units*. In other words these are those

entities whose responses to the experimental treatment are measured or observed. Individuals, organizational units, sales territories, or other entities may be the test units. Similar terminology is used on different component parts of the experiments.

Parts of Experiments: We can divide the experiments into seven parts and for each part there is a term. Not all experiments have all these parts, and some have all seven parts plus others. The following seven usually make up a true experiment.

1. Treatment or independent variable.
2. Dependent variable.
3. Pretest.
4. Posttest.
5. Experimental group.
6. Control group.
7. Assignment of subjects.

Treatment or independent variable: The experimenter has some degree of control over thee independent variable. The variable is independent because its value can be manipulated by the experimenter to whatever he or she wishes it to be. In experimental design the variable that can be manipulated to be whatever the experiment wishes. Its value may be changed or altered independently of any other variable.

In most experiments, a researcher creates a situation or enters into an ongoing situation, then modifies it. The treatment (or the stimulus or manipulation) is what the researcher modifies. The term comes from medicine, in which a physician administers a treatment to patients; the physician intervenes in a physical or psychological condition to change it. It is the independent variable or the combination of independent variables.
In experiments, for example, the researcher creates a condition or situation. Look at "the degree of fear or anxiety"; the levels are high-fear or low-fear situation. Instead of asking the subjects, as we do in surveys, whether they are fearful, experimenter puts the subjects into either in a high-fear or low-fear situation. They measure the independent variable by manipulating conditions so that some subjects feel a lot of fear and others feel little.

Researchers go to great lengths to create treatments. They want the treatment to have an impact and produce specific reactions, feelings, or behaviors.

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It also possible the researchers look at the alternative manipulations of the independent variable being investigated. In business research, the independent variable is often categorical or classificatory variable, representing some classifiable or qualitative aspects of management strategy. To determine the effects of training, for example, the experimental treatment that represents the independent variable is the training program itself.

Dependent Variable: The criterion or standard by which thee results are judged. It is assumed that the changes in the dependent variable are consequence of changes in the independent variable. For example, measures of turnover, absenteeism, or morale might be alternative choices for the dependent variable, depending on the purpose of the training.

The outcomes in the experimental research are the physical conditions, social behaviors, attitudes, feelings, or beliefs of subjects that change, in response to treatment. Dependent variables can be measured by paper-and-pencil indicators, observations, interviews, or physiological responses (e.g. heartbeat, or sweating palms).

Selection of dependent variable is crucial decision in the design of an experiment.

Pretests and Posttests: Frequently a researcher measures thee dependent variable more than once during an experiment. The *pretest* is the measurement of the dependent variable prior to the introduction of the treatment. The posttest is the measurement of the dependent variable after thee treatment has been introduced into the experimental situation.

Experimental and Control Groups: Experimental researchers often divide subjects into two or more groups for purposes of compassion. A simple experiment has only two groups, only one of which receives the treatment. The experimental group is the group that receives the treatment or in which the treatment is present.

The group that does not receive the treatment is called the "control group." When the independent variable takes on many different values, more than one experimental group is used.

In the simplest type of experiment, only two values of the independent variable are manipulated. For example, consider measuring the influence of a change in work situation, such as playing music over an intercom during working hours, on employee productivity. In the experimental condition (the treatment administered to the **experimental group**), music is played during working hours. In the control condition (the treatment administered to the **control group**), the work situation remains the same, without change. By holding conditions constant in the control group, the researcher controls for potential sources of error in the experiment. Productivity, (the dependent variable) in the two groups is compared at the end of the experiment to determine whether playing the music (the independent variable) has any effect.

Several experimental treatment levels can also be used. The music/productivity experiment, with one experimental and one control group, may not tell the researcher everything he or she wishes to know about the music/productivity relationship. If the researcher wished to understand the functional nature of the relationship between music and productivity at several treatment levels, additional experimental groups with music played for only 2 hours, only for 4 hours, and only for 6 hours might be studied. This type of design would allow the experimenter to get a better idea about the impact of music on productivity.

Assignment of Subjects/Test Units: Social researchers frequently want to compare. When making comparisons, the researchers want to compare the cases that do not differ with regard to variables that offer alternative explanations. Therefore the groups should be similar in characteristics in such a way that the change in the dependent variable is presumably the outcome of the manipulation of the independent variable, having no alternative explanations.

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Random assignment (Randomization) is a method for assigning the cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way. It is a mechanical method; the assignment is automatic, and thee researcher cannot make assignments on thee basis of personal preference or the features of specific cases.

Random assignment is random in statistical/mathematical sense, not in everyday sense. In everyday speech, random means unplanned, haphazard, or accidental, but it has a special meaning in mathematics. In probability theory, random describes a process in which each case has a known chance of being selected. Random selection allows the researcher calculate the odds that a specific case will be sorted into one group or the other. A random process is the one in which all cases have an exactly equal chance of ending up in one or the other group.

Random assignment or randomization is unbiased because a researcher's desire to confirm a hypothesis or a research subject's personal interests does not enter into the selection process. It also assures the researcher that repetitions of an experiment – under the controlled conditions – will show true effects, if

they exist. Random assignment of subjects allows the researcher to assume that thee groups are identical with respect to all variables except for experimental treatment.

Random assignment of subjects to the various experimental groups is thee most common technique used to prevent test units from differing from each other on key variables; it assumes that all the characteristics of these subjects have been similarly randomized. If the experimenter believes that certain extraneous variable may affect the dependent variable, he or she may make sure that the subjects in each group are matched on these characteristics. **Matching** the subjects on the basis of pertinent background information is another technique for controlling assignment errors.

Matching presents a problem: What are the relevant characteristics to match on, and can one locate exact matches? Individual cases differ in thousands of ways, and the researcher cannot know which might be relevant.

Three Types of Controls

1. **Manipulation of the Independent Variable:** In order to examine the causal effects of an independent variable on a dependent variable, certain manipulations need to be tried. Manipulation simply means control over the stimulus that is we create different levels of the independent variable to assess the impact on the dependent variable. Let us say we want to test the effects of lighting on worker production levels among sewing machine operators. To establish cause and effect relationship, we must measure the production levels of all the operators over a 15 day period with the usual amount of light they work with – say 60 watt bulbs. We might then want to split the group of 60 operators into three groups of 20 members each, and while allowing the subgroup to continue to work under the same conditions as before (60-watt electric light bulbs). We might want to manipulate the intensity of the light for the other two subgroups, by making one group work with 75 watt and the other with 100 watt light bulbs. After the different groups have worked with these varying degrees of light exposure for15 days, each group's total production for these 15 days may be analyzed to see the difference between the pre-experimental and the post experimental productions among the groups is directly related to the intensity of the light to which they have been exposed. If our hypothesis that better lighting increases the production levels is correct, the subgroups that did not have any change in the lighting (control group), should have no increase in production and thee other two groups should show increases, with the one having the most light (100 watts) showing greater increases than those who had the 75 watt lighting.

   In this case the independent variable, lighting, has been manipulated by exposing different groups to different degrees of changes in it. This manipulation of the independent variable is also known treatment, and the results of the treatment are called treatment effects.
2. **Holding Conditions Constant:** When we postulate cause-and-effect relationships between two variables X and Y, it is possible that some other factor, say A, might also influence the dependent variable Y. In such a case, it will not be possible to determine the extent to which Y occurred only because of X, since we do not know how much of the total variation of was caused by the presence of the other factor A. If the true effect of thee X is to be assessed, then the effect of A has to be controlled. This is also called as controlling the effect of contaminating factors or confounding factors.

3. **Control over the Composition of Groups:** If the experimental and control groups have such characteristics that could contaminate the results then the researcher may have to take note of such factors, if there are any. The group differences should not confound the effect of X variable that happens to be under study. The experimental and control groups need to be balanced. For this purpose the researcher may use **random selection** of the subjects and allocating to different groups. Finally the experimental and control groups should also be selected randomly. Another way to have identical groups is by following the procedure of

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matching. One could look at the possible characteristics of the subjects that could contaminate the effect of X variable, and try to distribute these evenly in all the groups. So pick up one subject and try to match it with another subject on the specified characteristics (age, gender, education, marital status) and put one subject in one group and the other in the other group. After the formation of groups, the researcher may randomly decide about experimental and control groups.

Random Assignment

Social researchers frequently want to compare. For example, a researcher has two groups of 15 students and wants to compare the groups on the basis of key differences between them (e.g. a course that one group completed). Or a researcher has five groups of customers and wants to compare the groups on the basis of one characteristic (e.g. geographic location). "Compare apples with apples, don't compare apples with oranges." It means that a valid comparison depends on comparing things that are fundamentally alike. Random assignment facilitates comparison in experiments by creating similar groups.

Random assignment is a method for assignment cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way. It is mechanical method; the assignment is automatic, and the researcher cannot make assignments on the basis of personal preference or the features of specific cases.

Random assignment is random in a statistical or mathematical sense, not in an everyday sense. In everyday speech, random means unplanned, haphazard, or accidental, but it has a specialized meaning in mathematics. In probability theory, *random* describes a process in which each case has a known chance of being selected. Random assignment lets a researcher calculate the odds that a specific case will be sorted into one group over another.

Random assignment or randomization is unbiased because a researcher's desire to confirm a hypothesis or a research subject's personal interest does not enter into selection process.

Matching

It implies to match the characteristics (such as age, sex) of the cases in each group. Matching is an alternative to random assignment, but it is an infrequently used one.
Matching presents a problem: What are the relevant characteristics to match on, and can one locate exact matches. Individual cases differ in thousands of ways, and the researcher cannot know which might be relevant. Therefore, randomization is preferred over matching. It takes care of the contaminating factors.

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LESSON 34

EXPERIMENTAL RESEARCH (Cont.)

Steps in Conducting an Experiment

Following the basic steps of the research process, experimenters decide on a topic, narrow it into a testable research problem or question, and then develop a hypothesis with variables. Once a researcher has the hypothesis, the steps of experimental research are clear. Broadly there are about 12 steps in conducting an experiment, which are as below:

1. Begin with a straightforward hypothesis that is appropriate for experimental research.
2. Decide on an experimental design that will test the hypothesis within practical limitations. The researcher decides the number of groups to use, how and when to create treatment conditions, the number of times to measure the dependent variable, and what the groups of subjects will experience from beginning till end.
3. Decide how to introduce the treatment or create a situation that induces the independent variable.
4. Develop a valid and reliable measure of the dependent variable.
5. Set up an experimental setting and conduct a pilot test of the treatment and dependent variable measures.
6. Locate appropriate subjects or cases.
7. Randomly assign subjects to groups (if random assignment is used in the chosen research design) and give careful instructions.
8. Gather data for the pretest measure of the dependent variable for all groups (if pretest is used in thee chosen design).
9. Introduce the treatment to the experimental group only (or to the relevant groups if there are multiple experimental groups) and monitor all groups.
10. Gather data for posttest measure of the dependent variable.
11. Debrief the subjects by informing them of the true purpose and reasons for the experiment. Ask subjects what they thought was occurring. Debriefing is crucial when subjects have been deceived about some aspect of the treatment.
12. Examine data collected and make comparisons between different groups. Where appropriate, use statistics and graphs to determine whether or not the hypothesis is supported.

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Types of Designs

Researchers combine parts of experiment (e.g. pretests, control groups, etc.) together into an experimental design. For example some designs lack pretests, some do not have control groups, and others have many experimental groups. Certain widely used standard designs have names.

Classical Experimental Design: All designs are variations of the classical experimental design, which has random assignment of subjects, a pretest and a posttest, an experimental group, and a control group.

Quasi-Experimental Designs:

One-shot Case Study Design: Also called the one-group posttest-only design, the one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment. For example, a researcher shows a group of students a horror film, then measures their attitude with a questionnaire. A weakness of this design is that it is difficult to say for sure that the treatment caused the dependent variable. If subjects were the same before and after the treatment, the researcher would not know it.

***One Group Pretest-posttest Design*:** This design has one group, a pretest, a treatment, and a posttest. It lacks a control group and random assignment. Continuing with the previous example, the researcher gives a group of students an attitude questionnaire to complete, shows a horror film, then has them

complete the same questionnaire second time. This is an improvement over the one-shot case study because the researcher measures the dependent variable both before and after the treatment. But it lacks the control group for comparison. The researcher cannot know whether something other than the treatment occurred between the pretest and the posttest to cause the outcome.

Two Groups Posttest-only Design: It has two groups, a random assignment of subjects, a posttest, and a treatment. It has all parts of the classical design except a pretest. Continuing with our previous example, the researcher forms two groups through randomization process. He shows group a horror film to one group i.e. the experimental group. The other group is not shown any film. Both groups then complete the questionnaire. The random assignment reduces the chance that the groups differed before the treatment, but without a pretest, a researcher cannot be as certain that the groups began the same on the dependent variable.

True Experimental Designs

Experimental designs, which have at least two groups, a random assignment of subjects to experimental and control groups, only experimental group is exposed to treatment, both groups record information before and after the treatment, are known as ex-post facto experimental designs.

Pretest and Posttest Experimental and Control Group Design: Two groups, one control group and the other experimental group, are formed randomly. Both the groups are exposed to pretest and posttest. The experimental group is exposed to treatment while the control group is not. Measuring the difference between the differences in the post- and pretests of the two groups would give the net effects of the treatment.

Experimental Group: Pretest (O1) X Posttest (O2)
Control Group: Pretest (O3) - Posttest (O4)

Randomization used for setting up the group.

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[(O2 – O1) – (O4 – O3)] = Treatment effect (could be anywhere between 0 to -1 or +1).

Solomon's Four Group Design: To gain more confidence in internal validity in experimental designs, it is advisable to set up two experimental groups and two control groups. One experimental group and one control group can be given the both pretest and the posttest. The other two groups will be given only the posttest. Here the effects of treatment can be calculated in several different ways as shown in figure 1:

Figure 1: Solomon's four group design

Group	Pretest	Treatment	Posttest
1. Experimental	O1	X	O2
2. Control	O3	-	O4
3. Experimental	-	X	O5
4. Control	-	-	O6

(O2 – O1) = E
(O4 - O3) = E
(O5 – O6) = E
(O5 - O3) = E
[(O2 –O1) – (O4 – O3)] = E
E = Effect

If all Es are similar, the cause and effect relationship is highly valid.

Interaction Effect

The effect of two variables together is likely to be greater than the individual effect of each put together. The idea of an interaction effect is familiar, especially in the area of medicine or illness. As an example, imagine that for a given population of 100 persons, all of the same age and sex, it was found that if all 100 smoked cigarettes the effect would be a lung cancer rate of 20 percent. Assume that for an identical group of 100 persons who did not smoke but lived in a smoggy environment, 10 percent would get lung cancer. Now consider a third identical group of 100 persons all of whom smoke and also live in a smoggy environment. The additive effect of both smoking and smog would be 20 percent plus 10 percent, or a total of 30 percent (30 people) having cancer. However, imagine that an actual medical survey of the population shows a cancer rate of 37 percent among persons experiencing both smoking and smog. This extra 7 percent can be computed residually as:

Interaction Effect = Total effect – (smoking effect + smog effect) = 37 percent

= 37 percent - (20 percent + 10 percent)
= 37 percent - 30 percent
= 7 percent

In experiments we have the pretests and posttests, in which case we use the same instrument for measuring the dependent variable, for example racial prejudice as an effect of a movie. In pretest is a questionnaire in which items forming the prejudice scale are dispersed at random among other items so that the subject does not know that his or her level of racial prejudice is being measured. Nevertheless, the measurement of this variable (prejudice) itself, by presenting questions about race relations may stimulate the subject's thinking and actually cause a change in his or her level of racial prejudice. Any pretest effect that occurs will be visible as part of extraneous change (change caused by the test stimulus) in the control group, as the pretest is also presented to the control group. Any change between the pretest and posttest for measuring the dependent variable in the control group may be attributed to the sensitization of the subjects with the instrument. In the experimental group of course a movie (an X variable) was shown due to which we expect a change in the racial prejudice of the subjects. But that is not all. The subjects in the experimental group were also exposed to the instrument for measuring the racial prejudice, hence they were also sensitized. Their posttest results include the combined effect of exposure to a movie and that of sensitization to the instrument. In other words the racial prejudice of the subjects in the experimental group exhibits the interaction effect of the treatment plus that of sensitization of the instrument.

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In order to calculate the interaction effect in the experiment we shall have two experimental groups and one control group created by using the randomization process. It may look like this:

Experimental group 1:	**Pretest (O1)**	**X**	**Posttest (O2)**
Control group:	Pretest (O3)	-	Posttest (O4)

Why O4 be different from O3? The difference may be due to sensitization. So let us figure it out. Let us take another experimental group and we do not pretest i.e. no sensitization with the instrument.

Experimental group 2:	**No pretest**	**X**	**Posttest (O5)**

Let us work out the results:

(O2- O1) = D
(O4- O3) = D/
(O5 – O3)= D// (Since all groups are identical, so we can use the pretest of any of the Other two groups)
Interaction effect = D – [D/ + D//]. Substituting it with our example of lung cancer →
37 - [10 + 20] = 37 – 30 = 7

There are many other experimental designs like the randomized block design, Latin square design, natural group design, and factorial design.

LESSON 37

USE OF SECONDARY DATA

Existing statistics/documents

Prior to the discussion of secondary data, let us look at the advantages and disadvantages of the use of content analysis that was covered in the last lecture. In a way content analysis is also the study of documents through which the writers try to communicate, though some of the documents (like population census) may simply contain figures.

Advantages

1. Access to inaccessible subjects: One of the basic advantages of content analysis is that it allows research on subjects to which the researcher does not have physical access. These could be people of old civilizations, say their marriage patterns. These could also be the documents form the archives, speeches of the past leaders (Quaid-e-Azam) who are not alive, the suicide notes, old films, dramas, poems, etc.

2. Non-reactivity: Document study shares with certain types of observations (e.g., indirect observation or non participant observation through one-way mirror) the advantage of little or no reactivity, particularly when the document was written for some other purpose. This is unobtrusive. Even the creator of that document, and for that matter the characters in the document, is not in contact with the researcher, who may not be alive.

3. Can do longitudinal analysis: Like observation and unlike experiments and survey, document study is especially well suited to study over a long period of time. Many times the objective of the research could be to determine a trend. One could pick up different periods in past and try to make comparisons and figure out the changes (in the status of women) that may have occurred over time. Take two martial periods in Pakistan, study the news papers and look at the reported crime in the press.

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4. Use Sampling: The researcher can use random sampling. One could decide on the population, develop sampling frame and draw sample random sample by following the appropriate procedure. For example how women are portrayed in weekly English news magazines. One could pick up weekly English news magazines, make a listing of articles that have appeared in the magazines (sampling frame), and draw a simple random sample.

5. Can use large sample size: Larger the sample closer the results to the population. In experimentation as well as in survey research there could be limitations due to the availability of the subjects or of the resources but in document analysis the researcher could increase the sample and can have more confidence in generalization. Let us assume that a researcher is studying the matrimonial advertisements in the newspapers over a long period of time, there should be no problem in drawing a sample as large as several thousand or more.

6. Spontaneity: The spontaneous actions or feelings can be recorded when they occurred rather than at a time specified by the researcher. If the respondent was keeping a diary, he or she may have been recording spontaneous feelings about a subject whenever he or she was inspired to do so. The contents of such personal recording could be analyzed later on.

7. Confessions: A person may be more likely to confess in a document, particularly one to be read only after his or her death, than in an interview or mailed questionnaire study. Thus a study of documents such as diaries, posthumously published autobiographies, and suicide notes may be the only way to obtain such information.

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8. Relatively low cost: Although the cost of documentary analysis can vary widely depending on the type of document analyzed, how widely documents are dispersed, and how far one must travel to gain access to them, documentary analysis can be inexpensive compared to large-scale surveys. Many a time's documents are gathered together in a centralized location such as library where the researcher can study them for only the cost of travel to the repository.

9. High quality: Although documents vary tremendously in quality, many documents, such as news paper columns, are written by skilled commentators and may be more valuable than, for example, poorly written responses to mailed questionnaires.

Disadvantages

1. Bias: Many documents used in research were not originally intended for research purposes. The various goals and purposes for which documents are written can bias them in various ways. For example, personal documents such as confessional articles or autobiographies are often written by famous people or people who had some unusual experience such as having been a witness to a specific event. While often providing a unique and valuable research data, these documents usually are written for the purpose of making money. Thus they tend to exaggerate and even fabricate to make good story. They also tend to include those events that make the author look good and exclude those that cast him or her in a negative light.

2. Selective survival: Since documents are usually written on paper, they do not withstand the elements well unless care is taken to preserve them. Thus while documents written by famous people are likely to be preserved, day-to-day documents such as letters and diaries written by common people tend either to be destroyed or to be placed in storage and thus become inaccessible. It is relatively rare for common documents that are not about some events of immediate interest to the researcher (e.g., suicide) and not about famous occurrence or by some famous person to be gathered together in a public repository that is accessible to researchers.

3. Incompleteness: Many documents provide incomplete account to the researcher who has had no prior experience with or knowledge of the events or behavior discussed. A problem with many personal documents such as letters and diaries is that they were not written for research purposes but were designed to be private or even secret. Both these kinds of documents often assume specific knowledge that researcher unfamiliar with certain events will not possess. Diaries are probably the worst in this respect, since they are usually written to be read only by the author and can consist more of "soul searching" and confession than of description. Letters tend to be little more complete, since they are addressed to a second person. Since many letters assume a great amount of prior information on the part of the reader.

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4. Lack of availability of documents: In addition to thee bias, incompleteness, and selective survival of documents, there are many areas of study for which no documents are available. In many cases information simply was never recorded. In other cases it was recorded, but the documents remain secret or classified, or have been destroyed.

5. Sampling bias: One of the problems of bias occurs because persons of lower educational or income levels are less likely to be represented in the sampling frames. The problem of sampling bias by educational level is more acute for document study than for survey research. It is a safe generalization that a poorly educated people are much less likely than well educated people to write documents.

6. Limited to verbal behavior: By definition, documents provide information only about respondent's verbal behavior, and provide no direct information on the respondent's nonverbal behavior, either that of the document's author or other characters in the document.

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7. Lack of standardized format: Documents differ quite widely in regard to their standardization of format. Some documents such as newspapers appear frequently in a standard format. Large dailies always contain such standard components as editorial page, business page, sports page, and weather report. Standardization facilitates comparison across time for the same newspapers and comparison across different newspapers at one point in time. However, many other documents, particularly personal documents have no standard format. Comparison is difficult or impossible, since valuable information contained in the document at one point in time may be entirely lacking in an earlier or later documents.

8. Coding difficulties: For a number of reasons, including differences in purpose for which the documents were written, differences in content or subject matter, lack of standardization, and differences in length and format, coding is one of the most difficult tasks facing the content analyst. Documents are generally written arrangements, rather than numbers are quite difficult to quantify. Thus analysis of documents is similar to analysis of open-ended survey questions.

9. Data must be adjusted for comparability over time: Although one of the advantages of document study is that comparisons may be made over a long period of time, since external events cause changes so drastic that even if a common unit of measure is used for the entire period, the value of this unit may have changed so much over time that comparisons are misleading unless corrections are made. Look at the change in measuring distance, temperature, currency, and even literacy in Pakistan.

Use of Secondary Data: Existing Statistics/Documents

Secondary Data

Secondary data refer to information gathered by someone other than the researcher conducting the present study. Secondary data are usually historical, already assembled, and do not require access to respondents or subjects. Many types of information about the social and behavioral world have been collected and are available to the researcher. Some information is in the form of statistical documents (books, reports) that contain numerical information. Other information is in the form of published compilations available in a library or on computerized records. In either case the researcher can search through collections of information with a research question and variables in mind, and then reassemble the information in new ways to address the research question.

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Secondary data may be collected by large bureaucratic organization like the Bureau of Statistics or other government or private agencies. These data may have been collected for policy decisions or as part of public service.

The data may be a time bound collection of information (population census) as well as spread over long periods of time (unemployment trends, crime rate). Secondary data are used for making comparisons over time in the country (population trends in the country) as well as across the countries (world population trends).

Selecting Topic for Secondary Analysis

Search through the collections of information with research question and variables in mind, and then reassemble the information in new ways to address the research question.

It is difficult to specify topics that are appropriate for existing statistics research because they are so varied. Any topic on which information has been collected and is publicly available can be studied. In fact, existing statistics projects may not neatly fit into a deductive model of research design. Rather researchers creatively recognize the existing information into the variables for a research question after first finding what data are available.

Experiments are best for topics where the researcher controls a situation and manipulates an independent variable. Survey research is best for topics where the researcher asks questions and learns about reported attitudes and behavior. Content analysis is for topics that involve the content of messages in cultural communication.
Existing statistics research is best for topics that involve information collected by large bureaucratic organizations. Public or private organizations systematically gather many types of information. Such information is collected for policy decisions or as a public service. It is rarely collected for purposes directly related to a specific research question. Thus existing statistics research is appropriate when a researcher wants to test hypotheses involving variables that are also in official reports of social, economic and political conditions. These include descriptions of organizations or people in them. Often, such information is collected over long periods. For example, existing statistics can be used by researcher who wants to see whether unemployment and crime rates are associated in 100 cities across a 20 year period.

As part of the trends, say in development, researchers try to develop social indicators for measuring the well being of the people. A *social indicator* is any measure of wellbeing used in policy. There are many specific indicators that are operationalization of well-being. It is hoped that information about social well being could be combined with widely used indicators of economic performance (e.g., gross national product) to better inform government and other policy making officials.

The main sources of existing statistics are government or international agencies and private sources. An enormous volume and variety of information exists. If you plan to conduct existing statistics research, it is wise to discuss your interests with an information professional – in this case, a reference librarian, who can point you in the direction of possible sources.
Many existing documents are "free" – that is, publicly available at libraries – but the time and effort it takes to research for specific information can be substantial. Researchers who conduct existing statistics research spend many hours in libraries or on the internet.
There are so many sources of existing statistics like: UN publications, UNESCO Statistical Yearbook, UN Statistical Yearbook, Demographic Yearbook, Labor Force Survey of Pakistan, and Population Census Data.

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Secondary Survey Data

Secondary analysis is a special case of existing statistics; it is reanalysis of previously collected survey or other data that was originally gathered by others. As opposed to primary research (e.g., experiments, surveys, and content analysis), the focus is on analyzing rather than collecting data.

Secondary analysis is increasingly used by researchers. It is relatively inexpensive; it permits comparisons across groups, nations, or time; it facilitates replication; and permits asking about issues not thought by the original researchers. There are several questions the researcher interested in secondary research should ask: Are the secondary data appropriate for the research question? What theory and hypothesis can a researcher use with the data? Is the researcher already familiar with the substantive area? Does the researcher understand how the data were originally gathered and coded?
Large-scale data collection is expensive and difficult. The cost and time required for major national surveys that uses rigorous techniques are prohibitive for most researchers. Fortunately, the organization, preservation, and dissemination of major survey data sets have improved. Today, there are archives of past surveys open to researchers (e.g., data on Population Census of Pakistan, Demographic Survey of Pakistan).

Reliability and Validity

Existing statistics and secondary data are not trouble free just because a government agency or other source gathered the original data. Researchers must be concerned with validity and reliability, as well as with some problems unique to this research technique.

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A common error is the *fallacy of misplaced concreteness*. It occurs when someone gives a false impression of accuracy by quoting statistics in greater detail than warranted by how the statistics are collected and by overloading detail. For example, in order to impress an audience, a politician might say that every year 3010,534 persons, instead of saying 3 million persons, are annually being added to the population of Pakistan.

Validity: Validity problems occur when the researcher's theoretical definition does not match that of the government agency or organization that collected the information. Official policies and procedures specify definitions for official statistics. For example, a researcher defines a *work injury* as including minor cuts, bruises, and sprains that occur on the job, but the official definition in government reports only includes injuries that require a visit to a physician or hospital. Many work injuries as defined by thee researcher will not be in the official statistics. Another example occurs when a researcher defines people *unemployed* if they would work if a good job was available, if they have to work part-time when they want full-time work, and if they have given up looking for work. The official definition, however, includes only those who are now actively seeking work (full or part-time) as unemployed. The official statistics exclude those who stopped looking, who work part-time out of necessity, or who do not look because they believe no work is available. In both the cases the researcher's definition differs from that in official statistics.

Another validity problem arises when official statistics are a proxy for a construct in which the researcher is really interested. This is necessary because the researcher cannot collect original data. For example, the researcher wants to know how many people have been robbed, so he or she uses police statistics on robbery arrests as a proxy. But the measure is not entirely valid because many robberies are not reported to the police, and reported robberies do not always result in an arrest.

Another validity problem arises because the researcher lacks control over how information is collected. All information, even that in official government reports, is originally gathered by people in bureaucracies as part of their job. A researcher depends on them for collecting organizing, reporting, and publishing data accurately. Systematic errors in collecting the initial information (e.g., census people who avoid poor neighborhoods and make-up information, or people who put a false age on their ID card); errors in organizing and reporting information (e.g., police department that is sloppy about filing crime reports and loses some); errors in publishing information (e.g., a typographical error in a table) all reduce measurement validity.

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Reliability: Stability reliability problems develop when official definition or the method of collecting information changes over time. Official definitions of work injury, disability, unemployment, literacy, poverty, and the like change periodically. Even if the researcher learns of such changes, consistent measurement over time is impossible.

Equivalence reliability can also be a problem. For example, studies of police department suggest that political pressures to increase arrests are closely related to the number of arrests. It could be seen when political pressures in one city may increase arrests (e.g., a crackdown on crime), whereas pressures in another city may decrease arrests (e.g., to show drop in crime shortly before an election in order to make officials look better).
Researchers often use official statistics for international comparisons but national governments collect data differently and the quality of data collection varies.

Inferences from Non-Reactive Data:
A researcher's ability to infer causality or to test a theory on the basis of non-reactive data is limited. It is difficult to use unobtrusive measures to establish temporal order and eliminate alternative explanations. In content analysis, a researcher cannot generalize from the content to its effects on those who read the text, but can only use the correlation logic of survey research to show an association among variables. Unlike the case of survey research, a researcher does not ask respondents direct questions to measure variables, but relies on the information available in thee text.

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LESSON 38

OBSERVATION STUDIES/FIELD RESEARCH

Observation studies are primarily part of qualitative research. Though qualitative and quantitative researches differ yet they compliment each other. Qualitative research produces *soft data*: impressions, words, sentences, photos, symbols. Usually it follows an interpretive approach, the goal of which is to develop an understanding of social life and discover how people construct meanings in natural settings. The research process follows a non-linear approach (spiral).

Quantitative research produces *hard data*: numbers. It follows a positivist approach to research in which the researcher speaks the language of variables and hypotheses. There is a much emphasis on precise measurement of variables and the testing of hypotheses. The researcher tries to establish causality. In most of the case there is a linear approach i.e. it follows sequential steps in doing research.

Participant/Non-Participant Observation

Observation studies can be participant or non-participant. In participant observation the researcher directly observes and participates in small scale social settings in the present time. Such a study is also referred to as **field research**, ethnography, or anthropological study. Here the researchers:

- Study people in their natural settings, or in situ.
- Study people by directly interacting with them.
- Gain an understanding of the social world and make theoretical statements about members' perspective.

The people could be a group who interact with each other on regular basis in a field setting: a street corner, a tea shop, a club, a nomad group, a village, etc.

Non-participant studies are such where the research tries to observe the behavior of people without interacting with them. It could be observing the behavior of shoppers in a departmental store through a mirror or on a closed circuit TV. Some body might be counting the number of vehicles crossing a particular traffic light.

Field researchers study people in a location or setting. It has been used to study entire communities. Field research has a distinct set of methodologies. Field researchers directly observe and interact with community members in natural settings to get inside their perspectives. They embrace an activist or social constructionist perspective on social life. They do not see people as a neutral medium through which social forces operate, nor do they see social meanings as something "out there" to observe. Instead they believe that people create and define the social world through their interactions. Human experiences are filtered through a subjective sense of reality, which affects how people see and act on events. Thus they replace the positivist emphasis on "objective facts" with a focus on the everyday, face-to-face social processes of negotiation, discussion, and bargaining to construct social meaning.

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Ethnography and Ethno-methodology

Two modern extensions of field research, ethnography and ethno-methodology, build on the social constructionist perspective.

Ethnography comes from cultural anthropology. *Ethno* means people or a folk distinct by their culture and graphy refers to describing something. Thus ethnography means describing a culture and understanding another way of life from the native point of view. It is just an understanding the culture of people from their own perspective.

Ethno-methodology implies how people create reality and how they interpret it. Ethno-methodologists examine ordinary social interaction in great detail to identify the rules for constructing social reality and

common sense, how these rules are applied, and how new rules are created. They try to figure out how certain meanings are attached to a reality.

Logic of Field Research

It is difficult to pin down a specific definition of *field research* because it is more of an orientation toward research than a fixed set of techniques to apply. A field researcher uses various methods to obtain information. A field researcher is a 'methodological pragmatist,' a resourceful, talented individual who has ingenuity and an ability to think on his or feet while in the field.

Field research is based on naturalism, which involves observing ordinary events in natural settings, not in contrived, invented, or researcher created settings.

A field researcher examines social meanings and grasps multiple perspectives in natural setting. He or she gets inside the meanings system of members and goes back to an outside or research viewpoint. Fieldwork means involvement and detachment, loyalty and betrayal, both openness and secrecy, and most likely, love and hate. The researcher switches perspectives and sees the setting from multiple pints of view simultaneously. Researchers maintains membership in the culture in which they were reared (research culture) while establishing membership in the groups which they are studying.

The researcher's direct involvement in the field often has an emotional impact. Field research can be fun and exciting, but it can also disrupt one's personal life, physical security, or mental well being. More than other types of research, it reshapes friendship, family life, self identity, or personal values.

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What Do the Field Researchers Do?

A field researcher does the following:

1. Observes ordinary events and everyday activities as they happen in natural settings, in addition to unusual occurrences.
2. Becomes directly involved with people being studied and personally experiences the process of daily life in the field setting.
3. Acquires an insider's point of view while maintaining the analytic perspective or distance of an outsider.
4. Uses a variety of techniques and social skills in a flexible manner as the situation demands.
5. Produces data in the form of extensive, written notes, as well as diagrams, maps, pictures to provide very detailed descriptions.
6. Sees events holistically (as a whole, not in pieces) and individually in their social context.
7. Understands and develops empathy for members in a field setting, and does not just record 'cold' objective facts.
8. Notices both explicit (recognized, conscious, spoken) and tacit (less recognized, implicit, unspoken) aspects of culture.
9. Observes ongoing social processes without upsetting, or imposing an outside point of view.
10. Copes with high levels of personal stress, uncertainty, ethical dilemmas, and ambiguity.

Steps in Field Research

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. Here is just the listing of these steps:

1. Prepare yourself, read the literature and defocus.
2. Select a site and gain access.

3. Enter the field and establish social relations with members.
4. Adopt a social role, learn the ropes, and get along with members.
5. Watch, listen, and collect quality data. .
6. Begin to analyze data, generate and evaluate working hypothesis.
7. Focus on specific aspects of the setting and use theoretical sampling.
8. Conduct field interviews with member informants.
9. Disengage and physically leave the setting.
10. Complete the analysis and write the report.

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LESSON 39

OBSERVATION STUDIES (Contd.)

Steps in Field Research

Background

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. These guideline steps are:

1. ***Prepare yourself, read the literature and defocus.*** As with all social and behavioral research, reading the scholarly literature helps the researcher learn concepts, potential pitfalls, data collection methods, and techniques for resolving conflicts. In addition field researcher finds diaries, novels, journalistic accounts, and autobiographies useful for gaining familiarity and preparing emotionally for the field. Field research begins with a general topic, not specific hypotheses. A researcher does not get locked into any initial misconceptions. He or she needs to be well informed but open to discovering new ideas.

   A researcher first empties his or her mind of preconceptions and defocuses. There are two types of defocusing. The first is casting a wide net in order to witness a wide range of situations, people, and setting – getting a feel of the overall setting before deciding what to include or exclude. The second type of defocusing means not focusing exclusively on the role of researcher. It may be important to extend one's experience beyond a strictly professional role.

   Another preparation for field research is self knowledge. A field researcher needs to know him or herself and reflect on personal experiences. He or she can expect anxiety, self doubt, frustration, and uncertainty in the field. Also all kinds of stereotypes about the community should be emptied.
2. ***Select a site and gain access.*** Although a field research project does not proceed by fixed steps, some common concerns arise in the early stages. These include selecting a site, gaining access to the site, entering the field, and developing rapport with members in the field.

   Field site is the context in which events or activities occur, a socially defined territory with shifting boundaries. A social group may interact across several physical sites. For example, a college football team may interact on the playing field, in the dressing room, at a training camp or at the place where they are staying. The team's field site includes all four locations.

   Physical access to a site can be an issue. Sites can be on a continuum, with open and public areas (e.g., public restaurants, airport waiting rooms) at one end and closed and private settings (e.g., private firms, clubs, activities in a person's home) at the other end. A researcher may find that he or she is not welcome or not allowed on the site, or there are legal and political barriers to access.

   Look for the gate keepers for getting an entry. A gatekeeper is someone with the formal authority to control access to a site. It can be a thug at the corner, an administrator of a hospital, or the owner of a business. In formal public areas (e.g., sidewalks, public waiting rooms) rarely have gatekeepers; formal organizations have authorities from whom permission must be obtained. Field researchers expect to negotiate with gatekeepers and bargain for access. Entry and access can be visualized as an *access ladder*. A researcher begins at the bottom rung, where access is easy and where he or she is an outsider looking for public information. The next access rung requires increased access. Once close on-site observations begin, he or she becomes a passive observer, not questioning what members of community say. With time in the field, the researcher observes specific activities that are potentially sensitive or seeks clarification of what he or she sees or hears. Reaching this access rung is more difficult. Finally, the researcher may try to shape interaction so that it reveals specific information, or he or she may want to see highly sensitive material. This highest rung of access ladder is rarely

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attained and requires deep trust. Such a situation may be applicable to a site of a public or private organization. In other situations just like entering the village community, the researcher may have to use different kind of access ladder. He or she may have to use local influential and some other contact persons who could introduce the researcher to local leaders and help building the rapport.

3. ***Enter the field and establish social relations with members.*** Present yourself in the field the way it is acceptable to the people to be studied. Develop relations and establish rapport with individual members. Here the researcher may have to learn the local language. A field researcher builds rapport by getting along with members in the field. He or she forges a friendly relationship, shares the same language, and laughs and cries with members. This is a step toward obtaining an understanding of members and moving beyond understanding to empathy – that is seeing and feeling events from another's perspective.

4. ***Enter the field: Adopt a social role, learn the ropes, and get along with members.*** At times, a researcher adopts an existing role. Some existing roles provide access to all areas of the site, the ability to observe and interact with all members, the freedom to move around, and a way to balance the requirements of researcher and member. There could be some limitations for the adoption of specific roles. Such limitations may be because of researcher's age, race, gender, and attractiveness. At other times, a researcher creates new roles or modifies the existing one. The adoption of field role takes time, and a researcher may adopt several different field roles over time.

   The role may also depend upon the level of involvement in the community's activities. The researcher may be a complete observer, observer as participant, participant as observer, and complete participant.

   As a researcher learns the ropes on the field site, he or she learns how to cope with personal stress, how to normalize the social research, and how to act like an "acceptable incompetent." A researcher is in the field to learn, not to be an expert. Depending on the setting, he or she appears to be friendly but naïve outsider, an acceptable incompetent who is interested in learning about social life of the field. An acceptable incompetent is one who is partially competent (skilled or knowledgeable) in the setting but who s accepted as a non-threatening person

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5. ***Observing and collecting data: Watch, listen, and collect quality data.*** A great deal of what field researchers do in the field is to pay attention, watch, and listen carefully. They use all the senses, noticing what is seen, heard, smelled, tasted, or touched. The researcher becomes an instrument that absorbs all sources of information.

   Most field research data are in the form of field notes. Good notes are the brick and mortar of field research. Full field notes can contain maps, diagrams, photographs, interviews, tape recordings, videotapes, memos, objects from the field, notes jotted in the field, and detailed notes written away from the field. A field researcher expects to fill many notebooks, or the equivalent in computer memory. He or she may spend more time writing notes than being in the field.

   Writing notes is often boring, tedious work that requires self discipline. The notes contain extensive descriptive detail drawn from memory. The researcher makes it a daily habit or compulsion to write notes immediately after leaving the field. The notes must be neat and organized because the researcher will return to them over and over again. Once written, the notes are private and valuable. A researcher treats them with care and protects confidentiality.

   Field researcher is supposed to collect quality data. What does the term high-quality data mean in the field research, and what does a researcher do to get it? For a quantitative researcher, high quality data are reliable and valid; they give precise, consistent measures of the "objective" truth for all researchers. An interpretive approach suggests a different kind of data quality. Instead of assuming one single, objective truth, field researchers hold that members subjectively interpret experiences within social context. What a member takes to be true results from social interaction and interpretation. Thus high quality field data capture such processes and provide an understanding of the member's viewpoint.

A field researcher does not eliminate subjective views to get quality data: rather, quality data include his or her subjective responses and experiences. Quality field data are detailed descriptions from the researcher's immersion and authentic experiences in the social world of members.

6. ***Begin to analyze data generate and evaluate working hypothesis.*** Right in the field try to look into the research questions and the kind of answers the researcher is getting. The analysis of the answers might help in the generation of hypotheses. Over time are such hypotheses being supported by further field research?
7. ***Focus on specific aspects of the setting and use theoretical sampling.*** Field researcher first gets a general picture, and then focuses on a few specific problems or issues. A researcher decides on specific research questions and develops hypotheses only after being in the field and experiencing it first hand. At first, everything seems relevant; later, however, selective attention focuses on specific questions and themes.

   Field research sampling differs from survey sampling, although sometime both use snowball sampling. A field researcher samples by taking a smaller, selective set of observations from all possible observations. It is called theoretical sampling because it is guided by the researcher's developing theory. Field researchers sample times, situations, types of events, locations, types of people, or context of interest.

   For example field researcher samples time by observing a setting at different times. He or she observes at all time of the day, on every day of the week, and in all seasons to get a full sense of how the field site stays the same or changes. Another example, when the field researcher samples locations because one location may give depth, but narrow perspective. Sitting or standing in different locations helps the researcher to get a sense of the whole site. Similarly the field researchers sample people by focusing their attention or interaction on different kinds of people (young, adult, old).

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8. ***Conduct field interviews with member informants.*** Field researchers use unstructured, non directive, in-depth interviews, which differs from formal survey research interviews in many ways. The field interview involves asking question, listening, expressing interest, and recording what was said.

   Field interview is a joint production of a research and a member. Members are active participants whose insights, feelings, and cooperation are essential parts of a discussion process that reveals subjective meaning. The interviewer's presence and form of involvement – how he or she listens, attends, encourages, interrupts, disagrees, initiates topics, and terminates responses – is integral to the respondent's account.

   Field research interviews go by many names: unstructured, depth, ethnographic, open ended, informal, and long. Generally, they involve one or more people being present, occur in the field, and are informal and nondirective.

   A comparison of the field research interview and a survey interview could be as below:

Survey Interview	Field Interview
1. It has clear beginning and end.	1. The beginning and end are not clear. The interview can be picked up later.
2. The same standard questions are asked of all respondents in the same sequence.	2. The questions and the order in which they are asked are tailored to specific people and situations.
3. The interviewer appears neutral at all times.	3.The interviewer shows interest in responses, encourages elaboration.
4. The interviewer asks questions, and the respondent answers.	4. It is like a friendly conversational exchange but with more interviewer questions.

5. It is almost always with one respondent alone.	5. It can occur in group setting or with others in area, but varies.
6. It has a professional tone and businesslike focus, diversions are ignored.	6. It is interspersed with jokes, aside, stories, diversions, and anecdotes, which are recorded.
7. Closed-ended questions are common, with rare probes.	7. Open-ended questions are common, and probes are frequent.
8. The interviewer alone controls the pace and direction of interview.	8. The interviewer and member jointly control the pace and direction of the interview.
9. The social context in which the interview occurs is ignored and assumed to make little difference.	9. The social context of the interview is noted and seen as important for interpreting the meaning of responses.
10. The interviewer attempts to mold the communication pattern into a standard framework.	10. The interviewer adjusts to the member's norms and language usages.

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9. ***Disengage and physically leave the setting.*** Work in the field can last for a few weeks to a dozen years. In either case at some point of work in the field ends. Some researchers suggest that the end comes naturally when the theory building ceases or reaches a closure; others feel that fieldwork could go on without end and that a firm decision to cut off relations is needed.
Experienced field researchers anticipate a process of disengaging and exiting the field. Depending on the intensity of involvement and the length o time in the field, the process can be disruptive or emotionally painful for both the researcher and the members.
Once researcher decides to leave – because the project reaches a natural end and little new is being learned, or because external factors force it to end (e.g., end of job, gatekeepers order the researcher out) – he or she chooses a method of exiting. The researcher can leave by quick exit (simply not return one day) or slowly withdraw, reducing his or her involvement over weeks. He or she also needs to decide how to tell members and how much advance warning to give. The best way to exist is to follow the local norms and continuing with the friendly relations.
10. ***Complete the analysis and write the report.*** After disengaging from the field setting the researcher writes the report. The researcher may share the written report with the members observed to verify the accuracy and get their approval of its portrayal in print. It may help in determining the validity of the findings. However, it may not be possible to share the findings with marginal groups like addicts, and some deviant groups.

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Ethical Dilemmas of Field research

The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has little time to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Let us look at some of these dilemmas:

Deception: Deception arises in several ways in field research: The research may be covert; or may assume a false role, name, or identity; or may mislead members in some way. The most hotly debated of the ethical issues arising from deception is that of covert versus overt field research. Some support it and see it as necessary for entering into and aiming a full knowledge of many areas of social life. Others oppose it and argue that it undermines a trust between researchers and society. Although its

moral status is questionable, there are some field sites or activities that can only be studied covertly. One may have to look into the cost and benefit equation; where the researcher is the best judge.

Covert research is never preferable and never easier than overt research because of the difficulties of maintaining a front and the constant fear of getting caught.

Confidentiality: A researcher learns intimate knowledge that is given in confidence. He or she has a moral obligation to uphold the confidentiality of data. This includes keeping information confidential from others in the field and disguising members' names in field notes.

Involvement with deviants: Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal activity. They might be getting 'guilty knowledge.' Such knowledge is of interest not only to law enforcement officials but also to other deviants. The researcher faces a dilemma of building trust and rapport with the deviants, yet not becoming so involved as to violate his or her basic personal moral standards. Usually, the researcher makes an explicit arrangement with the deviant members.

The powerful: Field researchers tend to study those without power in society (e.g., street people, the poor, children, and lower level workers). Powerful elites can block access and have effective gatekeepers. Researchers are criticized for ignoring the powerful, and they are also criticized by the powerful for being biased toward the less powerful.

Publishing field reports: The intimate knowledge that a researcher obtains and reports creates a dilemma between the right of privacy and the right to know. A researcher does not publicize member secrets, violate privacy, or harm reputations. Yet if he or she cannot publish anything that might offend or harm someone, some of what the researcher learned will remain hidden, and it may be difficult for others to believe the report if critical details are omitted.

Some researchers suggest asking members of the group under study to look at a report to verify its accuracy and to approve of their portrayal in print. For marginal groups (addicts), this may not be possible, but the researchers must always respect member privacy. On the other hand, censorship or self-censorship can be a danger. A compromise position is that truthful but unflattering material may be published only if it is essential to the researchers' larger arguments.

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LESSON 42

FOCUS GROUP DISCUSSION

A visitor to a locality stops by a house and inquires about the address of a resident he wants to see. May be he starts talking with a couple of persons asking for their help. In the meantime, some other passersby, or coming out of other houses join, showing their curiosity about the issue. They ask for some more information about the resident concerned, and then start discussing among them to come up with the exact identification of the resident. As an outcome of this discussion they would guide the visitor to reach the destination. This is quite a common feature in a folk society (village, neighborhood in a city) where we may start talking with a couple of persons and others come and join the conversation. This is an example of *informal* focus group discussion, which is built upon the social networks that operate in a natural setting. These social networks include both kinsfolk and other neighbors. In some cases the participants may be the local decision makers.

In research, focus group discussions (FGD) are a more formal way of getting groups of people to discuss selected issues. A focus group discussion is a group discussion of 6-12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic. There may be some disagreement about the exact number of participants in the discussion, as one comes across variations in numbers (6 to 10, 6 to 12, 6 to15, 8to 10, 5 to 7) in different books on research methods. The trend has been toward smaller groups due to some problems with the larger groups, which like:

- In a bigger group each participant's speaking time is substantially restricted. Dominant/submissive relationships are almost inevitable.
- Frustration or dissatisfaction among group members is likely to result because of some members' inability to get a turn to speak. This produces lower quality and quantity of data.
- Participants are often forced into long speeches, often containing irrelevant information, when they get to speak only infrequently.
- The tendency for side conversations between participants increases.

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In contrast, smaller group sessions are felt to provide greater depth response for each participant. The group is often more cohesive and interactive, particularly when participants are professionals, such as physicians or pharmacists.

The key factor concerning group size is generally the of group purpose. If the purpose of the group is to generate as many ideas as possible, a larger group may be most useful. If the purpose of the group is to maximize the depth of expression from each participant, a smaller group works better.

The Purpose of FGD

The purpose of an FGD is to obtain in-depth information on concepts, perceptions, and ideas of the group. An FGD aims to be more than a question-answer interaction (Focus group interview is different). Here the idea is that group members discuss the topic among themselves.

Formal Focus Groups

Formal groups are formally constituted, that is these are organized in advance by inviting the selected individuals to participate in the discussion on a specific issue. They are structured groups brought together in which the participants are expected to have similar background, age, sex, education, religion, or similar experiences. Similarity in background is likely to make them comfortable where they could express their viewpoint frankly and freely. If the big boss and his junior officer working in an organization together participate in an FGD, the junior officer may not be able to express his or her opinion freely in the presence of his/her boss. Similarly, in some situations the children may experience some inhibitions in expressing their views on a sensitive issue in the presence of their parents. A lot depends on the kind of issue that is to be discussed.

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The group is guided by a moderator/facilitator. The participants address a specific issue (talk freely, agree or disagree among them) within a specified time in accordance with clearly spelled out rules of procedure.

Designing a Focus group Study

As with other approaches to studying social phenomena, designing a focus group study requires careful thought and reflection. Given that focus groups can be used for a variety of purposes within social research, the design of focus group study will depend on its purpose. At one extreme, FGD is used at the exploratory stage of the study (FGD may help in the identification of variables, formulation of questions and response categories) and at the other extreme, when qualitative information is needed on issues about which the researchers have substantial background knowledge and a reasonable grasp of the issues. Here we are focusing on the latter type of design.

How to conduct FGD?

The following guideline may be provided for conducting FGD.

1. Preparation:

- Selection of topic, questions to be discussed. It is appropriate to define and clarify the concepts to be discussed. The basic idea is to lay out a set of issues for the group to discuss. It is important to bear in mind that the moderator will mostly be improvising comments and questions within the framework set by the guidelines. By keeping the questions open-ended, the moderator ca stimulates useful trains of thought in the participants that were not anticipated.
- Selecting the study participants: Given a clear idea of the issues to be discussed, the next critical step in designing a focus group study is to decide on the characteristics of the individuals who are to be targeted for sessions. It is often important to ensure that the groups all share some common characteristics in relation to the issue under investigation. If you need to obtain information on a topic from several different categories of informants who are likely to discuss the issue from different perspectives, you should organize a focus group for each major category. For example a group for men and a group for women, or a group for older women and group for younger women. The selection of the participants can be on the basis of purposive or convenience sampling. The participants should receive the invitations at least one or two days before the exercise. The invitations should explain the general purpose of the FGD.
- Physical arrangements: Communication and interaction during the FGD should be encouraged in every way possible. Arrange the chairs in a circle. Make sure the area will be quite, adequately lighted, etc., and that there will be no disturbances. Try to hold the FGD in a neutral setting that encourages participants to freely express their views. A health center, for example, is not a good place to discuss traditional medical beliefs or preferences for other types of treatment. Neutral setting could also be from the perspective of a place where the participants feel comfortable to come over and above their party factions.

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Conducting the session:

- One of the members of the research team should act as a "facilitator" or "moderator" for the focus group. One should serve as "recorder."
- ***Functions of the Facilitator:*** The facilitator should not act as an expert on the topic. His or her role is to stimulate and support discussion. He should perform the following functions:
  - **Introduce the session**: He or she should introduce himself/herself as facilitator and intro duce the recorder. Introduce the participants by name or ask them to introduce themselves (or develop some new interesting way of introduction). Put the participants at ease and explain the purpose of the FGD, the kind of information needed, and how the information will be used (e.g., for planning of a health program, an education program, et.).
  - **Encourage discussion:** The facilitator should be enthusiastic, lively, and humorous and show his/her interest in the group's ideas. Formulate questions and encourage as many participants as

possible to express their views. Remember there are no "right" or "wrong" answers. Facilitator should react neutrally to both verbal and nonverbal responses.

- **Encourage involvement:** Avoid a question and answer session. Some useful techniques include: asking for clarification (can you tell me more?); reorienting the discussion when it goes off the track (saying: wait, how does this relate to the issue? Using one participant's remarks to direct a question to another); bringing in reluctant participants (Using person's name, requesting his/her opinion, making more frequent eye contact to encourage his participation); dealing with dominant participants (avoiding eye contact or turning slightly away to discourage the person from speaking, or thanking the person and changing the subject).
- **Avoid being placed in the role of expert:** When the facilitator is asked for his/her opinion by a respondent, remember that he or she is not there to educate of inform. Direct the question back to the group by saying: "What do you think?" "What would you do?" Set aside time, if necessary, after the session to give participants the information they have asked.

Do not try to give comments on everything that is being said. Do not feel you have to say Something during every pause in the discussion. Wait a little and see what happens.

- **Control the timing of the meeting but unobtrusively:** Listen carefully and move the discussion from topic to topic. Subtly control the time allocated to various topics so as to maintain interest. If the participants spontaneously jump from one topic to the other, let the discussion continue for a while because useful additional information may surface and then summarize the points brought up and reorient the discussion.
- **Take time at the end of the meeting to summarize, check for agreement and thank the participants:** Summarize the main issues brought up, check whether all agree and ask for additional comments. Thank the participants and let them know that their ideas had been valuable contribution and will be used for planning the proposed research/intervention/or what ever the purpose of FGD was.

  Listen to the additional comments made after the meeting. Sometime some valuable information surfaces, which otherwise may remain hidden.

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LESSON 43

FOCUS GROUP DISCUSSION (Contd.)

Functions of the Recorder

The recorder should keep a record of the content of the discussion as well as emotional reactions and important aspects of group interaction. Assessment of the emotional tone of the meeting and the group process will enable the researcher to judge the validity of the information collected during the FGD. Record the following:

- Date, time, and place:
- Names and characteristics of participants:
- General description of the group dynamics (level of participation, presence of a dominant participant, level of interest):
- Opinions of participants, recorded as much as possible in their own words, especially for key statements: and
- Vocabulary used, particularly in focus group discussions that are intended to assist in developing questionnaire or other material as stipulated under the topic.

It is highly recommended that a tape/video recorder (with permission) be used to assist capturing information. Even if a tape/video recorder is used, notes should be taken as well, incase the machine malfunctions and so that information will be available immediately after the session.
A supplementary role for the recorder could be to assist the facilitator (if necessary) by drawing his/her attention to:

- Missed comments from participants, and
- Missed topics (the recorder should have a copy of the discussion guide, key probe questions during the FGD).

If necessary, the recorder could also help resolve conflict situations that facilitator may have difficulty handling.

Number and duration of sessions: The number of **focus group sessions** to be conducted depends upon project needs, resources, and whether new information is still coming from the sessions (that is, whether contrasting views from various groups in the community are still emerging).

One should plan to conduct at least two different focus group discussions for each subgroup (for example two for males and two for females).

For **duration**, a focus group session typically lasts up to an hour and a half. Generally the first session with a particular type of group is longer than the following ones because all of the information is new. Thereafter, if it becomes clear that all the groups have the same opinion on particular topics, the facilitator may be able to move the discussion along more quickly to other topics that still elicit new points of view.

3. Analysis of Results

- After each focus group session, the facilitator and the recorder should meet to review and complete the notes taken during the meeting. This is also the right moment to evaluate how the focus group went and what changes might be made when facilitating future groups.
- A full report of the discussion should be prepared that reflects the discussion as completely as possible using the participants' own words. List the key statements, ideas, and attitudes expressed for each topic of discussion.
- After the transcript of the discussion is prepared, code the statements right away, using the left margin? Write comments in the right margin. Formulate additional questions if certain issues are still unclear or controversial and include them in the next FGD.

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- Further categorize the statements for each topic, if required. Compare answers of different subgroups (e.g., answers of young mothers and answers of mothers of above childbearing age in the FGD on changes in weaning practices).
  The findings must be recorded in coherent manner. For example, if young women in all focus group discussions state that they start weaning some 3-6 months earlier than their mothers did and the women above childbearing age confirm this statement, one is likely to have a solid finding. If findings contradict each other, one may need to conduct some more focus group discussions or bring together representatives from two different subgroups to discuss and clarify the differences.
- Summarize the data in a matrix, diagram, flowchart, or narrative, if appropriate, and interpret the findings.
- Select the most useful quotations that emerged from the discussions to illustrate the main ideas.

4. Report Writing

- Start with a description of the selection and composition of the groups of participants and a commentary on the group process, so the reader can assess the validity of the reported findings.
- Present the findings, following a list of topics and guided by the objective(s) of the FGD. Include quotations whenever possible, particularly for key statements.

Uses of Focus Group Discussions

- The primary advantage of focus groups is its ability to quickly and inexpensively grasp the core issues of the topic. One might see focus group discussions as *synergistic* i.e. the combined effort of the group will produce a wider range of information, insights, and ideas than will the accumulation of separately secured responses of a number of individuals. Even in non-exploratory research, focus group discussions produce a lot more information far more quickly, and at less cost than individual interviews.
- As part of exploratory research, focus group discussions help the researcher to focus on the issue and develop relevant research hypotheses. In the discussions the relevant variables are identified, and relationships are postulated. Once the variables are identified, the same focus group discussions help in the formulation
  of questions, along with the response categories, for the measurement of variables.
- Focus group discussion is an excellent design to get information form non-literates.
- Focus groups discussions are a good means to discover attitudes and opinions that might not be revealed through surveys. This is particularly useful when the researcher is looking at the controversial issue, and the individual might be able to give his opinion as such but not discuss the issue in the light of other viewpoints. In focus group discussions there is usually a *snowballing* effect. A comment by one often triggers a chain of views from other participants.
- Focus group discussions are well accepted in the folk communities, as this form of communication already exists whereby the local communities try to sort out controversial issues.
- Focus group discussions generate new ideas, questions about the issues under consideration. It may be called *serendipity (*surprise ideas). It is more often the case in a group than in an individual interview that some idea will drop out of the blue. The group also affords the opportunity to develop the idea to its full significance.
- Focus group discussions can supplement the quantitative information on community knowledge, attitude, and practice (KAP), which may have already been collected through survey research.
- Focus group discussions are highly flexible with respect to topic, number of participants, time schedule, location, and logistics of discussion.
- Focus group discussions provide a direct link between the researcher and the population under study. In fact most of the focus group discussions are held close to peoples places of living and

work. It helps in getting the realistic picture of the issue directly from the people who are part of it.

- For some researchers, focus group discussions may be a fun. They enjoy discussing the issues directly with the relevant population.

Limitations

- Results of the focus group discussions cannot usually be used for generalization beyond the population from where the participants in FGD came. One important reason being the lack of their representative-ness about other populations.
- It is often seen that participants usually agree with the responses from fellow members (for different reasons). Without a sensitive and effective facilitator, a single, self-appointed participant may dominate the session. Researchers have to be cautious when interpreting the results.
- The moderator may influence focus group discussion and may bias the information.
- Focus group discussions may have limited value in exploring complex beliefs of individuals, which they may not share in open discussion.
- It is possible that focus group discussions may paint a picture of what is socially acceptable in the community rather than what is actually occurring or is believed. The picture may be given of what is ideally desirable and not what is really in practice. Participants may like to project a good image of their community to strangers; hence the information may be highly contaminated.

CASE STUDY

Case study is a comprehensive description and analysis of a single situation or a number of specific situations i.e. cases. It is an intensive description and analysis of a case. Researchers often use qualitative approach to explore the case in as rich a detail as possible. The examples could be a case study of a highly successful organization, a project (Orangi Pilot Project, Karachi), a group, a couple, a teacher, and a patient. In a way it is more like a clinical approach to study the case in detail.

If the researcher is looking at highly successful organization then he may have to look into all the factors that may have contributed to its success. The factors may relate to the availability of the financial resources, the management, the work environment, work force, the political atmosphere, and many more. All these factors may be considered as different dimensions for studying the organization. Similarly, one may do the case study of a happily married couple.

Data Sources

Usually the following sources are suggested:

- Naturalistic observations (ethnographic studies)
- Interviews
- Life histories
- Tests (Psychological, clinical)

In most of the cases the data sources may depend upon the nature of the case under investigation. If we are trying to do the case study of a community, then one shall be looking for naturalistic observations (ethnographic information), in-depth interviews with individuals, life histories of the people, and any thing, which may have previously been written about the community.

Preserve the unitary character of the object under study: The researcher tries to study the case as a whole by collecting the breadth of data about the totality of the unit. For the collection of such data a multidisciplinary approach may be used, which could help looking at the case from different

perspectives prior to coming to some conclusions. Hence it is not a segmental study; therefore effort is made to study it as a whole and while making the analysis try to present it as a unit.

Case Control studies

It is also possible to select two groups (taking them as cases), one with an effect (study group) and the other without effect (control group). Both the cases are similar except for the effect. One could look at the case of Manga Mandi village, where, a few years back, deformities in the bones of children were observed in one part of the village. Here one could explore the totality of the background of affected and unaffected parts of the locality, each being treated as a unit. One could develop hypothesis by having an in-depth analysis of the affected and unaffected parts.

Case study is empirical

Case study is empirical because:

- It investigates a contemporary phenomenon within its real life context. It is retrospective study in which the researcher follows the research process from effect to its cause. It is a study back in time. Just like a medical practitioner who is treating his patient as a case, tries to diagnose his/her ailment by taking the case history, doing the physical examination, and if necessary, doing some laboratory tests. On the basis of the triangulation of all this information the medical doctor traces the cause of patient's present ailment. The information is empirical.
- When the boundaries between the phenomenon and context are not clearly evident, the researcher tries to use multiple sources of evidence. One could say that the researcher is trying to look at the case by using multiple dimensions, and trying to come up with a finding that is empirical.

Limitations

Despite the fact that the case study may be considered empirical yet it lack rigor in its approach. Therefore it has limitations with respect to the reliability of the findings. Also one could question whether the case is representative of some population.

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LESSON 44

REPORT WRITING

Although every report is custom-made for the project it represents, some conventions of report format are universal. These conventions have developed over a long period of time, and they represent a consensus about what parts are necessary to a good research report and how they should be ordered. The consensus is not an inviolable law, though. Each report writing book suggests its own unique format and every report writer has to pick and choose the parts and the order that work best for the project at hand. Many companies and universities also have an in-house, suggested report formats or writing guides that researchers should be aware of.

Report format: The general plan of organization for the parts of a written or oral research report. The researchers tailor the format to the project. The format of a research report may need adjustment for two reasons: (1) to obtain the proper level of formality and (2) to decrease the complexity of the report. We shall look at the most formal type i.e. a report for a large project done within an organization or one done by a research agency for a client company. This sort of report is usually bound with a permanent cover and may be hundreds of pages long.

Students who are writing a thesis shall have to follow the format requirements of the university where they shall be submitting it. Thesis format is little different, and it shall be explained as we proceed.

The Makeup of the Report – the Report Parts

- **Prefatory parts**
  1. Title fly page
  2. Title page
  3. Letter of transmittal
  4. Letter of authorization
  5. Table of contents
  6. Executive summary

- **Main body**
  1. Introduction
  2. Methodology
  3. Results
  4. Conclusions and recommendations
  5. References

- **Appended parts**
  1. Data collection forms (questionnaires, checklist, interview guide, other forms)
  2. Detailed calculations
  3. General tables
  4. Other support material
  5. Bibliography, if needed

Let us now look at each one of the parts of the report.

Prefatory Parts

Title Fly Page: Only the title appears on this page. For the most formal reports, a title fly page precedes the title page. Most of the reports don't have it. May be it is more like the dustcover of some books.

Title Page: The title page should include four items: the title of the report, the name(s) of the person(s) for whom the report was prepared, the name(s) of person(s) who prepared it, and the date of release or presentation.

The title should be brief but include three elements: (1) the variables included in the study, (2) the type of relationship among the variables, and (3) the population to which the results may be applied. Redundancies such as "A report of," "A discussion of," and "A study of" add length to title but little else. Single-word titles are also of little value.

Addresses and titles of recipients and writers may also be included.

(For thesis follow the format as prescribed by the relevant university)

Letter of Transmittal: This element is included in relatively formal and very formal reports. Its purpose is to release or deliver the report to the recipient. It also serves to establish some rapport between the reader and the writer. This is one part of the formal report where a personal, or even a slightly informal, tone should be used. The transmittal letter should not dive into report findings except in the broadest terms. This letter may be like:

Virtual University Lahore

December 15, 2006

Mr. K. M. Khalil
Vice President for Marketing
…….
…….

Subject: Report on Employee Satisfaction and Organizational Commitment

Dear Mr. Khalil,

Here is a report on Employee Satisfaction and Organizational Commitment. The report was prepared according to your authorization letter of April 15, 2006.

……
……
……
……

We are grateful to you for your cooperation in this important study.

Sincerely,

……..
……..

Letter of Authorization: This is a letter to the researcher approving the project, detailing who has responsibility for the project and indicating what resources are available to support it. The letter not only shows who sponsored the research but also delineates the original request.

Researcher would not write this letter. In many situations, referring to the letter of authorization in the letter of transmittal is sufficient. If so, the letter of authorization need not be included in the report. In case the letter has to be included, exact copy of the original may be reproduced.

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Table of Contents: A table of contents is essential to any report. It should list the divisions and subdivisions of the report with page references. The table of contents is based on the final outline of the report, but it should include first-level subdivisions. For short reports it is sufficient to include only the main divisions. If the report includes many figures and tables, lists of these should immediately follow the table of contents. If lots of abbreviations have been used in the report, give a list of abbreviations, alphabetically arranged, after the list of figures/tables.

Executive Summary: It is vital part of the report. Studies have indicated that most managers always read a report's summary, whereas only a minority read the rest of the report. Thus the only chance a writer may have to make an impact be in summary.

An executive summary can serve two purposes. It may be a report in miniature – covering all the aspects in the body of the report, but in abbreviated form. Or it may be a concise summary of the major findings and conclusions, including recommendations. On the whole the summary briefly tells why the research project was conducted, what aspects of the problem were considered, what the outcome was, and what should be done.

The summary should be written only after the rest of the report is completed. It represents the essence of the report. Two to three pages are generally sufficient for a properly condensed summary. (For very big reports which run into number of volumes, like the one finds in the feasibility reports of big projects, the summary may be very big.) The summary should be written to be self-sufficient. In fact, it is not uncommon for a summary to be detached from the report and circulated by itself.

The summary contains four elements:

1. The objectives of the report are stated, including the most important background and specific purposes of the project.
2. The major results are presented. The key results regarding each purpose should be included.
3. The conclusions that are based on the results. There should be logical interpretation of the results which could lead to the stated conclusions.
4. The recommendations or suggestions for action, which are based on the conclusions. The recommendations must logically emerge from the results.

In many cases managers prefer not to have recommendations included in the report or summary. The consultant may have to go by the demand of the client.

Note: In many reports you may see that the executive summary comes first which is followed by the table of contents.

For students writing their thesis, in place of executive summary, they write an abstract of their thesis. This abstract is usually of one or two paragraphs. Abstract has information on the topic, the research problem, the basic findings, and any 'unusual' research design or data collection features.

Main Body

The main body constitutes the bulk of the report. It includes: Introduction, Methodology, Results, Conclusions, and Recommendations of the study.

Introduction: The introduction prepares the reader for the report by describing the parts of the project: background material, the problem statement, and research objectives of the study. In most projects, introduction can be taken from the research proposal submitted earlier by the consultant. The proposal itself was based on the terms of reference (TOR) supplied by the client.

Background could help in looking at the magnitude of the problem. It may include the results of exploration from an experience survey, focus group discussion, and secondary data from literature

review. The background includes definitions, qualifications and assumptions. It gives the reader the information needed to understand the remainder of research report.

Problem statement contains the need for the research project. The problem is usually represented by the research question raised by the client. It explains why the project was worth doing.

Research objectives address the purpose of the project. These objectives may be research questions and associated investigative questions. In correlational or causal studies, the hypothesis statement may be included. At the end of the study the researcher may see the extent to which these objectives have been addressed.

For Thesis: After introduction, for **s**tudents writing their thesis, it is recommended that they should have three separate chapters on review of literature, theoretical framework, and hypothesis or research question along with the operationalization of variables. These chapters may be in line with the steps in research that we discussed as part of the research process.

Methodology: Technical procedures for carrying out the study must be explained in a manner appropriate for the reader. It may be useful to supplement the material in this section with more detailed explanation in the appendix. This part of the report should address seven topics:

1. *Research design:* Was the study exploratory, descriptive, or causal? A specific strategy was used to conduct this study. Why was this particular design suited to the study?
2. *Data collection methods:* Did the data come from primary sources or secondary sources? How the primary data were collected – survey, experiment, observation? It is possible that multiple techniques may have been used – all these have to be explained.
3. *Sample design:* What was the target population? What sampling frame was used? What type of sampling was used? What was the selection procedure used?
4. *Instrument(s) of data collection:* What instrument(s) of data collection was (were) used? Why a particular instrument was selected? Include a copy of each instrument in the appendix.
5. *Fieldwork/data collection:* How many and what type of fieldworkers were used? What training and supervision did they receive? How was the quality control assured?
6. *Analysis:* How was the analysis carried out? How was the data reduction handled? Tell about the scoring scheme used. Outline the statistical methods applied for the analysis of the data.
7. *Limitations:* No report is perfect, so it is important to indicate the report's limitations. If there were problems with non-response error or sampling procedures, they should be discussed. The discussion of limitations should avoid overemphasizing the weaknesses. Its aim should be to provide a realistic basis for assessing the results.

Results: The presentation of results will occupy the bulk of the report. This section presents in some logical order those findings of the project that bear on the objectives. The results should be organized as a continuous narrative, designed to be convincing but not oversell the project. Summary tables and charts should be used to aid the discussion. Tables and charts may serve as points of reference to the data being discussed and free the prose from an excess of facts and figures. Comprehensive or detailed charts should be reserved for the appendix.

Conclusions and recommendations: The last part of the body of the report presents the conclusions and recommendations based on results. Findings state facts; conclusions represent inferences drawn from findings. A writer is sometimes reluctant to make conclusions and leaves the task to the reader. Avoid this temptation when possible. As the researcher, you are the one best informed on the factors that critically influence the findings and conclusions.

Recommendations emerge out of conclusions. These are actually suggestions for action in an applied research. The researcher may present several alternatives with justification. In academic research, the recommendations are often further study suggestions that broaden or test understanding of the subject area.

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The conclusions and recommendations are presented here in more detail than in the executive summary, with whatever justification is needed.

References: All citations used in the study must be given by arranging them alphabetically by the last name of the author.

For your thesis

For your thesis the following outline of chapters is suggested:

- Introduction
- Review of Literature
- Theoretical Framework
- Hypothesis and Operationalization of Concepts
- Research Design
- Analysis of Data
- Summary, Conclusions, and Recommendations
- References
- Appendixes

Appended Parts

Appendix: The appendix presents the "too …" material. Any material that is too technical or too detailed to go to the body should appear in appendix. This includes materials of interest only to some readers, or subsidiary materials not directly related to the objectives. Some examples of appendix material are data collection forms (instruments), detailed calculations, discussions of highly technical questions, detailed or comprehensive tables of results, and a bibliography (if appropriate).

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RESEARCH METHODS

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Ask questions like:

- Is the assistance of interviewer necessary?
- Are respondents likely to be interested in the issues being investigated?
- Will cooperation be easily attained?
- How quickly the information be easily attained?
- How quickly the information is needed?
- Will the study require a long complex questionnaire?
- How large is the budget?

The Criteria are:

- The cost, speed, anonymity, and the like may be different for each project.
- If none is a good fit, then use *mixed method*.

TOOLS FOR DATA COLLECTION

Basically Three Tools

1. Questionnaire
2. Interview schedule
3. Interview guide

Both 1 and 2 are list of questions.

Questionnaire handed over to the respondent, who reads and fills it.

I.S. remains with the interviewer, who reads the questions to the respondents, gets the response, and records it.

We will use *questionnaire* for both.

Interview guide is list of topics to be covered during an in-depth interview.

Guidelines for Questionnaire Design

- Questionnaire is an instrument for measuring variables.
- Properly worded questionnaire helps in minimizing many problems.
- Questionnaire design is a critical stage.
- Focus on: What should be asked? Phrasing of questions, sequencing the questions, and lay out of the questionnaire.

1. Questionnaire Relevancy

- Question relevant to the respondent.
- Questions relevant to the topic.
- Rationale for each item of information

2. Questionnaire Accuracy

- Make sure that the information is reliable and valid.
- Come down to the level of the respondent. Avoid jargon, slang, and abbreviations. *Marital status.*

3. Avoid Ambiguity, Confusion, and Vagueness

- People use their perspective to interpret a word/concept e.g. What is your income? People assign different meanings.
- Avoid using words with indefinite frame of reference.

4. Avoid Double-Barreled Questions

- D-B question consists of two or more questions joined together.
- Makes the answer ambiguous. “Does this Co. have pension and health insurance benefits?”

5. Avoid Leading Questions

- LQ: That leads the respondent to choose one response over the other by its wording.
- Make respondents feel that all responses are legitimate.
- “You don’t smoke, do you?” “Don’t you think that women should be empowered?”

6. Avoid Loaded Questions

- LQ: suggest socially desirable answer or are emotionally charged.
- Should the city government repair all broken roads?
- Question loaded with such material that may hit the emotions of people e.g. Behaviors associated with masculinity in Pakistani society – Asking a husband – Have your ever been beaten up by your wife? Can be embarrassing.

7. Avoid Burdensome Questions

- People forget.
- Certain question may make serious demand on respondent’s memory. How did you feel about your brother when you were 6 years old?

8. Arrange Questions Sequentially

- Make the questionnaire an integrated whole. No jumping back and forth.
- Make the opening questions simple and interesting. Arouses respondent’s curiosity.
- Funnel Technique: Moving from broader issues to specific one.

9. Use Filter questions if needed

- Asking question that doesn’t apply to the subject can be irritating or bias the answer.
- Use filter question: That screens out respondents not qualified the second question.
- Asking about bringing up of one’s children. Prior to that ask if a person is married. If yes is he/she a parent.

10. Layout of the Questionnaire

- Two issues: Overall physical layout, and format of questions and responses.
- Attractive in appearance is crucial in self administered questionnaire.
- Number each question. Never camp the questions.
- Make a cover sheet for each. Put the time and date of the interview, the interviewer name, the respondent ID number, interviewer's comments.
- Give instructions. Print in different style.
- Include a polite, professional cover letter on letterhead, identifying the researcher. Always end with "Thank you for your cooperation."

RESEARCH METHODS

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PILOT TESTING THE QUESTIONNAIRE

Pilot Testing

- Also called pre-testing: Small scale trial run of a particular component.
- Here focus is on trial run of questionnaire.

Involves

- Small group selected on convenience.
- Could be 25+
- Similar to the one to be sampled.
- Administering questionnaire exactly as planned often is not possible (Problems in mail questionnaire).
- Usually go for personal interviewing

What aspects to be evaluated?

- Reactions of Respondents
- Discovering errors in the instrument
- Checking the sampling procedure
- Staffing and activities of research team
- Evaluating the procedure for data processing and analysis

1. Reactions of the Respondents

- Availability of study population (timing).
- Acceptability of the questions asked. Experience stimulating. Were the questions sensitive/threatening? Acceptability of the wording.
- Willingness of the respondents to cooperate.

2. Discovering errors in the instrument

- Validity and reliability of the instrument. Tabulation of the results → Do these meet the measurement objectives? Determine the appropriateness of question.
- Time taken. Length of the questionnaire.
- Any need to revise the format of the tool? Placement of sensitive questions. Pre-testing helps in proper sequence, wording, translation, spacing, structuring answers, coding. Needing instructions.

3. Checking the Sampling Procedure

- The extent to which the instructions given are followed to reach the respondent. Locating problems and possible solutions.
- Time needed to locate the respondents.

4. Staffing and Activities of Research Team

- How successful the training has been? Pre-testing a period of extra training.
- What is the work output of each member?
- How well the research team works together?
- Is the logistical support adequate?

5. Evaluating the Procedure for Data Processing and Analysis

Make dummy tables. Use appropriate statistics, and make interpretations

FIELDWORK

Fieldworker



Fieldworker

- An individual responsible for gathering data in the field. Could be:
- A personal interviewer
- A telephone interviewer calling from central location
- An observer counting pedestrians in a shopping mall.
- Others involved in data collection process
- Cornerstone of fieldwork is the interviewer.
- *Here focus shall be on personal interviewers*

Interviewers

- In-house interviewers: hired by the researcher
- Field Interviewing Service: A research supplier that specializes in gathering data. Trained interviewers, and supervisors. Edit questionnaires in the field. Certify how the data were collected.
- Interviewers need to be healthy, outgoing, and of pleasing appearance i.e. well groomed and well dressed.

Role of interviewer

- The survey interview is a social relationship.
- It is a short term secondary social interaction between two strangers with the explicit purpose.
- Interviewer - Interviewee/respondent.
- Structured conversation – interviewer asks prearranged questions and records answers.
- The role of interviewer is difficult. They obtain cooperation and build rapport, yet remain neutral and objective.

RESEARCH METHODS

Lecture 26

SAMPLE AND SAMPLING TERMINOLOGY

Sample

- A subset, or some part, of a larger whole.
- Larger whole could be anything – bucket of water, a bag of sugar, a group of organizations, a group of students, a group of customers, a group of mid-level managers.

Why sample?

1. Saves cost, labor, and time

- To go for sample study is pragmatic.
- In case population is extremely small, then go for total study. Census another word – total enumeration.

2. Quality Management

- Professional fieldworkers – a scarce commodity.
- Instead of doing on large population with less qualified staff, do a sample study with quality fieldworkers.
- Easier to manage small group – quality control. Training, supervision, record keeping.

3. Accurate and Reliable Results

- Properly selected samples are accurate.
- Homogeneous population – only a small sample needed. Likely to be representative. Blood samples.
- Large population. More non-sampling errors – interviewer mistakes, tabulation errors. Low quality supervision.

4. No Alternative but Sampling

- For quality control testing may require the destruction of the items being tested e. g. Firecrackers, testing the life a bulb, Testing missiles.
- This is destructive testing.

5. Determine the Period of Study

Census study requires long time, may be a year. Seasonal variation. For example, Study of unemployment rate over a year. Results refer to which part of the year.

6. Determine the Confidence Level

- Calculate the sampling error – help in determining the confidence level in the data.
- Sampling type may facilitate the use of powerful statistical tests for analysis.

Sampling Terminology

Number of technical terms used that need explanation.

1. Element

- Unit about which information is collected and is the basis of analysis. Can be a person, a group, a family, an organization, a community.

2. Population

- Theoretically specified aggregation of study elements.
- Translating the abstract concept into workable concept. College students. Theoretical explanation.
- Pool of all available elements is population.

3. Target Population

- Out of conceptual variations, what exactly is the focus.
- Complete group of specific population elements relevant to project.
- Call it *Survey population* – aggregation of elements for selecting a sample. e.g. study of college students – college students from Govt. institutions, studying social sciences, aged 19 years, and with rural background

4. Sampling

- Process of using a small number of items. Estimate unknown characteristics of population.
- Process of selection – Depending upon the type of sample to be used.

5. *Sampling Frame*

- List of population elements. Listing of all college students meeting the criteria.
- Also called as *working population* – list that can be worked with operationally. Prepare the list of relevant college students.

6. Sampling Unit

- That element or set of elements considered for selection in some stage of sampling.
- Sampling can be single stage or multistage. Simple or complex.
- In single stage, sampling units are the same as elements.
- In multistage, different levels of sampling units may be employed. Sampling of Mohallahs, the of households, and then adults. Primary, secondary, final.

7. Observation Unit

- Unit of data collection from which information is collected.
- Unit of observation and unit of analysis can be same or different. [Interview head of household (UoO) and collect information about every member (UoA)]

8. Parameter

- Summary description of a given variable in population (Mean income of families in the city, mean age)
- Survey research involves the estimation of population parameters.

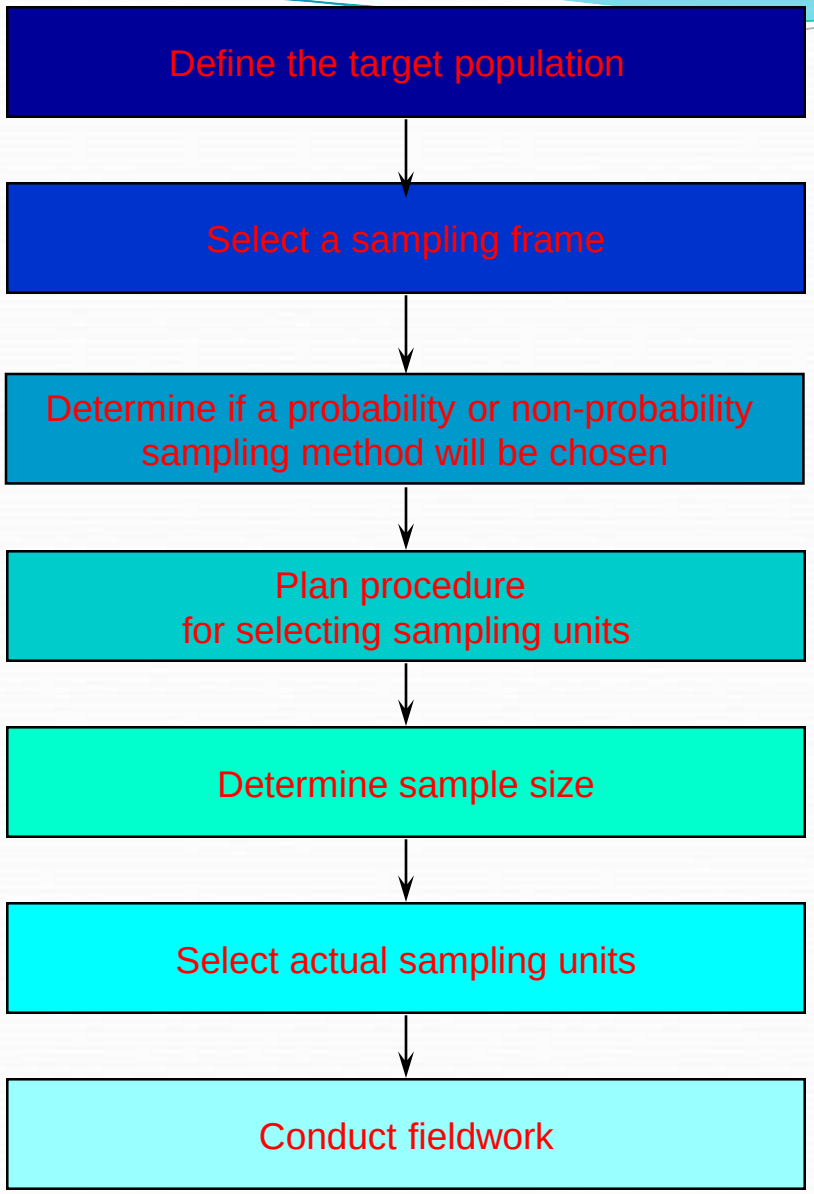
9. Statistic

- Summary description of a variable in survey sample. Mean income/age of the sample.
- Use it for estimation of population parameters

10. Sampling Error

- Probability samples seldom provide statistics exactly equal to parameters.
- Estimation of error to be expected for a given sample.

Stages in the Selection of a Sample



RESEARCH METHODS

RESEARCH METHODS

Lecture 27

NONPROBABILITY AND PROBABILITYSAMPLING

Major Alternatives to Sampling

- Probability
- Non-probability

Probability

- Every element in the population has a *known nonzero probability* of selection.
- Used when representativeness is crucial for wider gerenalizability.
- Can measure sampling error.

Non-Probability

- Probability of any element being chosen is unknown. Arbitrary – personal judgment.
- Time and other factors more crucial than generalizability.
- Cannot measure sampling error.
- Projecting data beyond sample is statistically inappropriate.

Types of Non-Probability Sampling

Convenience Sampling

- Also called *haphazard or accidental or grab sampling*.
- Obtaining units who are most conveniently available.
- Person-on-the street interviewed by TV people. Personal judgment.
- May not be representative. Least reliable.
- Used during exploratory phase

Purposive Sampling

- Researcher lays down the criteria for the subjects.
- Also called as *judgmental or expert opinion sample*. Selection depends upon the judgment of selector.

Quota Sampling

- Procedure that ensures certain characteristics of a population sample will be represented.
- Fix the quota. Use convenience sampling.

  Fieldworker bias in selection.

- Can be considered as proportionate stratified sampling – but on convenience basis.
- Speed of data collection, lower cost, and convenience.
- Ensures the inclusion of a subset when it is under-represented.

Snowball Sampling

- Also called: *network, chain referral, or reputational sampling.*
- Analogy of snowball.
- Begins with one/few, spreads out on the basis of links.
- Useful where respondents are difficult to identify.

Sequential Sampling

- Similar to purposive sampling.
- In purposive – get every possible case.
- In sequential – continuous evaluation of data. Stop when no new information is coming.

Theoretical Sampling

- Researcher selects cases based on his own insight.
- Used in observational studies

RESEARCH METHODS

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TYPES OF PROBABILITY SAMPLING

- Requires more work than non-random sampling.
- Researcher must identify sampling elements.
- Necessary to contact the sampled respondent. Call back several times.
- Likely to yield representative sample.
- Can determine the relationship between sample and population – Calculate the sampling error i.e. deviation between sample result and population parameter due to random process.

Simple Random Sample

- Assumption – homogeneous population.
- Develop an accurate sampling frame.
- Locate the exact element/s to be selected i.e. decide the sample size.
- Number all the elements in the sampling frame.
- Use the list/table of random numbers.

Table of Random Digits

37751	04998	66038	63480	98442	22245	83538	62351
50915	64152	82981	15796	27102	71635	34470	13608
99142	35021	01032	57907	80545	54112	15150	36856
70720	10033	25191	62358	03784	74377	88150	25567
18460	64947	32958	08752	96366	89092	23597	74308
65763	41133	60950	35372	06782	81451	78764	52645
83769	52570	60950	35372	06782	81451	84990	26400

- Take a random starting point.
- Characteristics of most random samples are close to the characteristics of population.
- Calculate the sampling error.

Systematic Random Sample

- Short cut for simple random sampling.
- Number each element in the sampling frame.
- Instead of using the table of random numbers, researcher develops a system for the selection of elements.

- Calculate the sampling interval. Basis of random selection.
- Sample size/Population size X 100 i.e. sample as % of population.
- Begin with a random start.
- Draw sample by choosing every Nth case.
- If elements are organized in some pattern in the sampling frame then problem. Possible periodicity in population.

Stratified Random Sampling

- In case population is heterogeneous, then stratify. Each stratum in itself becomes homogeneous.
- Researcher's decision about homogeneity of the population.
- Draw the sample from each stratum by using simple random sampling procedure.

Reasons for stratification

- To increase a sample's statistical efficiency. Each stratum gets represented. Reduces random sampling error. Makes the sample representative.
- Provides adequate data for analyzing the various subpopulations.
- Enables to use different research procedures in different strata.

Stratification Process

- Stratification based on the primary variable (Y). What characteristics of pop affect Y?
- Should increase homogeneity within stratum and heterogeneity between strata.
- Make sampling frame for each stratum.
- Serially number the elements.
- Use table of random numbers and select the sample from each stratum.

Proportionate vs. Disproportionate

- Proportionate: If the number of sampling units drawn from each stratum is in proportion to the relative population size of the stratum.
- Disproportionate: to ensure an adequate No. of sampling units in every stratum. Dictated by analytical considerations

Cluster Sampling

- Purpose: sample economically and retaining probability sampling.
- Heterogeneity within clusters but homogeneity between clusters.
- Random selection of clusters.
- Random selection of elements within the selected clusters.

Cluster sampling addresses two problems

- Researchers lack good sampling frame for dispersed population
- Reaching each sampled element is costly.

Multistage Sampling

- Researcher draws several samples in stages. For example:
- Cluster sampling is usually multistage →

  stage1– random sampling of big clusters

  stage 2– random sampling of small clusters within the big ones

  stage 3– random sampling of elements within the selected small clusters

City blocks – households – individuals

Double Sampling

- When further information is needed from a subset of the group from which some information has already been collected for the same study.
- Want to examine the matter in more detail.
- Sub-sample of the primary sample

Appropriate Sample Design

Depends upon a number of criteria like:

1. Degree of accuracy.
2. Resources.
3. Advance knowledge of population.
4. National vs. local project. Geographic proximity of population.
5. Need for statistical analysis. Projecting beyond the sample or the sample only.

RESEARCH METHODS

RESEARCH METHODS

Lecture 31

DATA PRESENTATION

Tables and Graphs

- Tabular and graphic presentation may take number of forms.
- Purpose is to facilitate the summarization and communication of the meaning.
- Bar charts, pie charts, curve diagrams, pictograms, and other forms can create strong visual impression.

Facilitation by Computer Technology

Commercial packages like:

- SAS, Statistical Package for the Social Sciences (SPSS), SYSTAT, Epi Info, MINITAB.
- User friendly packages are at the command of researcher.

Frequency Distribution

- Several useful techniques for displaying data.
- Easiest way to describe the numerical data is with a frequency distribution.
- Data of 400 students with different demographic characteristics.
- Let look at by gender.

Table 1: Frequency distribution of students by gender

Gender	Frequency	Percent
Male	300	75
Female	100	25
Total	400	100

Graphic Presentation

- Common types: histograms, bar charts, pie charts
- Graphic presentation lays emphasis on visual presentation over summary statistics.
- Summary statistics may obscure, conceal, or even misrepresent the underlying structure of data.

Interpretation

- Presented data have to be translated into some meaningful understanding.
- Explain the meaning for drawing inferences and conclusions.
- In order for interpretation, data have to be meaningfully analyzed.
- Statistics help for analysis.

Statistics

Different meanings:

- Set of collected numbers (No. of people living in a city)
- Branch of applied mathematics: used to manipulate and summarize the features of numbers.
- Use both. Descriptive statistics often used for univariate analysis.

Bivariate Tables

- Bivariate contingency table widely used.
- The table is based on cross tabulation.
- Cases are organized in the table on the basis of two variables at the same time.
- A contingency table is formed by cross-tabulating two variables.
- Contingent because the cases in each category of a variable get distributed into each category of second variable

Constructing bivariate and percentage table

Let us take two variables:

- Age of the respondents.
- Attitude toward women empowerment.
- Age of the respondents ranged from 25 to 70 years.
- The attitude index has three categories of ‘highly favorable,’ ‘Medium favorable,’ and ‘low favorable’

First let us make two univariate tables

- The age variable has so many categories that making a bi-variate table with that much categories becomes unwieldy and meaningless.
- Regroup (recode) the age categories into three i.e. under 40, 40 – 60, 61+

Univariate table for age

Table 2: Age of the respondents.

Age (Yrs.)	Frequency	Percent
Under 40	1000	33.3
40 – 60	1000	33.3
61 +	1000	33.3
Total	3000	100

Univariate table for attitude

Table 3: Attitude towards women empowerment

Attitude	Frequency	Percent
Hi Favorable	1100	37
Med Favorable	1050	35
Lo Favorable	850	28
Total	3000	100

Bivariate table

Table 4: Age by attitude towards women empowerment .

Level of attitude	Age (in years) under 40 F.	%	40 –60 F.	%	61 + F	%	Total F	%
Hi Favorable	600	60	300	30	200	20	1100	37
Med. Favorable	300	30	500	50	250	25	1050	35
Lo Favorable	100	10	200	20	500	50	850	28
Total	1000	100	1000	100	1000	100	3000	100

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The parts of the table

1. Give each table a number.
2. Give each table a title.
3. Label the row and column variables, and give name to each of the variable categories.
4. Include the totals of the columns and rows. These are called as marginals.
5. Each number or place that corresponds to the intersection of a category for each variable is cell of a table.
6. Missing information to be given under the table.

Percentaging

- Researchers convert raw count tables into percentages to see bivariate relationship.
- Three ways to percentage a bivariate table: by row, column, and for the total.
- Percentages by row and column are often used and show relationship

Right way to percentage

- Percentage by row or by column. Either could be appropriate.
- Decision based on researcher's hypothesis.
- Age affects attitude towards women empowerment.
- Look at the table, see where is the independent variable. See in which direction its values are being added up. Percentage in the same direction

Reading a table

- Once we know how table is made, reading it and figuring out what it says are much easier.
- Look at the title of table, the variable labels and background information.
- Look at the direction in which percentages have been computed – in rows or columns

Read percentaged tables to make comparisons

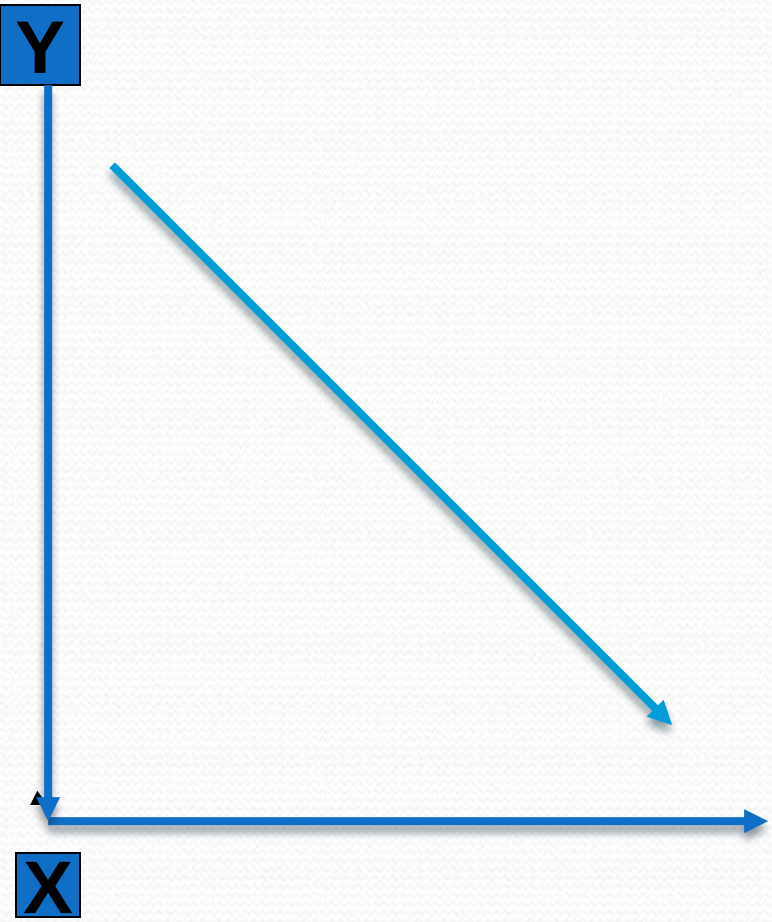
- Comparison are made in the opposite direction from that in which percentages are computed.
- Compare across rows if the table is percentaged down the column.

How to figure out relationship?

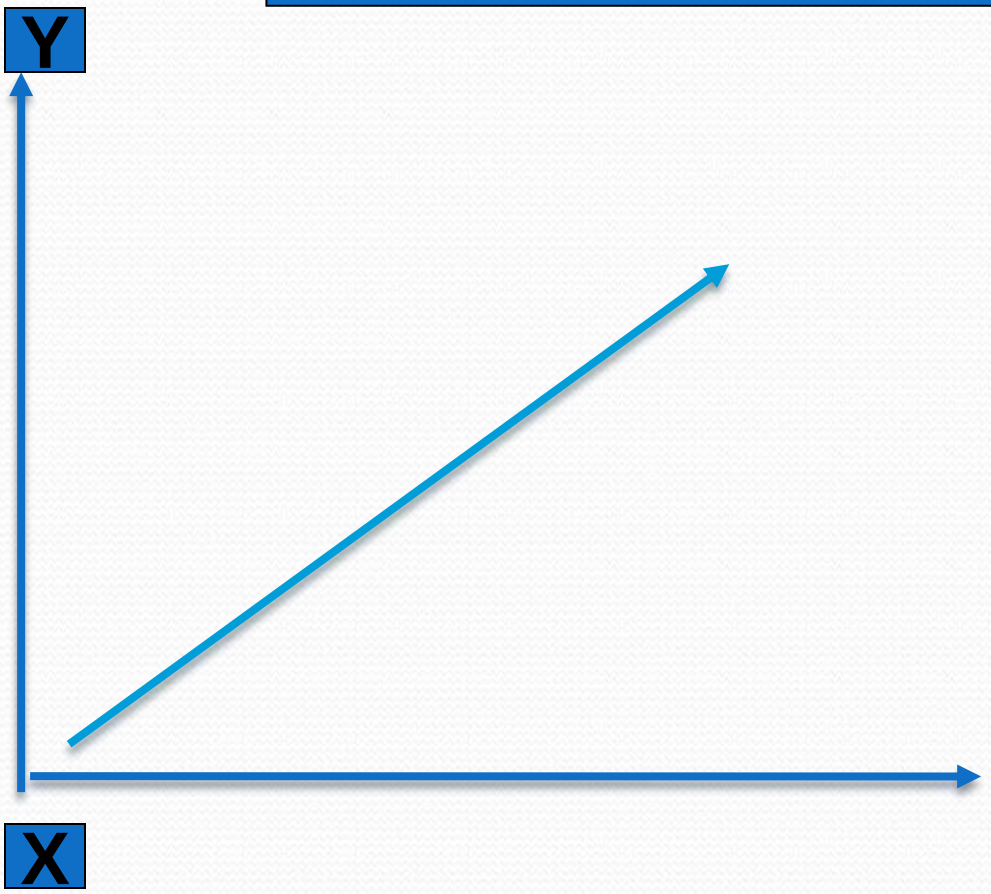
If no relationship:

- Percentages look approximately equal across row or columns.
- A linear relationship looks like larger percentage in diagonal cells.

Linear
Negative relationship



Linear
Positive relationship



Linear relationship

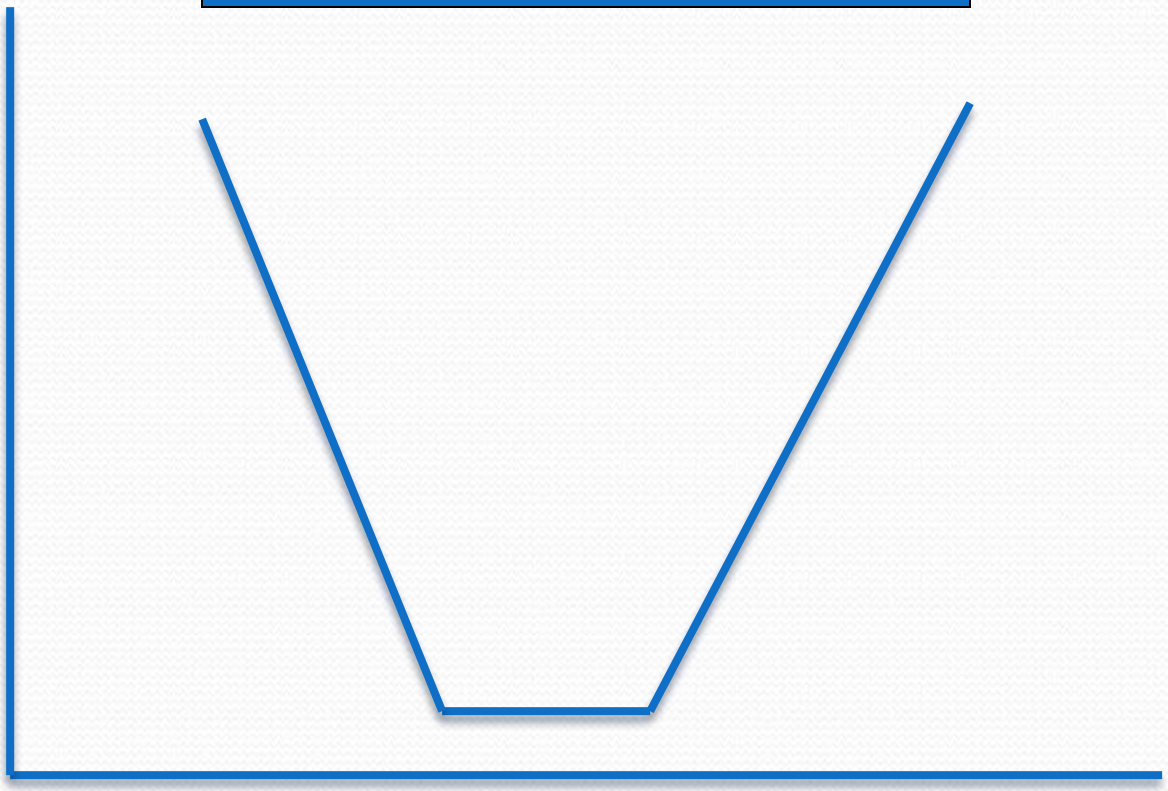
Table 4: Age by attitude towards women empowerment

Level of attitude	Age (in years) under 40 F.	%	40 –60 F	%	61 + F	%	Total F	%
Hi Favorable	600	60	300	30	200	20	1100	37
Med. Favorable	300	30	500	50	250	25	1050	28
Lo Favorable	100	10	200	20	500	50	850	28
Total	1000	100	1000	100	1000	100	3000	100

- **Larger percentages in the diagonal cells**

If there is curvilinear relationship, the largest percentages form a pattern across cells e.g. the largest cells might be the upper right, the bottom middle, and the upper left.

Curvilinear



It is easier to see relationship in a moderate sized table (9 cells) where most cells have some cases (at least 5 cases)

A simple way to see strong relationship is to circle the largest percentage in applicable row or column and see if a line appears

Table 4: Age by attitude towards women empowerment

.

Level of attitude	Age (in years) under 40 F.	%	40 –60 F.	%	61 + F	%	Total F	%
Hi Favorable	600	60	300	30	200	20	1100	37
Med. Favorable	300	30	500	50	250	25	1050	35
Lo Favorable	100	10	200	20	500	50	850	28
Total	1000	100	1000	100	1000	100	3000	100

Is the relationship genuine?

- Eliminate alternative explanations – explanations that can make the relationship *spurious*.
- Experimental researchers do this by choosing a research design that physically controls potential alternative explanations for results. Control prior to the start of experiment

In non-experimental research:

- A researcher controls for alternative explanations with statistics.
- Measures possible alternative explanations with *control variables*
- Examines the control variables with multivariate tables and statistics and decides whether a bivariate relationship is spurious.

Control for third factor

- Two variable table. Relationship between age of people and attitude towards women empowerment.
- To see the spuriousness of X and Y introduce third variable i.e. gender.
- Control the effect of gender i.e. the effect of gender is statistically removed.
- After control, does the bivariate relationship still persist?

Control for gender

- Under each category of male and female, negative relationship between age and attitude persists. Relationship is not spurious.
- If bivariate relationship weakens, or disappears, then age is not the factor affecting the attitude.
- Difference in attitude is due to gender.

Statistical control

- A measure of association tested for its genuineness by controlling third variable.
- Researchers are cautious in their interpretations.
- Look for net effect.
- Go for trivariate percentaged table or multiple regression

Trivariate Table

- Test the alternative explanation. 3rd factor.
- Control for third variable. Make a trivariate table.
- It has a bivariate table of XY for each category of control variable.
- The new tables may be called partials.
- No. of partial depends upon the No. of categories of control variable.
- Partial tables look like bivariate tables, but use subset of the cases. Break apart a bivariate table to form partials

Limitations

- Difficult to interpret if control variable has more than four categories.
- Total number of cases may be limiting factor because cases are divided into cells in partials. Thinning out of data. On average 5 cases per cell recommended.

Partial table for males

	Age (in years)							
Level of	Under 40		40—60		61+		Total	.
Attitude	F	%	F	%	F.	%	F.	%.
High	300	60	200	33	30	6	530	33
Medium	140	28	270	45	120	24	530	33
Low	60	12	130	22	350	70	540	34
Total	500	100	600	100	500	100	1600	100

Partial table for females

	Age (in years)							
Level of	Under 40		40—60		61+		Total	.
Attitude	F	%	F	%	F.	%	F.	% .
High	350	70	200	50	20	4	570	41
Medium	150	30	150	38	220	44	520	37
Low	-	-	50	12	260	52	310	22
Total	500	100	400	100	500	100	1400	100

Replication pattern

- When the partials replicate or reproduce the same relationship that existed in bivariate table prior to control
- Control has no effect.

The specification pattern

- When on partial replicate the same relationship but others do not.
- The researcher can specify in which partial there is strong relationship and where it is not.

The suppressor variable pattern

- When the bivariate table suggests independence of X and Y but the relationship appears in one or more partials.
- The control variable is suppressor – the true relationship appears in partials.

Multiple regression analysis

RESEARCH METHODS

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EXPERIMENTAL RESEARCH

Replication pattern

- When the partials replicate or reproduce the same relationship that existed in bivariate table prior to control
- Control has no effect.

The specification pattern

- When one partial replicate the same relationship but others do not.
- The researcher can specify in which partial there is strong relationship and where it is not.

The suppressor variable pattern

- When the bivariate table suggests independence of X and Y but the relationship appears in one or more partials.
- The control variable is suppressor – the true relationship appears in partials.

Multiple regression analysis

- Controls for many alternative explanations and variables simultaneously.
- Referred to course on statistics.
- Also there are a number of statistical tests that can be applied to test the hypothesis. Here again reference has to be made to course on statistics.

Logic of experiments in natural sciences

- Applied in experiments on human social behavior.
- Experiments are found in psychology, education, criminal justice, journalism, marketing, nursing, political science, social work, and sociology.
- Logic: *Control* the research situation, and evaluate the *causal* relationship among variables

In experiments

- The researcher manipulates a single variable and holds constant all others, extraneous variables.
- The way events are controlled in experiments is not possible in a survey.
- The researcher gains the confidence that his experimental treatment is the cause of the effect he measures.

In experiments

- Theoretically there is high degree of control on the research situation.
- X is manipulated and its effect on Y is measured.
- The effect of all other variables is controlled.
- Researcher creates an artificial situation or deliberately manipulates a situation.

To establish causal relationship between X and Y

All three conditions to be met:

1. X and Y should co-vary.
2. X should precede Y.
3. No other factor should possibly cause change in Y.

By controlling all other factors that can affect Y allows the researcher to say that X and X alone causes Y. Not possible to control other factors in an organization where some events occur naturally.

Parts of experiments:

No fixed number. Usually seven parts:

1. Treatment or independent variable
2. Dependent variable.
3. Pretest
4. Posttest
5. Experimental group.
6. Control group.
7. Assignment of subjects

Treatment or independent variable

- The experiment has some degree of control over X. Values can be manipulated.
- Treatment is what researcher modifies.
- Term comes from medicine: a physician administers a treatment to patients.
- Physician intervenes in a physical or psychological condition to change it. Hence it is X variable or the combination of many Xs.

Experimenter creates a situation

- Degree of fear – levels are high-fear or low fear.
- The researcher puts the subjects into either high-fear or low-fear situation.

Dependent variable

- The criterion or standard by which results are judged.
- Assumption: changes in Y are a consequence of changes in the X.
- The outcome of experimental research are the physical conditions, social behaviors, attitudes, or beliefs of subjects that change, in response to treatment.
- Measure Y by observations, interviews, or physiological responses (e.g. heartbeat)

Pretests and Posttests

- Frequently a researcher measures the Y more than once during an experiment.
- *Pretest* is the measurement of Y prior to treatment.
- *Posttest* is the measurement of Y after the treatment introduced into the experimental situation.

Experimental and Control groups

- Divide subjects into 2 or more groups for purposes of comparison.
- A simple experiment has only 2 groups, only one of which receives the treatment.
- Experimental group is the one that receives the treatment or in which treatment is present.
- Group that does not receive the treatment is control group.
- When X has many different values, more than one experimental group is used.

In a simple experiment

- Only two values of the X are manipulated. E.g. Consider measuring the influence of a change in work situation, such as playing music during working hours, on employee productivity.
- In the experimental condition (the treatment administered to the experimental group), music is played during working hours.
- In control group (treatment not administered) no change in work situation.
- Productivity in the two groups is compared at the end to determine the effect of X.

Several treatment levels

- The music/productivity experiment with one experimental and on control group may not tell the researcher everything about the relationship
- For understanding the functional nature of relationship between music and productivity at several treatment levels, additional experimental groups with music played for 2 hrs, only for 4 hrs, and only for 6 hrs.
- Allows the experimenter to get a better idea about the impact of music on productivity.

Assignment of Subjects/Test Units

- Social researchers frequently want to compare.
- Compare cases that do not differ.

Requires:

- Groups should be similar in characteristics
- Change in Y is presumably the outcome of the manipulation of X variable, having no alternative explanations.

Random assignment

- Ramdomization is a method for assigning the cases (individuals, organizations) to groups for making comparisons.
- It is a way to divide or sort a collection of cases into 2 or more groups in order to increase one's confidence that the groups do not differ in a systematic way.
- In a mechanical method; the assignment is automatic, and the researcher cannot make assignments on the basis of personal preference or the features of specific cases.

Random assignment (cont.)

- Random assignment is random in statistical/mathematical sense not in everyday sense.
- In every day speech, random means unplanned, haphazard, or accidental, but it has special meaning in mathematics.
- Random describes a process in which each case has a known chance of being selected.
- A random process is the one in which all cases have an exactly equal chance of ending up in one or the other group.
- It is unbiased.
- Makes the groups identical, except for treatment

Matching

- Matching the subjects on the basis of pertinent background information is another technique for controlling assignment errors.
- Matching presents a problem: What are the relevant characteristics to match on, and can one locate exact matches. Cases could differ in a number of ways.
- Randomization preferred. Takes care of contaminating factors.

Three types of controls

- Manipulation: control over stimulus/treatment.
- Holding conditions constant i.e. control over the environment (the confounding factors).
- Control over the composition of groups – balancing. Find out the way that individual differences do not confound the X variable under investigation. Randomization is the answer.

RESEARCH METHODS

Lecture 34

EXPERIMENTAL RESEARCH
(Continued)

Steps in conducing an Experiment:

- Begin with an hypothesis.
- Decide on an Exp design to test H.
- Decide how to introduce X.
- Develop a measure of Y.
- Set up an experiment and do pilot testing.
- Locate appropriate subjects.
- Randomly assign subjects to groups and give instructions.

Steps Cont.

- Gather data for the pretest of Y.
- Introduce the X to experimental group only and monitor all groups.
- Gather data for posttest of Y.
- Debrief the subjects by informing them of the true purpose of experiment.
- Examine data, make comparisons between groups. Test Hypothesis.

Types of Designs

- Researchers combine parts of an experiment (e.g. pretests, control groups) together into an experimental design.
- E.g. some designs lack pretests, some do not have control groups, others have many experimental groups.
- Classical experimental design has: random assignment, a pretest and posttest, an experimental group and a control group.
- Other designs are variations of classical design

Quasi Experimental Designs:

- One-shot case study design. One group posttest only design.
- One group experiment→
- Pretest (O1) X Posttest (O2). No control group for comparison. [O2 - O1] = Effect
- Two groups experiment → posttest with experimental and control group →

Exp. Group	X	O1
Control Group	-	O2

[O1 – O2] = Treatment Effect.

True Experimental Designs:

- Includes experimental and control group. Pretest and posttest to both groups. X only in experimental group. (Ex-post facto experimental design.)
- Exp: Pretest (O1) X Posttest (O2)

  Con: Pretest (O3) - Posttest (O4) Randomization for group set up.
- [(O2-O1) – (O4-O1)] = Treatment effect

Solomon 4 Group Design:

To gain more confidence, it is advisable to set up 2 exp groups and 2 cont groups. One exp and one control group be given both pretest and posttest. Other two are given posttest only.

Exp: Pretest (O1) X Posttest (O2)

Con: Pretest (O3) - Posttest (O4)

Exp: - X Posttest (O5)

Con: - - Posttest (O6)

(O2 – O1) = E (O2 – O4) = E

(O5 – O6) = E (O5 – O3) = E

[(O2 – O1) – (O4 – O3)] = E

If all Es are similar, the cause and effect relationship is highly valid.

Interaction Effect:

- Effect of 2 variables together is likely to be greater than the individual effect of each. For example:
- Population of smokers → 30% got lung cancer
- Population of nonsmokers but living in a smoggy climate → 10% got lung cancer.
- Pop of smokers + living in smoggy area → 45% got lung cancer instead of (30+10) 40%.
- Difference between 45-40= 5 is the interaction effect (smoking + smoggy climate)

In experiment

Interaction between treatment + sensitization due to the instrument.

Exp: Pretest (O1) X Posttest (O2)

Con: Pretest (O3) - Posttest (O4)

Why difference in O4 and O3? *Sensitization*.

Exp: - X Posttest (O5)

(O2 – O1) = D

(O4 – O3) = D/

(O5 – O3) = D//

D – [D/ + D//] = Interaction effect.

Further Experimental Designs:

- Randomized Block designs.
- Latin square Design.
- Natural Group Design.
- Factorial Design.

Validity in Experiments

- Validity refers to confidence in cause and effect relationship.
- Internal validity is high in Laboratory experiments.
- External validity (generalizability) is not sure.

Factors Affecting Internal Validity:

- History Effect: Other historical events may affect the X – Y relationship. In addition to advertisement --- something else happens (Virus, some legitimacy)
- Maturation Effect: With passage of time, biological and psychological maturity. Growing older, getting tired, feeling hungry
- Testing Effect: Pretests. Sensitization.
- Instrumentation Effect: Change in measuring instrument between pretest and posttest.

- Selection Bias Effect: Improper or unmatched selection of subject for groups.
- Statistical Regression: If extremes are taken then they tend to regress towards mean. Those who are at either end of the extreme would not truly reflect the cause and relationship.
- Mortality: Attrition of subjects. Subject loss. Random groups do not remain comparable.
- Mechanical Loss: Equipment failure

Factors Cont.

- Experimenter Expectancy: May indirectly communicate desired findings to subject.
- *The double blind experiment* is designed to control EE. Both the subjects and those in contact with them are blind to details of the experiment.

Ethical Issues in Lab Experiments:

- Putting pressure on subjects to participate.
- Asking demeaning question.
- Deceiving subjects by deliberately misleading them.
- Exposing participants to physical or mental stress.
- Not allowing subjects to withdraw.
- Using results to disadvantage the subjects
- Withholding benefits from control group.

RESEARCH METHODS

Lecture 37

Use of secondary data

Advantages:

1. Access to inaccessible subjects.

2. Non-Reactivity → unobtrusive research. Observations are unnoticed. Indirect observations. One way mirror. Subjects are not living. Creator may be dead.

3. Can do longitudinal analysis. Trend. Study over time.

4. Use sampling. Use random sampling – population sample. How women are portrayed in weekly news magazines?

5. Can use large sample size. More trust in generalization.

6. Spontaneity. Spontaneous actions or feelings recorded when they occurred. Diary records. Letters. Speeches.

7. Confessions. Persons more likely to confess in documents which are read after the death than in questionnaires. Diaries.

8. Relatively low cost. Travel costs .

9. High quality. Many documents (newspaper columns) written by skilled social commentators compared to respondents to a questionnaires.

Disadvantages

1. Bias. Documents written for purposes other than research. Goals may bias the information presented. Money making. Present only positive/negatives.

2. Selective survival. Preservation selective.

3. Incompleteness. Letters/diaries include references only. Private background only known to the author.

4. Lack of availability of documents. Documents remain classified.

5. Sampling bias. Only well educated people will write. Views of poorly educated people are not there.

6. Limited to verbal behavior.

7. Lack of standardized format. Newspapers may have it, not for personal documents.

8. Coding difficulties. Difficult to quantify.

9. Data must be adjusted for comparability over time. Unit/value may change.

USE OF SECONDARY DATA: EXISTINGSTATISTICS/ DOCUMENTS

Secondary Data

- Variety of data collected by others and available to researchers for further analysis.
- Data available in the form of statistical documents (books, reports). Also computerized records.
- Data collected by large bureaucratic organizations. Data gathered for policy decisions or as a public service.
- Time bound collection of information (pop. Census) as well as over long periods (unemployment, crime rates)
- Comparisons over time, across the countries.

Selecting Topic for Secondary Analysis

- Formulate research question, reassemble the data in new ways to address the research question.
- Research question has to be in line with the available data. So first find what is available, then frame the question.
- Look into trends. Develop social indicators.
- Secondary analysis may not fit neatly into a deductive model of research design.

Locating Data

- Main sources are government or international agencies and private sources.
- Many existing documents are “free” and available at libraries. Laborious job.
- UN publications. UNESCO Statistical Yearbook. United Nations Statistical Yearbook, Demographic Yearbook, Labor force survey of Pakistan, Population Census Data.

Secondary Survey data

- Secondary analysis is a special case of existing statistics; it is reanalysis of previously collected data by others.
- Focus is on analyzing rather than on collecting data.
- Relatively inexpensive. Collecting such huge data by a single researcher may be prohibitive.
- Permits comparisons across groups, nations or time;
- Facilitates replication;
- Permits asking about issues not thought by original researchers.

Reliability and Validity

- Existing statistics and secondary data are not trouble free. Researchers must be concerned about validity and reliability .
- A common error is the fallacy of misplaced concreteness. Someone gives a false impression of accuracy by quoting statistics in greater detail than warranted by how the statistics were collected.

Validity problems

- Researcher's theoretical definition may not match with government agency or organization that collected the data. *Unemployed.*
- Researcher lack control over how the data were collected. Data collection as part of job. No quality control – systematic errors. Typographical errors.

Reliability

- Stability reliability problems develop when official definition or method of collecting data changes over time.
- Definition of literacy changed many times. Definition of unemployment. Poverty.
- Equivalence reliability. Decimal system. Problems of comparability nationally and internationally.

Inferences from non-reactive data

- To infer causality and to test a theory on the basis of non-reactive data is limited.
- Cannot establish temporal order of variables. Cannot eliminate the interference of confounding factors.
- Not possible to generalize from content to its effects on those who read the text. Only logical correlation.

RESEARCH METHODS

RESEARCH METHODS

Lecture 38

QUALITATIVE RESEARCH

Observation Studies/ Field Research

Qualitative and Quantitative Research

- Differ but complement each other.
- Qualitative research produces *soft data*: impressions, words, sentences, photos, symbols. Interpretative approach. Non-linear approach (spiral)
- Quantitative research produces *hard data*: numbers. Follow positivist approach. Speak the language of variables and hypotheses. Emphasis on precise measurement of variables and testing hypotheses. Look for causality. Linear approach (sequential steps)

Observation Studies

- Observation can be participant or non-participant.
- In participant observation the researcher directly observes and participates in small scale social settings in the present time.
- Also referred to as field research, ethnography, or anthropological study.
- Direct, face to face interaction with “real people” in a natural setting.
- A group who interact with each other on regular basis in a fixed setting: street corner, tea shop, a club, nomads, village.

Ethnography and Ethno-methodology

- Ethno means people distinct by their culture. Graphy refers to description. Understanding the culture of people.
- Methodology: how people create reality as well as its interpretation. Meanings attached to a reality.

Logic of field research

- Observation of events in natural settings.
- Researcher uses various set of methods. Highly flexible use of multiple methods.
- He/she gets involved but also detached as a researcher. People culture vs. research culture.
- Direct involvement in field may have an impact on the researcher.

A field researcher does the following:

1.Observes ordinary events and everyday activities as they happen in natural settings, in addition to unusual occurrences.

2. Becomes directly involved with people being studied and personally experiences the process of daily life in the field setting.

3. Acquires an insider's point of view while maintaining the analytic perspective or distance of an outsider.

4. Uses a variety of techniques and social skills in a flexible manner as the situation demands.

5. Produces data in the form of extensive, written notes, as well as diagrams, maps, pictures to provide very detailed descriptions.

6. Sees events holistically (as a whole unit) and individually in their social context.

7. Understands and develops empathy for members in a field setting, and does not just record ‘cold’ objective facts.

8. Notices both explicit (recognized, conscious, spoken) and tacit (less recognized, implicit, unspoken) aspects of culture.

Cont.

9. Observes ongoing social processes without upsetting, or imposing an outside point of view.

10. Copes with high levels of personal stress, uncertainty, ethical dilemmas, and ambiguity.

Steps in Field research:

1. Prepare yourself, read the literature, and defocus. Empty your mind. No stereotypes
2. Select a field site and gain access. Look for gate keepers. Use access ladder.
3. Enter the field and establish social relations with members. Rapport
4. Adopt a social role, learn the ropes, and get along with members.

5. Watch, listen, and collect quality data.
6. Begin to analyze data, generate and evaluate working hypothesis.
7. Focus on specific aspects of the setting and use theoretical sampling. Take smaller, selective observations.
8. Conduct field interviews with member informants.
9. Disengage and physically leave the setting.
10. Complete the analysis and write the report.

RESEARCH METHODS

RESEARCH METHODS

Lecture 39

OBSERVATIONAL STUDIES (Contd.)

Steps in Field Research

Background

- Flexibility: naturalism and direct involvement demand researcher flexibility. Researcher himself/herself has to be flexible. No fixed steps to be followed.
- Good field researchers recognize and seize opportunities. Rapidly adjust.

1. Prepare yourself, read the literature, and defocus. Empty your mind. No stereotypes.

- Field project → some happening or personal interest. Generic topic (No hypothesis) Literature will help in learning concepts, methods, techniques for resolving conflicts.
- Not purely focusing on the role of researcher. Has to have experience.
- Know yourself. No preconceptions about self and others. Can you bear the strains.

2. Select a field site and gain access.

- Where to observe? Site: socially defined territory with shifting boundaries.
- Gain access. Look for gate keepers. Authority. May have to negotiate. Entry may be a process – use access ladder. Building trust helps in having access to observe sensitive events.

3. Enter the field and establish social relations with members.

- Develop rapport.
- Presentation of self. Acceptable to people.
- Researcher as measuring instrument. Has to be alert to happenings around. Maintain social relationships and personal feelings.

4. Relations in the field

- Adopt a social role, learn the ropes, and get along with members.
- Preexisting versus created roles. Limits to the roles chosen because of age, race, gender, attractiveness.
- Also the level of involvement. Complete observer, observer as participant, participant as observer, complete participant.

5. Observing and collecting data

- Watch, listen, and collect quality data.
- Field data are what the researcher experiences, remembers, and records in field notes.
- Maps and diagram. Machine recording

6. Begin to analyze data, generate and evaluate working hypothesis.

- Field research does not start with a hypothesis. Could be generated from the data.

7. Focus on specific aspects of the setting and use theoretical sampling.

- Researcher first gets general picture. Develops hypothesis. Then focuses on few specific problems.
- Take smaller, selective observations. Takes theoretical sampling. Guided by researcher's own theory. Sample times, situations, types of events, locations, types of people.

8. Conduct field interviews with member informants.

- Field researchers use unstructured, nondirective, in-depth interviews which differ from formal survey research interviews.

FR Interview vs. Survey Interview

Survey Interview

- It has a clear beginning and end.
- The same standard questions are asked and in same sequence.
- Interviewer appears neutral at all times.

Field Interview

- The beginning and end are not clear.
- Questions and order are tailored to specific people and situations.
- Interviewer shows interest in responses, encourages elaboration.

Survey Interview

- Interviewer asks questions, and the respondent answers.
- It is almost always with one respondent alone.

Field Interview

- It is like a friendly conversational exchange, with more interviewer questions.
- It can occur in group setting or with others in area.

Survey Interview

- Professional tone and business like focus; diversions ignored.
- Closed ended Qs are common; rare probes.
- Interviewer controls the pace and direction of interview.

Field Interview

- Interspersed with jokes, stories, diversions.
- Open ended Qs are common; probes frequent.
- Interviewer and member jointly control the pace and direction of interview

Survey Interview

- Social context of interview is ignored and assumed to make no difference.
- Interviewer attempts to mold the communication in a standard framework.

Field Interview

- Social context of interview is noted and seen as important for interpreting the meaning or responses.
- Interviewer adjusts to member's norms and language usage.

9. Disengage and physically leave the setting.

- At some point work in field ends.
- Can be an emotional ending depending upon the intensity of involvement. Can be disruptive and painful.
- Follow the local norms in the exit process.

10. Complete the analysis and write the report.

- Researcher may share the written report with the members observed to verify its accuracy and approve its portrayal in print.
- It may help in determining the validity of the study.
- (May not be possible with marginal groups like addicts, some deviant groups)

Ethical Dilemmas of Field Research

- Direct involvement of field researcher in the social lives of people raises many ethical dilemmas.

Let us look at these:

- Confidentiality
- Deception

Deception

- Research may be covert; or may assume a false role, name, or identity; or may mislead members in some way.
- Debatable. Cost benefit situation. Researcher the best judge.
- Covert research is never preferable and never easier than overt research.

Confidentiality

- Intimate knowledge of the people.
- Moral obligation to uphold the confidentiality of data. Also disguising the members' names in field notes and in report.

Involvement with deviants

- Research on deviants (illegal behavior)
- Guilty knowledge. Such knowledge is of interest to law enforcement agencies as well as to other deviants.
- Dilemma: building rapport with deviants, yet not so involved as to violate his/her basic personal moral standards. Some explicit arrangement with deviant members.

Publishing field reports

- Intimate knowledge that a researcher obtains and reports creates an ethical dilemma between the right of privacy and the right to know.
- Cannot publish that may offend or harm someone. Some things may remain hidden; the readers may not believe the report if critical details are omitted.

RESEARCH METHODS

RESEARCH METHODS

LECTURE 42

FOCUS GROUP DISCUSSION

Informal Focus Group Discussion

- Visiting a community. Stop by a house, chat with head of household.
- Others come and join the conversation.
- Informal GD is build upon the social networks that operates in a natural setting. Includes both kin and other neighbors. These might be the decision makers.

- **FGD are a more formal way of getting groups of people to discuss selected issues.**

Focus Group Discussion (FGD):

- A focus group discussion is a group discussion of 6-12 persons
- Guided by a facilitator,
- Participants talk freely and spontaneously about a certain issue.

The purpose of FGD:

- To obtain in-depth information on concepts, perceptions, and ideas of the group.
- More than an question-answer interaction.
- Group members discuss the topic among themselves.

Focus groups are:

- Formally constituted (organized in advance);
- Structured groups brought together (people from similar background, age, sex, education, religion, or similar experiences);
- 6-12 persons;
- Guided by a moderator/facilitator;

Cont.

- To address a specific issue (talk freely, agree or disagree),
- Within a fixed time frame, and
- In accordance with clearly spelled out rules of procedure.

Functions of the Facilitator/Moderator:

- Act as a coordinator, not as an expert.
- Introduce the session.
- Encourage discussion.
- Encourage involvement. Ask for clarifications; reorienting the discussion when it goes off the track; bringing in reluctant participants. Deal with dominant participant
- Build Rapport. Observe nonverbal communication.

How to conduct FGD?

1. Preparation

Selection of topic, questions to be discussed (open ended).

Selecting the study participants:

- Purposive or convenience sampling.
- Similar background. Age, sex, status.
- 6-12 persons in a session.
- Contact the participants. Invitations.

Making physical arrangements.

2. Conducting the session:

- One of the members of the research team should act as ‘facilitator’ for the focus group. One should serve as ‘recorder’ (rapporteur).

RESEARCH METHODS

Lecture 43

FOCUS GROUP DISCUSSION (Cont.)

Some guidelines for the facilitator:

- Observe verbal as well non-verbal communication. What are they saying? What does it mean to them? Empathize.
- Do not try to comment on everything that is said
- Control the timings unobtrusively.

Some guidelines for the Facilitator (Cont.)..

- Summarize the main issues at the end of FGD.
- Check whether all participants agree.
- Thank the participants.
- Listen for the added comments after meeting has closed.

Functions of the Recorder

- Keep a record of the content as well as emotional reactions and nature of group interactions.
- Record the following:

- Date, time, and place.

- Names and characteristics of participants.

- Description of group dynamics (level of participation, presence of a dominant participant, level of interest).

Functions of Recorder (Cont.)

- Opinions of the participants, as far as possible in their own words, especially the key statements.
- Emotional aspects (reluctance, strong feelings attached to certain opinions)
- Vocabulary used.

Functions of Recorder (Cont.)

- Assist the facilitator by drawing his/her attention to missed topics or missed comments from the participants.
- Help the facilitator resolve the conflict if necessary.
- Make sure a copy of the list of topics and key probe questions is available and referred to during the FGD.

Duration of FGD

- Between one hour to one hour and a half.

3. Analysis of Results

- After each FGD the facilitator and recorder meet to review and evaluate the discussion.
- The full report is prepared using the participants own words, listing the key statements, ideas and attitudes.
- Additional questions are formulated if needed.
- Answers of different sub-groups are compared.
- The findings must be recorded in a coherent way.
- The most useful quotations should be selected.

4. Report Writing

- Start with a description of the selection and composition of the group and participants, and a commentary on the group process.
- Present your findings, following your list of topics and guided by the objectives of FGD.
- Include questions whenever possible, particularly the key statements.

Uses of FGDs

- Produce a lot of information far more quickly, and at less cost than individual interviews.
- Help focus research and develop relevant research hypotheses. In-depth analysis of the problem and its causes.
- Help formulate appropriate questions for more structured, large scale surveys

- FGDs are excellent for obtaining information from illiterates.
- Means to discover attitudes and opinions that might not be revealed through surveys. Help explore controversial topics.
- Usually well accepted by the community as this form of communication already exists.
- Generate new ideas, questions (about a program)
- Supplement information on community KAP already available.

Cont.

Cont.

- Flexibility in discussion.
- Direct link with the population under study.
- Focus groups are good fun.

Limitations

- Results cannot usually be used for generalization.
- Participants often agree with responses from fellow members (for different reasons). Researcher to be cautious when interpreting the results.

Cont.

- The moderator may influence the participants (bias).
- FGDs have limited value in exploring complex beliefs of individuals.
- FGDs can paint a picture of what is socially acceptable in a community rather than what is actually occurring or believed. (real and ideal problems)

THE CASE STUDY

Case Study

- A comprehensive description and analysis of a single situation or number of specific situations i.e. cases.
- An intensive description and analysis of a case.
- Often use qualitative approach to explore the case in as rich a detail as possible.
- Explore new areas of inquiry.

Examples

Case study of a highly successful:

- Organization,
- A Project (OPP at Karachi),
- A group, a couple, a teacher, a patient.

Data sources:

- Naturalistic observations (ethnographic).
- Interviews.
- Life histories
- Tests (psychological

Preserve the unitary character of the object under study

- Study the unit as a whole. Collect the breadth of data about the totality of life.
- Data not necessarily related to one discipline. Look at from different perspectives. Triangulation.

Case control studies

- Select two groups (cases), one with an effect (study group) and the other without (control group). Similar except for effect. Case of Manga Mandi village.
- Deformities in the bones of children.
- Explore the totality of the background of affected and unaffected. Develop hypothesis out of fishing expedition.

CS is empirical because:

A. It investigates a contemporary phenomenon within its real life context (retrospective study: process from effect to cause. Back in time. Manga Mandi case);

B. When the boundaries between the phenomenon and context are not clearly evident; and

C. Multiple sources of evidence are used.

Case study limitations:

- Lacks rigor,
- Reliability, and
- Representativeness

RESEARCH METHODS

Lecture 44

REPORT WRITING

Every report is custom-made, yet some conventions of format.

- Many companies and universities also have in-house, suggested report formats or writing guides that researchers should be aware of.

Report Format:

- The general plan of organization for the parts of a written or oral research report.

Tailoring the format to the project -- 2 reasons:

- To obtain the proper level of formality.
- To decrease the complexity of the report.

We shall look at the most formal type i.e. a report for a large project done within an organization or one done by a research agency for a client company. Usually bound with a permanent cover. May be hundreds of pages.

The makeup of the report – the report parts:

- Prefatory parts.
- Main body.
- Appended parts.

Prefatory parts

- Title fly page.
- Title page.
- Letter of transmittal
- Letter of authorization.
- Table of contents
- Executive summary (Synopsis):
- - Objectives
- - Results
- - Conclusions
- - Recommendations

Main body

- Introduction:
- -- Background
- -- Objectives
- Methodology.
- Results → can be divided into different areas.
- Limitations.
- Conclusions and recommendations
- References.
- Acknowledgements.

Appended parts

- Data collection forms (questionnaires, check list, interview guide, other forms).
- Detailed calculations.
- General tables.
- Other support material.
- Bibliography, if needed.

Title fly page:

- Only the title appears on this page
- For the most formal reports, a title fly page precedes the title page.

Title Page:

- Title of the report. Give a brief but complete indication of the project purpose.
- The name/s of the person/s for whom the report was prepared. Titles and addresses to be given.
- The name/s of person/s who prepared it. Titles and addresses to be included.
- Date of release or presentation.

Letter of Transmittal:

- Included in relatively formal and very formal reports.
- Purpose to release or deliver the report.
- Serves to establish some rapport between the reader and the writer.
- The transmittal should not dive into report findings except in the broadest terms.

Virtual University, Lahore

December 15, 2013.

Mr. K. M. Khalil
Vice President for Marketing,
……..
Subject: Report on Study of Customer Satisfaction

Dear Mr. Khalil:

Here is a report on the study of customer satisfaction. The report was prepared according to your authorization letter dated April 15, 2013.
…….

We are grateful to you for your cooperation in this important study.

Sincerely

Letter of Authorization:

- This is the letter to the researcher approving the project, detailing who has responsibility for the project and indicating what resources are available.
- Researcher would not write this letter. Reference to this letter has already been made in letter of transmittal. Sufficient.
- Only in some cases exact copy of the original may be reproduced.

Table of Contents:

- Essential to any report.
- List the divisions and subdivisions of the report with page references.
- It is based on the final outline of the report.
- Should include the divisions and first level subdivisions. For short reports, only main divisions may be included.
- If report includes many figures and/or tables, list of these should immediately follow the table of contents.

Summary:

- Vital part of the report. Most managers always read a report's summary.
- The only chance the writer may have to make an impact on the management. Summary tells:

  -Why the research project was conducted?

  -What aspects of the problem considered?

  -What the outcome was?

  -What should be done?

Furthermore:

- Summary should be written after the report completion. It is the essence of the report.
- 2-3 pages. Properly condensed.
- Should be self sufficient. Often only summary is circulated.

1. Objectives stated, inclusive of background and purpose.
2. Major results are presented. Key results regarding each purpose.
3. Conclusions based on results. Interpretations.
4. Recommendations for action based on the conclusions. In many cases managers may not like recommendations in summary.

Main Body:

- Constitutes the bulk of the report. It includes: Introduction, methodology, results, and limitations of the study. It finishes with conclusions and recommendations based on results.
- Let us look at each:

Introduction:

- Explain why the project was undertaken and what is it aimed to discover. Based on the information provided in TOR. It is the:
- Background: Explain why the project was worth doing.
- What objectives? What research question to be answered? At the end of the study see that each objective has been addressed.

For thesis

- Review of literature (not looking for pieces of summaries)
- Theoretical framework and derivation of hypothesis (es)
- Hypothesis (es) or research questions. Operationalization of the variables

Methodology:

Technical procedures must be explained.

Supplement the material in this section with more details in the appendix. This part should address six topics:

1. *Research design.* Purpose of study → exploratory, descriptive, or explanatory. Why specific design suited to the study?

2. Data collection methods. Primary or secondary data used. How primary data were collected – survey, experiment, observation. Multiple techniques used – triangulation.

Methodology –contd.

3. Sample design: What was the target pop? Sampling frame. Type of sample. Selection process.

4. Instrument of data collection: What instrument and why? Copy in appendix.

5. Fieldwork/Data collection: how many, type of field workers used? Training/supervision How was quality control assured?

6. Analysis: How was analysis carried – score index applied, statistics used

1. When constructing a questionnaire, it is imperative to do each of the following EXCEPT _____.

   a. Use "leading" or "loaded" questions (Lesson no 23 Page no 79 Handouts)
   b. Use natural language
   c. Understand your research participants
   d. Pilot your test questionnaire

2. Pilot study is _____.

   a. A small scale study (Lecture no 24 Page no 81 Handouts)
   b. A study involving pilots
   c. A study to test the tool of data collection
   d. A large scale study

3. An expectant pause is a part of:

   a. Recording the responses
   b. Verbal communication
   c. Probing tactics (Lesson no 25 Page no 85 Handouts)
   d. In house training

4. The difference between parameter and statistics is called

   a. Random Error
   b. Sampling Error (Lesson no 26 Page no 90 Handouts)
   c. Bias
   d. Error

5. Coding categories should be:

   a. Mutually inclusive and dependent
   b. Mutually exclusive and independent (Lesson no 29 Page no 101 Handouts)
   c. Mutually exclusive and dependent
   d. Mutually inclusive and independent

6. "How much time is needed to locate the respondents?" is a part of _____ aspect which needs to be evaluated during pilot testing

   a. Reactions to Respondents (Lecture no 24 Page no 81 and 82 Handouts)
   b. Discovering errors in the instrument
   c. Sampling procedure can be checked
   d. Staffing and activities of research team can be checked

7. Asking neutral questions is a part of:

   a. Probing tactics ((Lesson no 25 Page no 85 Handouts)
   b. Recording the responses
   c. Verbal communication
   d. In house training

8. Which one of the following aspect needs to be evaluated during "Pilot Testing"?

   a. Reactions of Respondents
   b. Discovering errors in the instrument
   c. Sampling procedure can be checked
   d. All of the given options (Lesson no 24 Page no 81 and 82 Handouts)

9. How often do you watch TV drama?
   Occasionally
   Frequently
   Regularly
   I do not watch TV drama

   This question when asked from respondents

   a. It is a leading question
   b. It is a loading question
   c. Proper sequence is missing
   d. This is an ambiguous question (Lecture no 23 Page no 79 Handouts)

10. The summary description of a given variable in a population is called as:

    a. Statistic
    b. Sampling unit
    c. Observation unit
    d. Parameter (Lesson no 26 Page no 90 Handouts)

11. _____is a set of rules stating that certain numbers are assigned to variable attributes.

    a. Information
    b. Coding
    c. Data
    d. Coding procedure (Lesson no 29 Page no 100 Handouts)

12. What is the main purpose of pilot testing?

a. To accumulate responses
b. To identify respondent
c. To identify potential problems (Lesson no 24 Page no 81 Handouts)
d. To evaluate the research

13. _____ is that unit about which information is collected and which provides the basis of analysis.

a. Population
b. Sampling frame
c. Element (Lecture no 26 Page no 89 Handouts)
d. Sampling unit

14. _____ is the list of elements from which the sample is drawn.

a. Population
b. Sampling frame (Lesson no 26 Page no 89 Handouts)
c. Sampling unit
d. Observation unit

15. In cluster sampling, subgroups are:

a. Internally homogeneous
b. Externally heterogeneous
c. Externally homogeneous (PPT 28) & Internally heterogeneous
d. None of the given

16. In which of the following Tests, we give extra training to the interviewer in case of big research funded by an organization?

a. Test Run
b. Pilot test (Lecture no 24 Page no 81 Handouts)
c. Post Hoc Test
d. T-Test

17. The small scale trial run of a particular item is called _____.

a. Test run
b. Pilot test (Lecture no 24 Page no 81 Handouts)
c. Post hoc test
d. t-test

18. _____ means the verbal prompts made by fieldworker when the respondent must be motivated to communicate.

a. Probing (Lecture no 25 Page no 85 Handouts)
b. Recording the responses
c. Verbal communication
d. In house training

19. "Willingness of the respondents to co-operate" is a part of _____ aspect which needs to be evaluated during pilot testing.

    a. Reactions of Respondents (Lecture no 24 Page no 81 Handouts)
    b. Discovering errors in the instrument
    c. Sampling procedure can be checked
    d. Staffing and activities of research team can be checked

20. Which of the following is **NOT** a form of non-probability sampling?

    a. Snowball sampling
    b. Stratified sampling (Lesson no 27 Page no 91 and 92 Handouts)
    c. Quota sampling
    d. Purposive sampling

21. Which of the following can be referred as a Reputational sampling?

    a. Purposive Sampling
    b. Quota Sampling
    c. Convenience Sampling
    d. Snowball Sampling (Lecture no 27 Page no 92 Handouts)

22. "Is the logistical support adequate?" is a part of _____ aspect which needs to be evaluated during pilot testing?

    a. Reactions of Respondents
    b. Discovering errors in the instrument
    c. Sampling procedure can be checked
    d. Staffing and activities of research team can be checked (Lecture no 24 Page no 82 Handouts)

23. When constructing a questionnaire, there are few principles which must never be ignored. Which of the following is **NOT** one of those principles?

    a. Use "leading" or "loaded" questions (Lesson no 23 Page no 78,79 and 80 Handouts)
    b. Avoid double-barreled questions
    c. Avoid double negatives
    d. Avoid using multiple items to measure a single construct

24. Assigning numbers or other symbols to the responses, to group them into limited number of categories is called _____.

    a. Information
    b. Coding (Lecture no 29 Page no 100 Handouts)
    c. Data
    d. Editing

25. Which of the following options is first step to conduct an experiment?

    a. Choose an experimental design
    b. Develop a hypothesis (Page 118)
    c. Develop valid and reliable measure
    d. Locate appropriate subjects

26. Which of the following best defines the population element/unit of study?

    a. The subject on which the measurement is being taken
    b. The total collection of elements about which we wish to make some inferences
    c. The count of all elements in a population
    d. Some selected elements form a population through which we can draw conclusions about all population (Page 89)

27. Which of the following issues should NOT be mentioned in an introductory statement of questionnaire?

    a. The information will be kept confidential
    b. What the findings of the research are?
    c. Who is funding the research?
    d. Why the respondent has been chosen?

28. Which of the following is Not something a researcher will have to consider when thinking about their sample size?

    a. Time and cost
    b. Non-response
    c. Length of questionnaire
    d. Heterogeneity of population

29. The final stage in the coding process is

    a. Data entry
    b. Data cleaning (Lecture no 29 Page no 101 Handouts)
    c. Production coding
    d. Code construction

30. _____ sampling techniques, the researcher asks the participants to identify other potential research participants.

a. **Snowball (Lecture no 27 Page no 92 Handouts)**
b. Convenience
c. Purposive
d. Quota

31. You wish to make a precise estimate about the characteristics of a population of individuals. You should use:

a. Quota sampling
b. Probability sampling (Page 94)
c. Accidental sampling
d. Snowball sampling

32. Which of the following is NOT true about stratified random sampling?

a. It involves a random selection process from identified subgroups
b. Proportions of groups in the sample must always match their population proportions (Page 95)
c. Disproportional stratified random sampling is especially helpful for getting large enough subgroup samples when subgroup comparisons are to be done
d. Proportional stratified random sampling yields a representative sample

33. Which of the following would generally require the largest sample size?

a. Cluster sampling (Page 96)
b. Simple random sampling
c. Systematic sampling
d. Proportional stratified sampling

34. Which of the following techniques yields a simple random sample?

a. Choosing volunteers from an introductory psychology class to participate
b. Listing the individuals by ethnic group and choosing a proportion from within each ethnic group at random
c. Numbering all the elements of a sampling frame and then using a random number table to pick cases from the table. (Page 94)
d. Randomly selecting schools, and then sampling everyone within the school.

35. The faculty members of a university consist of 85% staff with masters as highest qualification. Rest of 15% can be divided as 10% M.phill, 4% Phd. And 1% post doctorate as highest qualification. A researcher wants the sample of each category in a proportion as given above. Which of the following is the best choice for him to take sample?

    a. Quota sampling (Page 92)
    b. Convenience sampling
    c. Simple random sampling
    d. Snowball sampling

36. A census taker often collects data through which of the following?

    a. Standardized tests
    b. Interviews
    c. Secondary data
    d. Observations

37. The recorded measures of certain phenomenon, is called _____.

    a. Information
    b. Editing
    c. Data (Lesson no 29 Page no 98 Handouts)
    d. Coding

38. A researcher wanted to study the satisfaction level of private school teachers in Lahore. For this purpose he selected two private schools from population of all private schools in Lahore and then took sample of 40 teachers randomly from those schools. Which of the following sampling technique he is using?

    a. Purposive sampling (Page 91)
    b. Convenience sampling
    c. Simple random sampling
    d. Stratified random sampling

39. Which of the following is more operational with the help of coding system?

    a. External Appearance
    b. Count Behaviors
    c. Time Duration
    d. Content analysis

40. "Have you ever had or been treated for skin allergy?
Yes or No"
Which type of question it is?

a. Leading question
b. Loaded question
c. Double barreled question (Lesson no 23 Page no 79 Handouts)
d. Ambiguous question

41. Which of the following cannot be done through pilot testing?

a. Structuring open ended questions (Page 81)
b. Determining population
c. Measuring the reliability of questions
d. Determining the acceptability of questions

42. Pilot testing can be beneficial in:

a. Structuring open ended questions
b. Determining population
c. Collecting sample
d. Analysis of data (Page 82)

43. "What is the level of satisfaction in your organization?" Which of the following represents the problem associated with this question? **(Page 79)**

a. This question may lead respondent to a particular answer
b. It is an open ended question
c. This question contains multiple hidden questions
d. It is vague as respondent may interpret it in his own way

44. Which of the following is an advantage of open questions?

a. They allow respondents to answer in their own terms. (Page 85)
b. They prevent respondents from giving unusual answers.
c. They allow respondents to put less effort into their answer.
d. They allow respondents to code their answers in advance

45. Which of the following will give a more "accurate" representation of the population from which a sample has been taken?

a. A large sample based on the convenience sampling technique
b. A small sample based on simple random sampling
c. A large sample based on simple random sampling (Page 94)
d. A small cluster sample

46. In which of the following sampling methods People are available such as volunteer or can be easily recruited?

a. Simple random sampling
b. Cluster sampling
c. Systematic sampling
d. Convenience sampling (Page 91)

47. Which of the following is an example of an issue that would be checked in a pilot testing?

a. Completion of all questions on a questionnaire
b. Legible writing
c. Single responses to all multiple choice questions
d. Logical answers between questions (Page 92)

48. In data collection process, when a researcher has to conduct an in depth interview, which of the following too necessary?

a. Observation
b. Questionnaire
c. Interview Guide (Lecture no 23 Page no 78 Handouts)
d. Interview schedule

49. When does random assignment occur?

a. When subjects have an equal chance to be assigned to any condition and/or group in a study
b. When a subject can choose its group and study
c. When a subject cannot choose its group and study
d. None of the given options

50. Statistics is used by researchers to:

a. Analyze the empirical data collected in a study (Page 106)
b. Make their findings sound better
c. Operationally define their variables
d. Ensure the study comes out the way it was intended

51. All of the following are the instruction to record answers to open ended questions EXCEPT:

a. Record responses during the interview
b. Use the interviewer's own words (Page 86)
c. Include everything that pertains to the question objectives
d. Include all your probes

52. If, after pilot testing, a researcher finds that questionnaire is unable to reveal all the facts the researcher needed. Which type of the following error is expected if questionnaire is not modified?

a. Researcher may not be able to capture the required variables (Page 82)
b. Sampling issues may arise
c. Data analysis may get wrong
d. Respondents may lose interest

53. The list of all units in a population is called

a. Random sampling **b. Sampling Frame (Page 89)**
c. Parameter d. Statistic

54. A collection of related records is called _____.

a. Data b. Information
c. Knowledge **d. File (Lecture no 29 Page no 100 Handouts)**

55. A numerical value calculated from population is called

a. Bias b. Sampling Error
c. Error **d. Parameter (Page 90)**

56. A researcher has divided population in three geographical areas and has taken random sample from the population divided on the basis of level of education, age and salary package by making groups of persons having particular characteristics in those specified areas. Which of the following sampling technique he is doing?

a. Simple random sample
b. Stratified random sampling (Page 95)
c. Cluster sampling
d. Systematic sampling

57. In stratified sampling, strata are:

a. Internally homogenous (Page 95) & Externally heterogeneous
b. Internally heterogeneous
c. Externally homogenous
d. None of the given option

58. The first and most important condition for simple random sampling is:

a. Population must be homogenous (Page 95)
b. Target population must be greater than 30
c. There must be no budget constraints
d. All of the given options

(Page no 66 of 86 Is from lesson no 31 Not included in quiz 2)

59. Which of the following provides insight into the complexity of using interviews?

a. Potential influence of gender or cultural differences between interviewer and interviewee
b. The accuracy of the interviewee's responses
c. The different semantic meanings of the language being used
d. Accuracy, influences and language and

60. How do you find comfort at your work place?
Very good
Good
Fair
Poor

This question is poorly structured. In your opinion what is the problem in its design?

a. Ambiguity and vagueness (Lecture no 23 Page no 79 Handouts)
b. Loaded question
c. Leading question
d. Double barreled question

61. All of the following are principles of asking interview questions, EXCEPT:

a. Ask the question in the order in which they are presented in the questionnaire.
b. Repeat questions that are misunderstood or misinterpreted.
c. Read each question rapidly (Page 84)
d. Ask questions exactly as they worded in the questionnaire

62. The logic of purposive sampling is:

a. That a random sample can generalize to a population.
b. That a few information-rich participants studied in depth yield many insights about a topic. (Page 91)
c. To include all participants, even though they are not all relevant to the problem.
d. To use participants because the researcher has access to them.

63. Which of the following concerns is MOST likely to become a problem if Mr. Baber uses purposive sampling techniques?

a. Generalizability from the sample to the population. (Page 91)
b. Statistically equal groups
c. Subjects participating in the study.
d. Selection of participants who are thought to be informative, articulate, and experienced with the research topic and setting.

64. Which of the following is pivotal in capturing accurate variables researcher wants to study?

a. Identifying the area of research
b. Question design and development
c. Selection of respondents
d. Selection of sampling techniques

65. A researcher wants to study the impact of average hour of study in a week on CGPA of MBA students in virtual university of Pakistan. He focused on specifically the students specialized in finance. The students in VU is an example of which of the following?

a. Sample
b. Element
c. Population
d. Target population (Page 89)

66. A random sampling technique which chooses number of subgroups from population for in-depth study is called

a. Cluster sampling
b. Simple random sample
c. Systematic sampling
d. Stratified random sampling (Page 95)

67. Non probability sampling is used when which of the following condition fulfills?

a. Researcher thinks that every element should be given equal chance of selection
b. Researcher is least concerned about the generalizability of results (Page 91)
c. Results should accurately represent total population
d. Researcher needs to reduce sampling bias

68. Which of the following is/are principle(s) of questionnaire construction?

a. Use multiple items to measure abstract constraints
b. Avoid double-barreled questions (Page 79)
c. All of the given options
d. Consider using multiple methods when measuring abstract constructs

69. A sample can be defined as:

a. The count of all elements in a population
b. The total collection of elements about which we wish to make some inferences
c. Some selected elements from a population through which we can draw conclusions about all population (Page 89)
d. The subject on which the measurement is being taken

Question # 10 of 10 (Start time: 09:19:08 PM, 25 August 2020)

Total Marks: 1

"How much time is needed to locate the respondents?" is a part of _____ aspect which needs to be evaluated during pilot testing

Select the correct option

- Reactions of Respondents
- Discovering errors in the instrument
- Sampling procedure can be checked
- Staffing and activities of research team can be checked

Question # 5 of 10 (Start time: 09:13:40 PM, 25 August 2020) Total Marks: 1

Which of the following options is first step to conduct an experiment?

Select the correct option

- Choose an experimental design
- Develop a hypothesis
- Develop valid and reliable measure
- Locate appropriate subjects

Asking neutral questions is a part of:

Select the correct option

- Probing tactics 
- Recording the responses
- Verbal communication
- In house training

Question # 9 of 10 (Start time: 04:30:39 PM, 25 August 2020) Total Marks: 1

Which of the following best defines the population element/unit of study?

Select the correct option

- The subject on which the measurement is being taken
- The total collection of elements about which we wish to make some inferences

Correct

- The count of all elements in a population
- Some selected elements from a population through which we can draw conclusions about all population

Question # 1 of 10 (Start time: 09:09:11 PM, 25 August 2020) Total Marks: 1

Which one of the following aspect needs to be evaluated during "Pilot Testing"?

Select the correct option

- Reactions of Respondents
- Discovering errors in the instrument
- Sampling procedure can be checked
- All of the given options



Question # 8 of 10 (Start time: 09:03:17 PM, 25 August 2020)

Which of the following issues should NOT be mentioned in an introductory statement of questionnaire?

Select the correct option

- The information will be kept confidential
- What the findings of the research are?
- Who is funding the research?
- Why the respondent has been chosen?

Question # 1 of 10 (Start time: 08:17:28 PM, 25 August 2020)

Which of the following is not something a researcher will have to consider when thinking about their sample size?

Select the correct option

○	Time and cost
○	Non-response
○	Lenigth of questionnaire
○	Heterogeneity of population

Question # 10 of 10 (Start time: 09:05:43 PM, 25 August 2020)

_____ is a set of rules stating that certain numbers are assigned to variable attributes

Select the correct option

- Information
- Coding
- Data
- Coding procedure

Correct

The final stage in the coding process is

Select the correct option

- Data entry
- Data cleaning
- Production coding
- Code construction

Question # 2 of 10 (Start time: 09:28:07 PM, 25 August 2020)

_____ sampling techniques, the researcher asks the participants to identify other potential research participants

Select the correct option

- Snowball
- Convenience
- Purposive
- Quota

Question # 5 of 10 (Start time: 09:32:05 PM, 25 August 2020)

You wish to make a precise estimate about the characteristics of a population of individuals. You should use:

Select the correct option

- ○ Quota sampling
- ○ Probability sampling
- ○ Accidental sampling
- ○ Snowball sampling

Question # 4 of 10 (Start time: 10:24:00 AM, 25 August 2020)

When constructing a questionnaire, it is imperative to do each of the following EXCEPT _____.

Select the correct option

- Use "leading" or "loaded" questions
- Use natural language
- Understand your research participants
- Pilot your test questionnaire

Question # 7 of 10 (Start time: 08:51:28 PM, 25 August 2020) Total Marks: 1

How often do you watch TV drama?
Occasionally
Frequently
Regularly
I do not watch TV drama

This question when asked from the respondents

Select the correct option

- It is a leading question 
- It is a loaded question
- Proper sequence is missing
- This is an ambiguous question

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 08:45:34 PM, 25 August 2020)

Total Marks: 1

In non-experimental research, a researcher controls for alternative explanations with;

Select the correct option

- Statistics
- Instruments
- Laboratories
- None of the given options

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 08:54:09 PM, 25 August 2020) Total Marks: 1

The summary description of a given variable in a population is called as:

Select the correct option

- Statistic
- Sampling unit
- Observation unit
- Parameter



Question # 4 of 10 (Start time: 08:10:23 PM, 25 August 2020) Total Marks: 1

An expectant pause is a part of:

Select the correct option

- Recording the responses
- Verbal communication
- Probing tactics
- In house training

Question # 4 of 10 (Start time: 09:12:21 PM, 25 August 2020) Total Marks: 1

Asking neutral questions is a part of:

Select the correct option

- Probing tactics
- Recording the responses
- Verbal communication
- In house training

Question # 8 of 10 (Start time: 09:17:18 PM, 25 August 2020) Total Marks: 1

What is the main purpose of pilot testing?

Select the correct option

- To accumulate responses
- To identify respondent
- To identify potential problems 
- To evaluate the research

Question # 7 of 10 (Start time: 10:47:40 AM, 25 August 2020)

Total Marks: 1

Coding categories should be:

Select the correct option

- Mutually inclusive and dependent
- Mutually exclusive and independent
- Mutually exclusive and dependent
- Mutually inclusive and independent

Click to Save Answer & Move to Next Question

MC170200593: Misbah Aslam

Time Left 78 sec(s)

STA630:QUIZ 3

Quiz Start Time: 11:02 PM, 08 August 2019

Question # 6 of 10 (Start time: 11:04:21 PM, 08 August 2019)

Total Marks: 1

The final stage in the coding process is

Select the correct option

- Data entry
- Data cleaning
- Production coding
- Code construction

ABU SULTAN

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 11:02:43 PM, 08 August 2019) Total Marks: 1

Which of the following is not true about stratified random sampling?

Select the correct option

- It involves a random selection process from identified subgroups
- Proportions of groups in the sample must always match their population proportions
- Disproportional stratified random sampling is especially helpful for getting large enough subgroup samples when subgroup comparisons are to be done
- Proportional stratified random sampling yields a representative sample

ABU SULTAN



Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 09:59:20 PM, 08 August 2019) Total Marks: 1

Which of the following is not something a researcher will have to consider when thinking about their sample size?

Select the correct option

- ○ Time and cost.
- ○ Non-response.
- ○ Length of questionnaire.
- ● Heterogeneity of population.



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Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 09:14:57 PM, 08 August 2019)

Total Marks: 1

Which of the following would generally require the largest sample size?

Select the correct option

- Cluster sampling
- Simple random sampling
- Systematic sampling
- Proportional stratified sampling

ABU SULTAN

Question # 8 of 10 (Start time: 09:16:39 PM, 08 August 2019) Total Marks: 1

Which of the following techniques yields a simple random sample?

Select the correct option

- Choosing volunteers from an introductory psychology class to participate
- Listing the individuals by ethnic group and choosing a proportion from within each ethnic group at random.
- Numbering all the elements of a sampling frame and then using a random number table to pick cases from the table.
- Randomly selecting schools, and then sampling everyone within the school.

ABU SULTAN

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 08:49:13 PM, 08 August 2019) Total Marks: 1

Assigning numbers or other symbols to the responses, to group them into limited number of categories is called _____.

Select the correct option

- Information
- Coding 
- Data
- Editing

ABU SULTAN

Question # 6 of 10 (Start time: 08:44:46 PM, 08 August 2019) Total Marks: 1

The faculty members of a university consist of 85% staff with masters as highest qualification. Rest of 15% can be divided as 10% M.phill, 4% Phd. And 1% post doctorate as highest qualification. A researcher wants the sample of each category in a proportion as given above. Which of the following is the best choice for him to take sample?

Select the correct option

- ○ Quota sampling ✓ **ABU SULTAN**
- ○ Convenience sampling
- ○ Simple random sampling
- ○ Snowball sampling

Click to Save Answer & Move to Next Question

STA630:QUIZ 3

Quiz Start Time: 05:48 PM, 08 August 2019

Question # 1 of 10 (Start time: 05:48:41 PM, 08 August 2019)

Total Marks: 1

_____ sampling techniques, the researcher asks the participants to identify other potential research participants

Select the correct option

- Snowball 
- Convenience
- Purposive
- Quota

ABU SULTAN

Question # 3 of 10 (Start time: 09:13:09 PM, 08 August 2019) Total Marks: 1

A census taker often collects data through which of the following?

Select the correct option

- Standardized tests
- Interviews
- Secondary data
- Observations

ABU SULTAN

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 08:42:53 PM, 08 August 2019) Total Marks: 1

The recorded measures of certain phenomenon, is called _____.

Select the correct option

- Information
- Editing
- Data
- Coding

ABU SULTAN

Question # 10 of 10 (Start time: 06:00:48 PM, 08 August 2019)

Total Marks: 1

A process of changing data from their original form to a format that is more suitable to perform a data analysis is called as_____

Select the correct option

- Data creation
- Information transformation
- Information creation
- Data transformation ✓

ABU SULTAN

Question # 8 of 10 (Start time: 05:57:12 PM, 08 August 2019) Total Marks: 1

A researcher wanted to study the satisfaction level of private school teachers in Lahore. For this purpose he selected two private schools from population of all private schools in Lahore and then took sample of 40 teachers randomly from those schools. Which of the following sampling technique he is using?

Select the correct option

- Purposive sampling Correct
- Convenience sampling
- Simple random sampling
- Stratified random sampling

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 04:45:55 PM, 08 August 2019) Total Marks: 1

In cluster sampling, subgroups are:

Select the correct option

- Internally homogenous ✓ ABU SULTAN
- Externally heterogeneous
- Externally homogenous
- None of the given option

Question # 1 of 10 (Start time: 04:37:28 PM, 08 August 2019) Total Marks: 1

Which of the following is more operational with the help of coding system?

Select the correct option

- External Appearance
- Count Behaviors
- Time Duration
- Content analysis ✓

ABU SULTAN

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 08:05:05 PM, 26 August 2020) Total Marks: 1

"Have you ever had or been treated for skin allergy?
Yes
No"
Which type of question it is?

Select the correct option

- Leading question
- Loaded question
- Double barreled question
- Ambiguous question

Correct

Question # 6 of 10 (Start time: 05:21:29 PM, 27 August 2020) Total Mark

Which of the following can not be done through pilot testing?

Select the correct option

- Structuring open ended questions
- Determining population
- Measuring the reliability of questions
- Determining the acceptability of questions

Question # 9 of 10 (Start time: 10:42:52 PM, 27 August 2020) Total Marks: 1

_____ sampling techniques, the researcher asks the participants to identify other potential research participants

Select the correct option

- Snowball — Correct
- Convenience
- Purposive
- Quota

Question # 10 of 10 (Start time: 11:03:11 PM, 27 August 2020) Total Marks: 1

When constructing a questionnaire, there are few principles which must never be ignored. Which of the following is NOT one of those principles?

Select the correct option

- Do not use "leading" or "loaded" questions
- Avoid double-barreled questions
- Avoid double negatives
- Avoid using multiple items to measure a single construct

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 11:21:12 PM, 27 August 2020) Total Mar

"Is the logistical support adequate?" is a part of _____ aspect which needs to be evaluated during pilot testing

Select the correct option

- Reactions of Respondents
- Discovering errors in the instrument
- Sampling procedure can be checked
- Staffing and activities of research team can be checked

Question # 8 of 10 (Start time: 09:39:16 PM, 27 August 2020) Total Marks: 1

How often do you watch TV drama?
Occasionally
Frequently
Regularly
I do not watch TV drama

This question when asked from the respondents was

Select the correct option

- ○ It is a leading question
- ○ It is a loaded question
- ○ Proper sequence is missing
- ○ This is an ambiguous question

Click to Save and see a View to next question

Question # 7 of 10 (Start time: 08:07:51 PM, 26 August 2020)

Total Marks: 1

Pilot testing can be beneficial in:

Select the correct option

- Structuring open ended questions
- Determining population
- Collecting sample
- Analysis of data



Question # 8 of 10 (Start time: 08:46:42 PM, 26 August 2020) Total Marks

"What is the level of satisfaction in your organization?" Which of the following represents the problem associated with this question?

Select the correct option

- This question may lead respondent to a particular answer
- It is an open ended question
- This question contains multiple hidden questions
- It is vague as respondent may interpret it in his own way

Correct

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 01:54:46 PM, 27 August 2020) Total Marks

Which of the following is an advantage of open questions?

Select the correct option

- They allow respondents to answer in their own terms.
- They prevent respondents from giving unusual answers.
- They allow respondents to put less effort into their answer.
- They allow respondents to code their answers in advance.

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 02:06:59 PM, 27 August 2020) Total Marks

Which of the following will give a more "accurate" representation of the population from which a sample has been taken?

Select the correct option

- A large sample based on the convenience sampling technique
- A small sample based on simple random sampling
- A large sample based on simple random sampling
- A small cluster sample

In which of the following sampling methods People are available such as volunteer or can be easily recruited?

Select the correct option

- Simple random sampling
- Cluster sampling
- Systematic sampling
- Convenience sampling

Question # 9 of 10 (Start time: 03:07:21 PM, 27 August 2020) Total Mark

Which of the following is not true about stratified random sampling?

Select the correct option

- It involves a random selection process from identified subgroups
- Proportions of groups in the sample must always match their population proportions
- Disproportional stratified random sampling is especially helpful for getting large enough subgroup samples when subgroup comparisons are to be done
- Proportional stratified random sampling yields a representative sample

Question # 8 of 10 (Start time: 03:06:04 PM, 27 August 2020) Total Mark

Which of the following is an example of an issue that would be checked in a pilot testing?

Select the correct option

- Completion of all questions on a questionnaire
- Legible writing
- Single responses to all multiple choice questions
- Logical answers between questions (i.e., attributes should be rated high if purchase interest is high)

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 03:39:39 PM, 27 August 2020) Tot

In data collection process, when a researcher has to conduct an in depth interview, which of the following too necessary?

Select the correct option

- Observation
- Questionnaire
- Interview Guide
- Interview schedule

Question # 4 of 10 (Start time: 04:49:36 PM, 27 August 2020) Total Marks: 1

When does random assignment occur?

Select the correct option

- When subjects have an equal chance to be assigned to any condition and/or group in a study
- When a subject can choose its group and study
- When a subject cannot choose its group and study
- None of the given options

Question # 1 of 10 (Start time: 08:42:27 PM, 27 August 2020)

Coding categories should be:

Select the correct option

- Mutually inclusive and dependent
- Mutually exclusive and independent
- Mutually exclusive and dependent
- Mutually inclusive and independent

Question # 4 of 10 (Start time: 08:44:48 PM, 27 August 2020) Total Marks

The difference between parameter and statistics is called

Select the correct option

- Random Error
- Sampling Error
- Bias
- Error

Question # 5 of 10 (Start time: 05:31:26 PM, 27 August 2020) Total Mark

Statistics is used by researchers to:

Select the correct option

- Analyze the empirical data collected in a study
- Make their findings sound better
- Operationally define their variables
- Ensure the study comes out the way it was intended

Question # 2 of 10 (Start time: 05:38:54 PM, 27 August 2020) Total Marks: 1

Which of the following is an advantage of open questions?

Select the correct option

- They allow respondents to answer in their own terms.
- They prevent respondents from giving unusual answers.
- They allow respondents to put less effort into their answer.
- They allow respondents to code their answers in advance.

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 03:17:19 PM, 27 August 2020) Total Marks

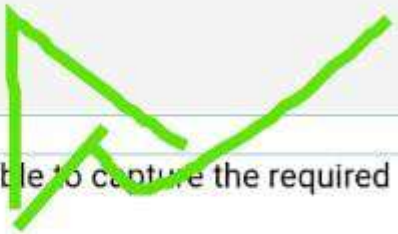
All of the following are the instruction to record answers to open ended questions EXCEPT:

Select the correct option

- Record responses during the interview
- Use the interviewer's own words
- Include everything that pertains to the question objectives
- Include all your probes

Question # 1 of 10 (Start time: 03:22:29 PM, 27 August 2020) Total Marks: 1

If, after pilot testing, a researcher finds that questionnaire is unable to reveal all the facts the researcher needed. Which type of following error is expected if questionnaire is not modified?



Select the correct option

- Researcher may not be able to capture the required variables
- Sampling issues may arise
- Data analysis may get wrong
- Respondents may lose interest

Question # 10 of 10 (Start time: 03:44:53 PM, 27 August 2020)

The list of all units in a population is called

Select the correct option

- Random sampling
- Sampling Frame
- Parameter
- Statistic

Question # 9 of 10 (Start time: 05:07:38 PM, 27 August 2020)

The small scale trial run of a particular item is called _____.

Select the correct option

- Test run
- Pilot test
- Post hoc test
- t-test

Correct

Question # 10 of 10 (Start time: 05:08:14 PM, 27 August 2020) Total Marks

Which of the following techniques yields a simple random sample?

Select the correct option

- Choosing volunteers from an introductory psychology class to participate
- Listing the individuals by ethnic group and choosing a proportion from within each ethnic group at random.
- Numbering all the elements of a sampling frame and then using a random number table to pick cases from the table.
- Randomly selecting schools, and then sampling everyone within the school.

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 05:17:41 PM, 27 August 2020)

Asking neutral questions is a part of:

Select the correct option

- Probing tactics
- Recording the responses
- Verbal communication
- In house training

Question # 4 of 10 (Start time: 05:20:29 PM, 27 August 2020)

A collection of related records is called _____.

Select the correct option

- Data
- Information
- Knowledge
- File



Click to Save Answer & N

Question # 10 of 10 (Start time: 05:24:34 PM, 27 August 2020)

A numerical value calculated from population is called

Select the correct option

- Bias
- Sampling Error
- Error
- Parameter

Click to Save Answer & M

Question # 1 of 10 (Start time: 05:28:21 PM, 27 August 2020) Total Marks

A researcher has divided population in three geographical areas and has taken random sample from the population divided on the basis of level of education, age and salary package by making groups of persons having particular characteristics in those specified areas. Which of the following sampling technique he is using?

Select the correct option

- Simple random sample
- Stratified random sampling
- Cluster sampling
- Systematic sampling

Assigning numbers or other symbols to the responses, to group them into limited number of categories is called _____.

Select the correct option

- Information
- Coding
- Data
- Editing

Question # 4 of 10 (Start time: 01:37:47 PM, 27 August 2020) Total Marks

The summary description of a given variable in a population is called as:

Select the correct option

- Statistic
- Sampling unit
- Observation unit
- Parameter

Question # 1 of 10 (Start time: 01:52:18 PM, 27 August 2020)

In stratified sampling, strata are:

Select the correct option

- Internally homogenous
- Internally heterogeneous
- Externally homogenous
- None of the given option

Click to Sav

Question # 4 of 10 (Start time: 01:55:26 PM, 27 August 2020)

Total Mark

The first and most important condition for simple random sampling is:

Select the correct option

- Population must be homogenous — Correct
- Target population must be greater than 30
- There must be no budget constraints
- All of the given options

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 01:58:05 PM, 27 August 2020)

Total Marks

Ways to manipulate and summarize numbers that represent data from research project is termed as;

Select the correct option

- Statistics
- Interpretation
- Description
- None of the given options

Question # 2 of 10 (Start time: 01:59:39 PM, 27 August 2020) Total Ma

Which of the following provides insight into the complexity of using interviews?

Select the correct option

- Potential influence of gender or cultural differences between interviewer and interviewee
- The accuracy of the interviewee's responses
- The different semantic meanings of the language being used
- Accuracy, influences and language used

Click to Save Answer & Move to Next Question

Which one of the following aspect needs to be evaluated during "Pilot Testing"?

Select the correct option

- Reactions of Respondents
- Discovering errors in the instrument
- Sampling procedure can be checked
- All of the given options

Question # 5 of 10 (Start time: 08:18:47 PM, 26 August 2020) Total Marks: 1

_____ is the list of elements from which the sample is drawn.

Select the correct option

- Population
- Sampling frame
- Sampling unit
- Observation unit

Question # 5 of 10 (Start time: 08:32:40 PM, 26 August 2020)

A researcher wanted to study the satisfaction level of private school teachers in Lahore. For this purpose he sele
Lahore and then took sample of 40 teachers randomly from those schools. Which of the following sampling tec

Select the correct option

- Purposive sampling
- Convenience sampling
- Simple random sampling
- Stratified random sampling

Question # 5 of 10 (Start time: 08:43:15 PM, 26 August 2020)

An expectant pause is a part of:

Select the correct option

- Recording the responses
- Verbal communication
- Probing tactics
- In house training

Question # 4 of 10 (Start time: 05:10:18 PM, 26 August 2020)

Coding categories should be:

Select the correct option

- Mutually inclusive and dependent
- Mutually exclusive and independent
- Mutually exclusive and dependent
- Mutually inclusive and independent

Question # 2 of 10 (Start time: 10:05:51 PM, 26 August 2020)

_____ means the verbal prompts made by fieldworker when the respondent must be motivated to communicate

Select the correct option

- Probing
- Recording the responses
- Verbal communication
- In house training

Question # 4 of 10 (Start time: 10:08:17 PM, 26 August 2020)

What is the main purpose of pilot testing?

Select the correct option

○	To accumulate responses
○	To identify respondent
○	To identify potential problems
○	To evaluate the research

Question # 6 of 10 (Start time: 10:10:20 PM, 26 August 2020)

How do you find comfort at your work place?
Very good
Good
Fair
Poor

This question is poorly structured. In your opinion what is the problem in its design?

Select the correct option

○	Ambiguity and vagueness
○	Loaded question
○	Leading question
○	Double barreled question

Question # 1 of 10 (Start time: 05:17:57 PM, 26 August 2020) Total Marks: 1

All of the following are the principles of asking interview questions, EXCEPT:

Select the correct option

- Ask the question in the order in which they are presented in the questionnaire.
- Repeat questions that are misunderstood or misinterpreted.
- Read each question rapidly
- Ask questions exactly as they are worded in the questionnaire

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 05:25:09 PM, 26 August 2020) Total Marks: 1

"Willingness of the respondents to co-operate" is a part of _____ aspect which needs to be evaluated during pilot testing.

Select the correct option

- Reactions of Respondents
- Discovering errors in the instrument
- Sampling procedure can be checked
- Staffing and activities of research team can be checked

Question # 7 of 10 (Start time: 05:22:34 PM, 26 August 2020) Total Marks: 1

The logic of purposive sampling is:

Select the correct option

- That a random sample can generalize to a population.
- That a few information-rich participants studied in depth yield many insights about a topic.
- To include all participants, even though they are not all relevant to the problem.
- To use participants because the researcher has access to them.

Click to Save Answer & Move to Next Question

Which of the following concerns is MOST likely to become a problem if Mr. Baber uses purposive sampling techniques?

Select the correct option

- Generalizability from the sample to the population.
- Statistically equal groups.
- Subjects participating in the study.
- Selection of participants who are thought to be informative, articulate, and experienced with the research topic and setting.

Question # 7 of 10 (Start time: 08:51:28 PM, 25 August 2020) Total Marks: 1

How often do you watch TV drama?
Occasionally
Frequently
Regularly
I do not watch TV drama

This question when asked from the respondents

Select the correct option

- ○ It is a leading question
- ○ It is a loaded question
- ○ Proper sequence is missing
- ○ This is an ambiguous question

Question # 10 of 10 (Start time: 05:25:41 PM, 26 August 2020) Total Marks

Which of the following is pivotal in capturing accurate variables researcher wants to study?

Select the correct option

- Identifying the area of research
- Question design and development
- Selection of respondents
- Selection of sampling techniques

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Time Left 87 sec(s)

STA630:Quiz 2

Quiz Start Time: 11:32 PM, 24 August 2020

Question # 9 of 10 (Start time: 11:42:18 PM, 24 August 2020)

Total Marks: 1

Which of the following is NOT a form of non-probability sampling?

Select the correct option

- Snowball sampling
- Stratified sampling
- Quota sampling
- Purposive sampling

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 11:36:47 PM, 24 August 2020) **Total Marks: 1**

A researcher wants to study the impact of average hour of study in a week on CGPA of MBA students in virtual university of Pakistan. He focused on specifically the students specialized in finance. The total of number of MBA students in VU is an example of which of the following?

Select the correct option

- Sample
- Element
- Population
- Target population

Click to Save Answer & Move to Next Question

sec(s)

STA630:Quiz 2

Quiz Start Time: 11:32 PM, 24 August 2020

Question # 2 of 10 (Start time: 11:33:13 PM, 24 August 2020)

Total Marks: 1

Which of the following can be referred as a Reputational sampling?

Select the correct option

- Purposive Sampling
- Quota Sampling
- Convenience Sampling
- Snowball Sampling

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 11:32:01 PM, 24 August 2020) Total Marks: 1

In which of the following Tests, we give extra training to the interviewer in case of big research funded by an organization?

Select the correct option

- Test Run
- Pilot Test
- Post Hoc Test
- T- Test

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 08:48:50 PM, 24 August 2020)

Total Marks: 1

Assigning numbers or other symbols to the responses, to group them into limited number of categories is called _____.

Select the correct option

- Information
- Coding
- Data
- Editing

Question # 2 of 10 (Start time: 07:15:44 PM, 11 August 2021) Total Marks: 1

Which of the following experimental designs has two groups, both the groups are exposed to pretest and posttest, the experimental group is exposed to treatment while the control group is not?

Select the correct option

- Classical Experimental Design
- Solomon's Four Group Design
- Two Groups Posttest-only Design
- Pretest and Posttest Experimental and Control Group Design

VERIFIED

Easiest way to describe the numerical data of one variable is with a:

Select the correct option

- Pie chart.
- Bar charts.
- Frequency distribution.
- Cumulative percentage distribution table.

pass VERIFIED

Question # 2 of 10 (Start time: 09:18:46 PM, 11 August 2021)

Which of the following is a measure in which a researcher adds or combines several distinct indicators of a construct into a single score?

Select the correct option

- Weighting
- Unidimensionality
- Scale
- Index

pass VERIFIED

STA630 QUIZ 3

Question # 1 of 10 (Start time: 09:17:32 PM, 11 August 2021)

A study that involves no researcher intervention is called _____.

Select the correct option

- A quasi-experimental study
- A non involvement study
- An experimental study
- An observational study

pass VERIFIED

Question # 3 of 10 (Start time: 09:19:54 PM, 11 August 2021)

Statistics is used by researchers to:

Select the correct option

- Operationally define their variables
- Make their findings sound better
- Ensure the study comes out the way it was intended
- Analyze the empirical data collected in a study

pass VERIFIED

Question # 5 of 10 (Start time: 09:22:09 PM, 11 August 2021)

The bivariate table derived from trivariate table is called:

Select the correct option

- Partials
- Correlation tables
- Sub table
- Bivariates

pass VERIFIED

Question # 4 of 10 (Start time: 09:21:30 PM, 11 August 2021)

Ways to manipulate and summarize numbers that represent data from research project is termed as:

Select the correct option

- None of the given options
- Interpretation
- Description
- Statistics

pass VERIFIED

STA630:QUIZ 3

Question # 6 of 10 (Start time: 09:23:46 PM, 11 August 2021)

_____ is a measure in which the researcher captures the direction of a variable construct.

Select the correct option

- Collapsing
- Combining
- Index
- Scale

pass VERIFIED

STA630:QUIZ 3

Question # 7 of 10 (Start time: 09:25:04 PM, 11 August 2021)

In non-experimental research, a researcher controls for alternative explanations with;

Select the correct option

- Laboratories
- Statistics
- Instruments
- None of the given options

pass VERIFIED

Question # 9 of 10 (Start time: 09:27:34 PM, 11 August 2021)

The purpose of experimental research is to allow the researcher to control the research situation to evaluate which type of relationship?

Select the correct option

- Correlationship
- Linear relationship
- Causal relationship
- Multivariate relationship

VERIFIED

STA630 QUIZ 3

Question # 8 of 10 (Start time: 09:25:50 PM, 11 August 2021)

The group that receives the experimental treatment condition is called as _____

Select the correct option

- Participant group
- Control group
- Independent group
- Experimental group

pass VERIFIED

Question # 1 of 10 (Start time: 07:14:43 PM, 11 August 2021) Total Marks: 1

_____ is a measure in which the researcher captures the direction of a variable construct

Select the correct option

- Scale
- Index
- Combining
- Collapsing

VERIFIED

Question # 4 of 10 (Start time: 07:17:40 PM, 11 August 2021)

Bar charts or graphs are used for:

Select the correct option

- Continuous variables
- Categorical variables
- Ordered variables
- Discrete variables

pass VERIFIED

STA630:QUIZ 3

Question # 10 of 10 (Start time: 09:29:26 PM, 11 August 2021)

_____ variable(s) should be changed at a time when conducting a single-case design.

Select the correct option

- 2
- 1
- 3
- 4

Question # 3 of 10 (Start time: 07:16:53 PM, 11 August 2021) Tota

Which of the following can be referred as semantic analysis?

Select the correct option

- Accretion
- Latent Coding
- Count Behaviors
- Intensity

pass VERIFIED

Question # 5 of 10 (Start time: 07:18:28 PM. 11 August 2021) **Total**

_____ is a measure in which the researcher captures the intensity of a variable construct.

Select the correct option

- Index
- Collapsing
- Scale
- Combining

Question # 8 of 10 (Start time: 07:22:11 PM, 11 August 2021)

Statistics is used by researchers to:

Select the correct option

- Analyze the empirical data collected in a study
- Ensure the study comes out the way it was intended
- Operationally define their variables
- Make their findings sound better

pass VERIFIED

Question # 7 of 10 (Start time: 07:20:45 PM, 11 August 2021) Total Marks:

Which of the following control techniques that is available to researcher, controls for both known and unknown variables?

Select the correct option

- Building the extraneous variable into the design
- Random assignment
- Matching
- Analysis of covariance

pass VERIFIED

The group that receives the experimental treatment condition is called as _____.

Select the correct option

- Participant group
- Control group
- Independent group
- Experimental group

pass VERIFIED

Question # 6 of 10 (Start time: 07:19:18 PM, 11 August 2021) Total Marks: 1

Which of the following is the process of marking segments of data with symbols, descriptive words or category?

Select the correct option

- Coding
- Concurring
- Segmenting
- Coloring

pass VERIFIED

Question # 1 of 10 (Start time: 10:08:31 PM, 11 August 2021) Total Marks: 1

'Making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions' is definition of;

Select the correct option

- Summary statistics
- Graphical explanation
- Visual representation
- Interpretation

pass VERIFIED

Question # 3 of 10 (Start time: 10:10:35 PM, 11 August 2021)

Total Marks: 1

When you are confident that the experimental manipulation produced the changes you measured in the dependent variable, your study probably has good _____ validity.

Select the correct option

- Internal
- External
- Causal
- Construct

pass VERIFIED

Question # 2 of 10 (Start time: 10:09:40 PM, 11 August 2021) Total Marks: 1

_____ is a measure in which a researcher adds or combines several distinct indicators

Select the correct option

- Index
- Collapsing
- Scale
- Combining

pass VERIFIED

Question # 6 of 10 (Start time: 10:16:14 PM, 11 August 2021)
Total Marks: 1

Which of the following means that all the items in a scale or index fit together or measure a single construct?

Select the correct option

- Weighting
- Scale
- Unidimensionality
- Index

pass VERIFIED

Question # 4 of 10 (Start time: 10:12:48 PM, 11 August 2021)
Total Marks: 1

When studying the simultaneous responses to two categorical questions we should set up a:

Select the correct option

- Cumulative percentage distribution table.
- Frequency distribution table.
- Contingency table.
- Histogram.

pass VERIFIED

Question # 8 of 10 (Start time: 10:18:22 PM, 11 August 2021) Total Marks: 1

A process of changing data from their original form to a format that is more suitable to perform a data analysis is called as_____

Select the correct option

- Information creation
- Information transformation
- Data creation
- Data transformation

pass VERIFIED

Question # 7 of 10 (Start time: 10:17:15 PM, 11 August 2021) Total Marks: 1

Control in experimental research is BEST described as an attempt to:

Select the correct option

- Limit threats to external validity
- Equate the subjects on the treatment variable
- Limit threats to internal validity
- Increase the generalize ability of findings

pass VERIFIED

Question # 9 of 10 (Start time: 10:18:53 PM, 11 August 2021)
Total Marks: 1

Which of the following is an example of a Likert scale?

Select the correct option

- 0 to 100
- Strongly agree/agree/neutral/disagree/strongly disagree
- Male/female
- Agree/disagree

pass VERIFIED

Question # 10 of 10 (Start time: 10:20:33 PM, 11 August 2021) Total Marks: 1

Which of the following control techniques that is available to researcher, controls for both known and unknown variables?

Select the correct option

- Building the extraneous variable into the design
- Analysis of covariance
- Random assignment
- Matching

pass VERIFIED

Question # 2 of 10 (Start time: 12:30:11 PM, 12 August 2021)

Which of the following experimental designs has only one group, a pretest, a treatment, and a posttest?

Select the correct option

- Pretest and Posttest Experimental and Control Group Design
- None of the given options
- One Group Pretest-posttest Design
- One-shot case study Design

pass VERIFIED

estion # 3 of 10 (Start time: 12:30:59 PM, 12 August 2021)

Making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions' is definition o

ect the correct option

- Interpretation
- Summary statistics
- Visual representation
- Graphical explanation

pass VERIFIED

Click to Save

Question # 1 of 10 (Start time: 12:28:27 PM, 12 August 2021)

Which of the following control techniques that is available to researcher, controls for both known and unknown varia

Select the correct option

- Analysis of covariance
- Random assignment
- Matching
- Building the extraneous variable into the design

pass VERIFIED

Question # 4 of 15 (Start time: 12:32:11 PM, 12 August 2021)

Which of the following experimental designs has two groups, both the groups are exposed to pretest and posttest, the experimental group while the control group is not?

Select the correct option

- Solomon's Four Group Design
- Classical Experimental Design
- Pretest and Posttest Experimental and Control Group Design
- Two Groups Posttest-only Design

pass VERIFIED

Question # 6 of 10 (Start time: 12:35:12 PM, 12 August 2021)

Which of the following can be referred as semantic analysis?

Select the correct option

- Count Behaviors
- Accretion
- Latent Coding
- Intensity

pass VERIFIED

Question # 5 of 10 (Start time: 12:33:08 PM, 12 August 2021)

To establish that variable X cause's variable Y, all of the following conditions should be met, EXCEPT;

Select the correct option

- X (the presumed causal factor) should precede Y
- Other factors should possibly cause the change in the dependent variable Y
- Both X and Y should co-vary
- None of the given options

pass VERIFIED

Which of the following means that all the items in a scale or index fit together or measure a single const

Select the correct option

○	Unidimensionality
○	Scale
○	Index
○	Weighting

Question # 7 of 10 (Start time: 12:36:39 PM, 12 August 2021)

Which of the following term refers to a statistical method that can be used to statistically equate groups on a pretest or some othe

Select the correct option

- Differential influence
- Matching
- Analysis of covariance
- Experimental control

pass VERIFIED

Question # 10 of 10 (Start time: 12:39:40 PM, 12 August 2021)

The purpose of experimental research is to allow the researcher to control the research situation to evaluate which type of relationsh

Select the correct option

- Linear relationship
- Correlationship
- Multivariate relationship
- Causal relationship

pass VERIFIED

Click to Save Answ

Question # 2 of 10 (Start time: 11:22:48 PM, 11 August 2021) Total Marks: 1

Which of the following refers to the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable?

Select the correct option

- Specification pattern
- Negative linear relationship
- Suppressor variable pattern
- Replication pattern

pass VERIFIED

Question # 1 of 10 (Start time: 11:21:09 PM, 11 August 2021) Total Marks: 1

A study that involves no researcher intervention is called _____.

Select the correct option

- An observational study
- An experimental study
- A non involvement study
- A quasi-experimental study

pass VERIFIED

Question # 4 of 10 (Start time: 11:26:30 PM, 11 August 2021) Total Marks: 1

Which of the tables is based on cross-tabulation and the cases are organized in the table on the basis of two variables at the same time?

Select the correct option

- MINITAB Table
- SPSS Table
- Frequency Distribution Table
- Bivariate Tables

pass VERIFIED

Question # 3 of 10 (Start time: 11:24:40 PM, 11 August 2021) Total Marks: 1

When studying the simultaneous responses to two categorical questions we should set up a:

Select the correct option

- Histogram.
- Frequency distribution table.
- Contingency table.
- Cumulative percentage distribution table.

pass VERIFIED

Question # 6 of 10 (Start time: 11:29:16 PM, 11 August 2021) Total Marks: 1

Statistics is used by researchers to:

Select the correct option

- Make their findings sound better
- Ensure the study comes out the way it was intended
- Analyze the empirical data collected in a study
- Operationally define their variables

pass VERIFIED

Question # 5 of 10 (Start time: 11:27:22 PM, 11 August 2021) Total Marks: 1

Which of the following control techniques that is available to researcher, controls for both known and unknown variables?

Select the correct option

- Analysis of covariance
- Building the extraneous variable into the design
- Matching
- Random assignment

pass VERIFIED

Question # 8 of 10 (Start time: 11:31:54 PM, 11 August 2021)
Total Marks: 1

Which of the following elements are not required for structuring Observation in Non reactive research?

Select the correct option

- Direction
- Frequency
- Accretion
- Intensity

pass VERIFIED

Question # 7 of 10 (Start time: 11:31:11 PM, 11 August 2021) Total Marks: 1

To establish that variable X cause's variable Y, all of the following conditions should be met, EXCEPT;

Select the correct option

- Both X and Y should co-vary
- X (the presumed causal factor) should precede Y
- None of the given options
- Other factors should possibly cause the change in the dependent variable Y

pass VERIFIED

Question # 10 of 10 (Start time: 11:35:13 PM, 11 August 2021) Total Marks: 1

_____ is a measure in which a researcher adds or combines several distinct indicators

Select the correct option

- Collapsing
- Combining
- Index
- Scale

pass VERIFIED

Question # 9 of 10 (Start time: 11:33:06 PM, 11 August 2021) Total Marks: 1

Which of the following is true?

Select the correct option

- A study is internally valid but may not be externally valid
- None of the given options
- A study must be both internally and externally valid
- The internal and external validity of studies are not related in any way

pass VERIFIED

Question # 2 of 10 (Start time: 11:40:13 PM, 11 August 2021) Total Marks: 1

Which of the following is a measure in which the researcher captures the intensity, direction, level, or potency of a variable construct?

Select the correct option

- Unidimensionality
- Weighting
- Scale
- Index

pass VERIFIED

Question # 1 of 10 (Start time: 11:38:24 PM, 11 August 2021) Total Marks: 1

Frequency distribution technique can be used for which of the following scales?

Select the correct option

- Ordinal
- Interval
- Nominal
- All of the given options

pass VERIFIED

Question # 4 of 10 (Start time: 11:42:04 PM, 11 August 2021) Total Marks: 1

All of the following options are ethical issues in lab experiment EXCEPT;

Select the correct option

- Exposing participants to physical or mental stress
- Using research results to disadvantage the participants
- Debriefing the participants accurately after the experiment is over
- Deceiving subjects by deliberately misleading them

pass VERIFIED

Question # 3 of 10 (Start time: 11:41:07 PM, 11 August 2021) Total Marks: 1

In experimental research, a specific event in the external environment occurring between the first and second measurement that is beyond the control of the experimenter and that affects the validity of an experiment, is termed as;

Select the correct option

- Testing Effect
- Instrumentation Effect
- History Effect
- Maturation Effect

pass VERIFIED

Question # 6 of 10 (Start time: 11:44:34 PM, 11 August 2021) Total Marks: 1

A form of the nonequivalent comparison-group design is recommended when:

Select the correct option

- It is not possible to identify two groups
- It is not possible to control for a basic history effect
- All of the above
- It is not possible to randomly assign participants to groups

pass VERIFIED

Question # 5 of 10 (Start time: 11:43:43 PM, 11 August 2021) Total Marks: 1

Ways to manipulate and summarize numbers that represent data from research project is termed as;

Select the correct option

- Interpretation
- None of the given options
- Statistics
- Description

pass VERIFIED

Question # 8 of 10 (Start time: 11:47:09 PM, 11 August 2021) Total Marks: 1

Which of the following refers to the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable?

Select the correct option

- Negative linear relationship
- Replication pattern
- Suppressor variable pattern
- Specification pattern

pass VERIFIED

Question # 7 of 10 (Start time: 11:45:58 PM, 11 August 2021) Total Marks: 1

_____ is a measure in which the researcher captures the level of a variable construct.

Select the correct option

- Combining
- Scale
- Collapsing
- Index

pass VERIFIED

Question # 9 of 10 (Start time: 11:48:15 PM, 11 August 2021) Total Marks: 1

Easiest way to describe the numerical data of one variable is with a:

Select the correct option

- Cumulative percentage distribution table.
- Pie chart.
- Frequency distribution.
- Bar charts.

pass VERIFIED

Question # 10 of 10 (Start time: 11:49:32 PM, 11 August 2021) Total Marks: 1

A linear relationship looks like larger percentages in the:

Select the correct option

- None of the above
- Diagonal cell
- Vertical cell
- Horizontal cell

pass VERIFIED

Question # 1 of 10 (Start time: 12:28:27 PM, 12 August 2021)

Which of the following control techniques that is available to researcher, controls for both known and unknown varia

Select the correct option

- Analysis of covariance
- Random assignment
- Matching
- Building the extraneous variable into the design

pass VERIFIED

stion # 3 of 10 (Start time: 12:30:59 PM, 12 August 2021)

Making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions' is definition o

ect the correct option

- Interpretation
- Summary statistics
- Visual representation
- Graphical explanation

pass VERIFIED

Click to Save

Question # 2 of 10 (Start time: 12:30:11 PM, 12 August 2021)

Which of the following experimental designs has only one group, a pretest, a treatment, and a posttest?

Select the correct option

- Pretest and Posttest Experimental and Control Group Design
- None of the given options
- One Group Pretest-posttest Design
- One-shot case study Design

pass VERIFIED

Question # 4 of 10 (Start time: 12:32:11 PM, 12 August 2021)

Which of the following experimental designs has two groups, both the groups are exposed to pretest and posttest, the experimental group while the control group is not?

Select the correct option

- Solomon's Four Group Design
- Classical Experimental Design
- Pretest and Posttest Experimental and Control Group Design
- Two Groups Posttest-only Design

pass VERIFIED

Question # 6 of 10 (Start time: 12:35:12 PM, 12 August 2021)

Which of the following can be referred as semantic analysis?

Select the correct option

- Count Behaviors
- Accretion
- Latent Coding
- Intensity

pass VERIFIED

Question # 5 of 10 (Start time: 12:33:08 PM, 12 August 2021)

To establish that variable X cause's variable Y, all of the following conditions should be met, EXCEPT;

Select the correct option

- X (the presumed causal factor) should precede Y
- Other factors should possibly cause the change in the dependent variable Y
- Both X and Y should co-vary
- None of the given options

pass VERIFIED

Which of the following means that all the items in a scale or index fit together or measure a single const

Select the correct option

- Unidimensionality
- Scale
- Index
- Weighting

pass VERIFIED

Question # 7 of 10 (Start time: 12:36:39 PM, 12 August 2021)

Which of the following term refers to a statistical method that can be used to statistically equate groups on a pretest or some othe

Select the correct option

- Differential influence
- Matching
- Analysis of covariance
- Experimental control

pass VERIFIED

Click to Save

Question # 10 of 10 (Start time: 12:39:40 PM, 12 August 2021)

The purpose of experimental research is to allow the researcher to control the research situation to evaluate which type of relationsl

Select the correct option

- Linear relationship
- Correlationship
- Multivariate relationship
- Causal relationship

pass VERIFIED

Click to Save

In bivariate relationship, if an alternative explanation explains a relationship, then relationship is;

Select the correct option

○	Controlled
○	None of the given options
○	Spurious
○	Factual

pass VERIFIED

STA 630-Research Methods NEW LECTURE-WISE QUESTIONS AND ANWERS FOR FINAL TERMS PREPARED BY VIRTUALIANS.PK

LESSON NO-23

Q-explain funnel technique?

ANSWER=Funnel technique is basically used while doing literature review and in questionnaire construction. Funnel technique in research refers to move from general to specific i.e. ask general questions first from respondents to put them at ease and then funneling down towards narrowed (specific) questions. You must have looked funnel whose one end is broad and other end is narrow, the technique is adopted likewise in academics.

Q- What are open ended questions Open-ended?

ANSWER-Open-ended questions are questions in which respondent is free to answer in a way he/she likes. Options to answer the question are not given rather only vacant lines are present in which respondent writes his answer. Open ended questions are useful in opinion seeking, exploratory research when problem is totally new and researcher wants to have clarity of it, and taking suggestion.

Structured questions are also called closed-ended questions. Options are given and respondent has to choose his/her response from them.

Closed ended questions include following:

1.Multiple choices.

2.Dichotomous questions.

3.Ratingscales.

Closed ended questions are useful when researcher wants to have a desired positive or negative frame of mind of respondents; also when researcher wants to get answer that is limited and can be answered in a few words

Q-.why questionnaire Design? It is essential for data collection tools?

ANSWER-While conducting a research, a researcher sets some questions and to answer those questions he/she needs some evidences. Those evidences may be in the form of data. To collect the data various techniques are used like making observations, conducting interviews, performing experiments and conducting surveys. In surveys, to find the perception of participants, researcher needs a standardized tool against which he/she can collect data. The standardized tool is called questionnaire and to make it standardized, researcher has to design that questionnaire according to predefined rules. For all studies, it is not necessary to design questionnaire. Sometimes researcher uses already developed questionnaires and in some case, he/she develops new questionnaire.

Q-Where we apply the questionnaire tool of data collection?

ANSWER-Questionnaire refers to a list of questions which is used to get responses of the customers by handling the questionnaire directly to respondents. It is used when respondent's convenience is of importance. They can fill the questionnaire when they get time. Moreover as it is cost efficient, researcher uses it when there is this constraint. When observing the body language of respondent is not of high importance, questionnaire can be used also.

LESSON NO-24

Q-explain make dummy tables?

Answer-During analysis process, researcher checks the presentation of results of statistical tests by developing some table formats to evaluate the readability and understandability of the information provided in those tables. Those tables are called dummy tables. For more understanding, read the content available in following link.

Q-explain Pilot Testing and also the dummy table?

ANSWER-In surveys, to find the perception of participants, researcher needs a standardized tool against which he/she can collect data. The standardized tool is called questionnaire and to make it standardized, researcher has to design that questionnaire according to predefined rules. To check whether all the predefined rules are incorporated in the developed questionnaire, he/she selects few of the respondents from targeted population and asked them to fill the questionnaire and then researcher applies statistical tests. This process is called pilot testing.

Pilot testing is not limited to questionnaire development; researcher can use this technique to test whole methodology as well.

After applying the statistical tests, researcher also checks the presentation of results of statistical tests by developing some table formats to evaluate the readability and understandability of the information provided in those tables. Those tables are called dummy tables

LESSON NO 25

Q) What is interviewing in two days era?

ANSWER-

In fastest growing trends of Social Media - Interviewing Technique may be implemented through social media websites and online Questionnaires

Lesson no 26

Q-explain in easy words the theoretical sampling?

Answer-Theoretical sampling term was coined by Barney Glaser and Anselm Strauss in 1967. Theoretical sampling refers to the process of choosing new research insights, sites or cases to compare with ones that have already been studied. It is one of the tools of qualitative research

For example, a field researcher may be observing a group of people on a site during week days. Theoretically, the researcher may question whether the people at the same at other times like on weekends/ nights or when other aspects of site change. He or she could then sample other time periods (e.g. nights and weekends) to get full picture of group people and compare whether outcomes or actions are same or dissimilar. Hope this reply will make you understand the concept.

Q-WHAT S is sampling technology?

ANSWER-As you know, Sample is a subset of the population. It comprises some member selected from it. In other words, some, but not all, elements of the population would form the sample. If 200 members are drawn from a population of 1000 middle level managers, these 200 employees form the sample of the study.

The process to select the sufficient number of elements from the target population is known as Sampling. Sampling (terminology) is used for selecting the sample from a population. Hope this reply will make you understand the term Sampling.

Q- EXPLAIN parameter and statistics

ANSWER-A parameter is a characteristic of a population such as the population mean, the population standard deviation etc. But we cannot calculate them without conducting the census of the whole population. Therefore to cope with this situation we apply Statistics on the selected sample. A statistic is a characteristic of a sample. The Sample Statistics---- the sample mean, standard deviations, variance in the sample and other statistical analysis enables us to make an educated guess about a population parameter based on a statistic computed from a sample.

Q-Explain about sampling frame why in actual practice sample will be drawn from a list of population elements that is often different from the target population?

ANSWER-The second part of your query is about sampling frame. It is a complete list of the units or elements of the population. Sometime it is not available and we have to prepare it by ourselves for example list of students of colleges in Lahore. Sometime sampling frame is already made and readily available for example telephone directory as it has the data of all the telephone users. But the telephone directory is not updated on regular basis and it might not be updated for the last five year, if we draw sample from the directory in the current year, the sample may be different from the sample which we require for our research. Due to such problems, sample drawn from a list of population elements is often different from the target population that has been defined in term of our research. Hope this reply will make you understand the term Sampling.

Q-explain Sampling Terminology?

ANSWER-Sample is a subset of the population. It comprises some member selected from it. In other words, some, but not all, elements of the population would form the sample. If 200 members are drawn from a population of 1000 middle level managers, these 200 employees form the sample of the study.

A sample is thus a subgroup of the population. By studying the sample, the researcher should be able to draw conclusion that would be generalizable to the whole population of the interest.

The process to select the sufficient number of elements from the target population is known as Sampling.

Q-Explain stratified random sampling?

Answer-Another type of random or probability sampling is which has great importance when our population is not homogenous and have different groups in the same population. Each groups has its own characteristics and different from each other. Each group is known as strata. Each group or strata is internally homogenous but different from other groups. When we divide the whole heterogeneous population into the number of homogenous groups, after that we take a random sample individually from the each group or strata. For example, let us assume we want to determine the spending habits of a hotel's guests. Knowing that the business clientele behaves quite different from the leisure guests, you might divide the whole clientele into groups or strata. The clientele consists of 80% business class and 20% leisure guests, using the thumb of rule about sample size determination; we need at least 100 respondents from our leisure guest strata and need four times (400) as many business clientele with respect to leisure guests to get proportionate stratified sampling (80% and 20%). Disproportionate sampling is only undertaken if particular strata are too small to present the population and for getting clear picture in analysis. Hope this reply will make you understand concept.

Q-explain bias?

ANSWER-Bias is infect an inclination of a presenter while offering a partial perspective and a refusal to even consider the possible merits of alternative points of view on a particular topic. You did not clearly mention in what context you are asking for like interviewee biasness, interviewer biasness, researcher biasness, sampling biasness etc. Hope this reply will make you understand concept.

Q-EXPLAIN Sampling technique selection process

ANSWER-Sampling technique selection process depends upon the nature of the research, situation, researcher's convenience and availability of the data. Probability and non-probability both can be used in qualitative and quantitative types of research.

Cluster sampling offers more homogeneity among groups and heterogeneity among groups' members therefore each cluster is comprised of heterogeneous characters and may be similar with the other cluster units. For example Ad hoc committee drawn from various departments to offer inputs to company president to enable him to make decision on products development, budget allocation, marketing strategies ; this is the example of a cluster.

Lesson no-28

Q-explain Cluster Sampling?

Answer-In Cluster sampling, the target population is first divided into mutually exclusive and collectively exhaustive subpopulations or cluster. Groups of elements that, ideally, would have heterogeneity among the members within each group are chosen for study in cluster sampling .Cluster sampling offers more

Homogeneity among groups and heterogeneity among groups' members therefore each cluster is comprised of heterogeneous characters and may be similar with the other cluster units. For example Ad hoc committee drawn from various departments to offer inputs to company president to enable him to make decision on products development, budget allocation, marketing strategies ; this is the example of a cluster. If all the elements in each selected cluster are included in the sample is called one-stage cluster sampling. If a sample of elements is drawn probabilistically from each selected cluster, the procedure is two-staged cluster sampling.

Lesson NO: 29

Q-how make table of random sample? What is its importance and for what sampling it is used?

Answer-Random number tables are created by random number generator using computational or physical device. It is used in simple random sampling, sampling frame is made and unique number is given to each item. Number is selected the by picking number from random number table, and then item which is allocated the same number is selected. Computer software for random sampling is suitable for large populations. It is important as it helps to select sample without any bias.

Q-What is meant by multivariate sample?

ANSWER-The term multivariate sample is ambiguous. Kindly mention the source from where you read that term. However, in the context of research, multivariate analysis is the term mostly used. Multivariate analysis is based on the statistical principle of multivariate statistics, which involves observation and analysis of more than one statistical outcome variable at a time

Q-explain stratified random sample in easy words with example

ANSWER-Stratified sampling is a two stage step. In first step heterogeneous groups are divided into homogeneous groups. These strata are mutually exclusive and collectively exhaustive. From each stratum an item is selected by using simple random sampling.

An item by using simple random sampling can be done in two ways:

Proportionate sampling: sample size is proportionate to the size of population in each stratum.

Disproportionate sampling: sample size is not proportionate to the size of population in each stratum.

In the following example total population is 1000. A total sample of 200 is to be extracted.

For proportionate sampling sample will be extracted from each stratum according to size of strata, for this 1000 will be divided by 200 i.e. 1000/200=5. Each stratum will then be divided by factor "5"

Proportionate sampling:

A	B	C	D	E	TOTAL
300	200	150	250	100	1000
60	40	30	50	20	200

Disproportionate sampling:

Same example is taken for disproportionate sampling, but sample will not be taken proportionately.

A	B	C	D	E	TOTAL
300	200	150	250	100	1000
50	30	45	25	50	200

The choice of selecting proportionate vs. disproportionate sampling depends on the population. If a stratum is very small and proportional taking of items from stratum will not make sample representative then it's better to take disproportionate sampling.

Q- Explain systematic random sample?

Answer-

In systematic random sampling, first item is selected through simple random sampling and then every nth item is chosen from the sampling frame.

Systematic sampling involves only a slight variation from simple random sampling. If the population contains N ordered elements and a sample size n is desired, one merely finds the ratio of N/n and rounds to the nearest integer to obtain the sampling interval. For example, if there are 600 members of the population and one desires a sample of

60, the sampling interval (i.e. N/n=600/60=10) is 10. A random number is then selected between 1 and 10, inclusively. Suppose the selected number turns out to be 4. The analyst then takes as the sample elements 4, 14, 24, and so on.

Lesson No-30

Q-explain scoring and score index with illustration

ANSWER-Scoring means that each response by an individual is given a value. For example in Likert scale value of 1 to 5 is given to responses from strongly disagree to strongly agree respectively. For score index, on each statement of the scale respondent could get a minimum score of 1 and a maximum score of 5, on 10 statements a respondent could get a minimum score of (10 X 1=10) and a maximum score of (10 X 5=50). So in this way score index of a scale ranges from 10 to 50.

Q-explain weighting with example.

ANSWER-Weight means that certain value is assigned to items to measure the total score of an index. Take an example of assigning equal weight i.e. each item is assigned equal value. If each item is assigned a weight of 2, then every item is multiplied by 2 and after this total sum is attained by adding all weighted score.

Q-explain unidimensionality with example

ANSWER-Unidimensionality means that all the items of a scale or index measure the same concept. For example; motivation is a concept, but in order to measure motivation of an individual, there are three dimensions included; job loyalty, interest in work, acceptance of responsibility. All these three dimensions measure the single concept motivation; therefore; unidimensionality exists there.

Q-explain indexes and scales with illustration

ANSWER-All scales are indices, but not all indexes are scales. In order to be precise a scale is a measure of the intensity of an attitude or emotion. Scales are usually formed at ordinal level of data; most of the times level of agreement and disagreement is measured with the help of a scale. Whereas; an index is a measure in which a researcher adds several indicators of a construct into a single score. The total score is usually a sum of multiple indicators. Indexes are often measured at the interval or ratio level. Clear distinction of scale and index can be done with the help of example of exams. The exam given to measure the competence of a student can be thought of a scale, but student's GPA can be taken as an index because it is a combination of a number of separate, independent marks obtained in different subjects.

Q-**explain summative score with example.**

ANSWER-Summative score refers to aggregate score of two or more items of questionnaire in order to form a new variable. For example there are 4 items in a questionnaire related to job satisfaction, for finding out the total job satisfaction; the values of all those four items will be added together. The aggregate sum which is called summative score will provide us with total job satisfaction.

Lesson no-31

Q-What is cluster sampling?

ANSWER- In cluster sampling, first population is divided into different clusters then with random sampling procedure clusters are selected and data is taken from clusters rather than individuals.

For example; a researcher needs to select a sample of 200 households in a particular city for in-home interviews. If the researcher selects a sample of 200 households via simple random sampling, they will be scattered across the city. Cluster sampling might be implemented in this situation by selecting 20 residential blocks in the city and randomly choosing 10 households on each block to interview.

Q- Explain constructing percentage table with example.

ANSWER-Constructing percentage table means to develop a percentage table. For example you want to know how many students have passed research methods course in previous semester. Assume you have hypothetical data of 1000 students with four categories i.e. “excellent”, “average”, “below average” and “Fail”. As per data, 500 students passed the course excellently, 250 students lie in average category, 200 students lie in below average category and 50 students failed the course. In order to construct a percentage table you have to draw percentages of each category. You can construct a percentage table in such a way;

Student performance in Research Methods course

Passing criteria		Frequency	Percentage
Excellent	500	500/1000*100=50	
Average	250	25	
Below average	200	20	
Fail	50	5	
Total	1000	100	

Q-explain bivariate table with example

ANSWER-Bivariate tables are one that presents the joint distribution of two variables.

Let's take an example: suppose in a survey, the data is collected from the respondents in order to find out that customers' choice of cellular phone of three brands; A, B, C; is dependent upon certain features i.e. battery time, design, price, durability. Here 3 brands is one variable; whereas, features is the second one.

Following presents the bivariate table:

Features	A	B	C	Total
Battery time	24	10	15	49
Design	29	18	12	59
Price	18	21	18	57
Durability	20	9	13	42
Total	91	58	58	207

Lesson No-32

Q-explain statistical control with example.

ANSWER-Once you found the relationship; positive linear or negative linear, next thing is to get confirmed that relationship is genuine and there is no spuriousness in it. In order to see

whether relationship is spurious or not researcher has to eliminate alternative hypothesis or explanation from study. In experimental study, researcher actually controls all other alternative explanations but in survey study, researcher controls such alternative explanations through statistics. In statistics control, another variable is taken which is called as control variable and its effect is checked on variables which are under observation. If control variable changes the nature of relationship then it means relationship between dependent and independent variable is not genuine but if control variable does not effect on relationship among variables then it means relationship is genuine.

For example, a researcher is studying IQ level of students of different levels and he found that students of higher level have more IQ. Now he wants to check whether such relation is genuine or not, therefore, he takes gender as control variable. If gender causes any difference in relationship then it means identified relation is spurious and if it does not then it means relation is genuine.

Q-explain linear relationship, negative, positive and curvilinear relationship with example.

ANSWER-A relationship that occurs between variables when change in one variable creates a change of same amount in other variable is called linear relationship. Linear relationship can either be negatively or positively related. Positive linear relationship means that increase in one variable creates an increase of same amount in other variable, whereas, relation will be negative if change in one variable creates decrease in other variable. A curved relationship is called curvilinear, because it approximates a curved line. In curvilinear relationships, the data points increase together up to a certain point (like a positive relationship) and then as one increase, the other decreases (negative relationship). It can also happen the other way round.

Q-explain about control variable with example.

ANSWER-Control variable is the variable which is held constant in a research in order to study the relationship between two variables. In some researches there are certain variables that can influence dependent – independent relationship and can modify the overall result findings. Thus, in order to have actual results researchers keep those variables constant.

Q- **What is the gist?**

ANSWER-Gist means essence or idea of something.

Q-what is 3rd factor? And trivariate table?

ANSWER-Trivariate table consist of three variables. Basically in lesson 32, trivariate tables are constructed to check the influence of control variable on bivariate relationship whether relationship is spurious or not. Trivariate tables consist of multiple bivariate tables.

Third factor is basically third variable.

Q-what is linear and diagonal relationship?

ANSWER-Linear relationship between variables depict that variables are either directly proportional or inversely proportional to each other. Linear relationships can be shown by straight line drawn in diagonal cells. The term diagonal means line joining two nonconsecutive vertices of a square which is a linear line. Thus, when linear relationship between variables is drawn, it forms a shape of diagonal. The term diagonal relation is not used in statistics rather linear relationship is used.

Lesson NO-33

Q-what is purpose of control group? Define in meaningful word the pretest and posttest?

ANSWER-Control groups are also called non-experimental groups. They are not exposed to experimental treatment. For example, if a researcher wants to test the new drug, the control group will not get any drug, whereas, experimental group does.

Pre-testing is done before and post-testing is done after. In pretesting subjects are measured on some dependent variable, without giving any treatment. Whereas, in posttest, after applying treatment, level of change is measured.

Q-what is statistical control, language of experiment and control grouping or

ANSWER- to see whether relationship is spurious or not researcher has to eliminate alternative hypothesis or explanation from study. In experimental study, researcher actually controls all other alternative explanations but in survey study, researcher controls such alternative explanations through statistics. In statistics control, another variable is taken which is called as control variable and its effect is checked on variables which are under observation. If control

variable changes the nature of relationship then it means relationship between dependent and independent variable is not genuine but if control variable does not effect on relationship among variables then it means relationship is genuine.

While doing experiment, there are two groups; experimental and control group. Control group is the group which is not exposed to conditions of experiment, whereas experimental group is the group which is exposed to treatment,

Lesson No-34

Q-explain quasi experiment design.

ANSWER-Quasi experimental design is one in which experimental group is exposed to treatment and effect is measured. It is weakest of all experimental designs as it does not measure the true cause and effect relationship. This is so because it does not measure the change by comparing the status of dependent variable as it was prior to treatment.

Q- Explain interaction effect explain with example.

ANSWER-Interaction effect refers to combined effect of independent variable on dependent variable as compared to individual impact of independent on dependent variable when taken alone.

This is used in experimental deigns. Take an example; a group of 50 individuals is given medicine to get cure of throat infection, second group of 50 is given herbal tea. A third group of 50 individuals is given both medicine and herbal tea for cure of throat infection. In order to measure combined impact of medicine and herbal tea, interaction effect will be calculate which is as follows:

Interaction effect= total effect – (medicine effect + herbal tea effect)

Q-What is pretest and protest

ANSWER-

Posttest and pretest are parts of experiment. Pretest refers to the measurement of dependent variable before experiment and posttest refers to the measurement of dependent variable after experiment. For example, a training session is conducted for the employees to enhance communication skills in them. You as a researcher are supposed to take data of employees before training program about their communication skills, which is pretest, and then data after training session to make comparison either training program is effective or not, which is posttest.

Q-What is alternative explanation of variables?

ANSWER-Alternative explanations are those explanations that can make dependent-independent relationship spurious or fake. This alternative explanation must be eliminated form study so that actual relationship can be studied.

Lesson No-35

Q-what is anonymity and confidentiality of researcher and sponsor in research.

ANSWER-Confidentiality means that information about a client's general business activities or the results of a client's project should not be disclosed to a third party. The researcher should not even disclose the name of a client unless permission is received in advance. Whereas, anonymity means that identity of the respondents is not even known by the researcher. It means respondent's identity remains anonymous to everyone. Sponsor refers to an individual/entity that is able to guarantee that the resources will be made available for the project in research.

Q-statistical regression explain with example

ANSWER-f an experimental group has extreme score on dependent variable, then this causes the values to get closer to average or mean after getting experimental treatment. The extreme values can be high or low. If extreme consists of high values then low values will get close to average, and vice versa. Existence of statistical regression is a threat to internal validity.

For example, a researcher wants to know the impact of training regarding salesmanship on employees. He should not choose extremely low or extremely high abilities. This is because, the employees with very low score will show high improvement and scores will get closer to mean. This is also known as "regressing towards mean".

Q-explain maturation effect and testing effect

ANWER-.Maturation and testing effects are two of the 8 effects that can jeopardize the internal validity. Maturation effect refers to the effect on cause and effect relationship due to the passage of time.

On the other hand, testing effect refers to the influence of pretest on posttest. Respondents may change the responses in posttest just because they have given pretest. It is how internal validity of measurement is at risk.

Q-what is pretest and posttest and what is interaction in experiment? What is Solomon table? Explain with examples.

ANSWER-

Solomon four group design is one of the types of experimental designs. In this design 4 groups are taken; two experimental and two control. One experimental group exposed to pretest – treatment – posttest and control group exposed to pretest – posttest. While on the other hand, second experimental group exposed to treatment – posttest and control group exposed only to posttest.

For example; you want to check the effect of bright lighting and high temperature on employees' performance. You make 4 groups; two experimental groups and two control groups. Two experimental groups are placed in room of bright light and high temperature whereas; two control groups is placed in normal light and room temperature. Posttest and pretest will be taken in such a way;

Effect of bright lighting and high temperature on employees' performance

Group	Pretest	Bright light and high Temperature	Posttest
Experimental	01	Treatment given	02
Control	03	-	04
Experimental	-	Treatment given	05
Control	-	-	06

Effect will be analyzed in such a way;

(O2 – O1) = E

(O4 - O3) = E

(O5 – O6) = E

(O5 - O3) = E

[(O2 –O1) – (O4 – O3)] = E

E = Effect

Lesson No-36

Q-What is meant by inferences?

ANSWER-Inference is last step of conducting content analysis research which is to draw conclusion of the analysis. Though it is quite difficult to give conclusion in content analysis because it only reflects the meaning inferred from content and not the true intention of those who actually created that content.

Q-amplainVariables and constructing coding categories with example.

ANSWER-

Variables and constructing coding categories is one of the steps of conducting content analysis research. In this step, researcher need to first identify variable that he wants to study. For

example variable is empowerment more specifically women empowerment. Now in order to operationally measure the variable researcher has to develop some codes. Codes are usually developed on the base of intensity of occurring. Defined codes help researcher, after reading or watching the content, to measure whether women are being empowered or not. For example, you watch a movie to analyze whether women are empowered or not. In a movie you see the stories of 5 women and each takes her own decision independently. You can say after analyzing the movie that women are highly empowered in it. Codes are helpful in such a way that researcher after analyzing content, categorize the content with respect to the defined code.

Q- Explain manifest and latent coding with example.

ANSWER-Dictionary meaning of manifest is clearly apparent and manifest coding is to code categories that are visible and apparent. For example, you as a researcher coded smiley face as happiness, and where you see smiley face, while analyzing the content, you perceive it as happiness. Dictionary meaning of latent is hidden and latent coding is coding in which researcher tries to comprehend the hidden meaning of content. For example, researcher reads full content (story or any phrase) and decides whether written material is about happiness or not.

Coder's knowledge about social meaning can influence latent coding hence it tends to be less reliable than manifest coding.

Q-Explain measurement and coding with example.

ANSWER-Measurement means taking possible observable dimensions of a construct are to be taken while measuring it. For example we want to know about spending habits of consumers. Here we have to define consumers. We classify them in terms of frequency of purchases made or by dividing them in terms of social class. Coding means labeling each category formed in measurement process. We can label social class as:

Classification	code
Upper class	3
Middle class	2

Lower class 1

Q-What is meant by construct?

ANSWER-A construct also called a concept is a name, term or terminology which a researcher gives to evidences found in reality. The task of the researcher is to find out the reality; in finding out this reality he observes various objects, attributes, or processes. These objects, attributes, or processes when given a name are called construct. For example, the researcher wants to find out how much employees are willing to work in an organization. This attribute of willingness is given a name "motivation", now motivation is a construct.

Lesson No.37

Q- Explain inferences form non reactive

ANSWER-

Firstly the concept is Inferences from Non-Reactive Data. Write proper concept if you are intended to ask.

Inference means conclusion which refers to deduce meaning after analyzing the content. In content analysis, researcher's inference play pivotal role in whole analysis. It is difficult to generalize the deduced inferences in content analysis research as there is less objectivity in it.

Q- Explain reliability and validity

ANSWER-Reliability and validity are two important criteria of good measurement. In any research whether, communication or non-communication, reliability and validity is way important. Validity of instrument refers to measure exactly what it is intended to measure and reliability of instrument refers to have consistent result repeatedly.

Q-Explain selecting topic for secondary analysis

ANSWER-In order to conduct secondary analysis you first have to select topic. As there is varied type of secondary data available hence, it is very important to neatly select topic and search for secondary data on same topic and analyze later on.

Q-explain use of secondary data and existing statistic documents

ANSWER-Secondary data refer to information gathered by someone other than the researcher.

You are guided to listen specifically lesson no. 8, 9 and 37 for better understanding.

Q-What is meant spontaneity?

ANSWER-

Spontaneity is one of the advantages of content analysis research. Spontaneity refers to spontaneous reaction of respondents which means natural or unplanned reaction. In content analysis, researcher can observe spontaneous reactions of respondents.

Q- Explain existing statistics document

ANSWER-Existing statistic document refers to secondary data means documents or report that are readily available. For example, books, research articles, dissertations, government reports.

Lesson No.38

Q-explain ethnography and ethno-methodology.

ANSWER-Ethnography is the study of culture from the perspective of people of that culture. The word ethnography is the combination of two Greek words; Ethnos and graph. Ethnos means people and graph means to write and describe something.

Ethno-methodology refers to practices and realities created by people for social interaction and how people interpret those realities. Ethno methodology is the study of those practices and methods that people use in their daily life for social interaction in their society.

Q- Explain logic of field research with example

ANSWER-Field research is a research in which researcher directly observes and interacts with community members in natural settings to get inside their perspectives. Questionnaire survey filled in field i.e. natural setting is the example of field research.

The logic of doing field research is; it's participatory kind of data collection in which respondents' perspectives is taken and analyzed.

Q-explain subject terminology.

ANSWER-Random assignment is a technique to make groups in order to compare them. In experimental research, researchers frequently want to compare one group with the other to draw results on the base of facts. In random assignment method, groups are made in a systematic way. It is a mathematical way to divide cases into groups and researcher cannot make assignments on the base of personal preferences. In random assignment, each individual has equal chance of being selected. Researcher makes a list of all subjects and randomly divide them into two groups i.e. control and experimental group. It is how random assignment is done.

Lesson No-39

Q-what is meant by deviants and explain involvement with deviants

ANSWER-In field research, deviant means those people who are involved in illegal activities. Involvement with deviants refers to how would researcher get involve and build rapport, trust with the deviants

Q-explain focus specification of the setting and are theoretical sampling.

ANSWER-The statement written by you refers to researcher suggesting to focus on only aspects which are of interest to the study. By theoretical sampling we mean the process of choosing new research sites or cases to compare with ones that have already been studied

Q-explain enter the field adopt a social role learn the ropes and get along with members

ANSWER-It means that while conducting the field research, the researcher has to adopt the culture and role of the people he is observing. In short, do as the Romans do.

Q-explain select site and gain access

ANSWER-It means select the area for the field research and go and do field research there.

Q-explain repair you read the literature and defocus

ANSWER-It means to get ready for field research by reading literature to know what already has been done in the topic of your research.

Lesson No-40

Q- Is the report and thesis same thing?

ANSWER-Thesis is a document of research work and findings of a researcher that follows systematic research process, whereas report is not necessarily a research work. For example you attended training and trainer asked you to submit a report presenting what you learnt. That document will not be thesis rather only a report. A thesis can be report but report is not always thesis.

Lesson No.41

Q-explain recollections in detail?

ANSWER-Recollection is one of the sources of historical data. Recollections are basically memories which are in form autobiographies, interviews and diary of a person about his past life or experience. It is documented or recorded material of individual's life which is considered as memory that help in historical research.

Q-EXPLAIN unique features of H.C research

ANSWER- unique features of H-C research are as follows;

- H-C researcher reconstructs what occurred from the evidence.
- H-C researchers interpret the evidence.
- H-C researchers are usually more aware of events occurring prior to the time studied and events that occurred after the period studied.
- H-C researcher does not use deterministic approach.

Q-EXPLAIN H.C research Similarities to field research

ANSWER- Basically comparison of historical research with field research has been made then similarities are identified. Six similarities are identified: First; both researches involve interpretation made by researcher thus researcher's point of view is unavoidable. Second; both researches collect and examine a lot of data. Third; both researches use grounded theory. Fourth; both researches involve researcher's own perspective that may differ from other. Fifth; both researches are systematic and consider time as an essential element. Sixth; in both researches generalization is limited due to incomplete historical and cross-cultural knowledge.

Q-TELL forms of economic system

ANSWER-In broader level there are three economic systems; Capitalism, socialism and Islamic economic system.

In historical comparative research, researchers also try to develop a concept by comparing the patterns and trends of past with relevant to the present. For example Karl Max, an economist, studied economic systems and examined that economic system determines the societal system. Basically economic system is an example quoted in lesson you can take any other example such as women empowerment and can make historical comparison.

Q-explain what is bracketing with regard to Historical comparative research?

ANSWER-

Bracketing means that the H-C researcher when reads the data available to him, reads it with his own perception and mind. He may not get the real message that the writing conveys, so this holds back the actual knowledge which could have been conveyed. In other words, this puts boundary to the knowledge.

Lesson No-42

Q-what is FGD?

ANSWER-Designing FGD refers to planning of FGD which needs very careful thought. Researcher has to design for what purpose he is going to conduct Focus Group Discussion. At

times, FGD is conducted for exploratory study and sometimes, it is conducted for having in depth qualitative information for a particular topic.

Q-define purpose of FGD in detail.

ANSWER-

The purpose of an FGD is to obtain in-depth information about concepts by taking perceptions, and ideas of the group. In FGD, participants not only answer lists of questions rather discuss the topic. Researcher acts the role of moderator and only moderates discussion without leading it.

Lesson No-43

Q-How can differentiate history effect and mechanical loss in experimental research,

ANSWER-The experiment should not be very long. History effect refers to the longer the time of interval between observations greater the chance that external factors will impact the dependent variable. E.g. a manufacturer of cheese products wants to launch an advertising campaign to see that how much increase would occur with it but meanwhile a rumor spreads that some virus is spreading through milk. This external factor will drastically decrease the demand of the cheese products. Whereas, mechanical loss refers to a problem due to equipment failure. Such as camera breaks while observing the respondents. This loss will create bias.

Q-what is societal change due to historical comparative?

ANSWER-Societal change refers to how change occurs in societies. This change can be change in nature, social institutions, or social behavior.

Q-EXPLAIN participant and non participant observation, also meanings

ANSWER-In participant observation, researcher observes and studies people by directly interacting with them for example, field researches whereas; in non-participant observation, researcher tries to observe the behavior of people without interacting with them. For example, observing the behavior of shoppers in a departmental store through a mirror or on a closed circuit TV.

Q-what are the important varieties of non reactive observation?

ANSWER-Varieties of non-reactive observation include Physical traces, Archives and Observations. Physical traces include Erosion and Accretion. Archives include Running Records (Regularly produced public records) and other Records (Private records). Observations include External appearance, Count Behaviors and Time Duration

Q-what is importance of debriefing in research?

ANSWER-Debriefing refers to the explanation of any deception with the participants and all the reasons for using deception in the context of the study's goals. Debriefing retains the goodwill of the respondent and providing incentive to participate in future research projects.

Q-define case study empirical.

ANSWER-Case study is empirical means that with case study reality can be tested through it as it investigates a phenomenon within its real-life context.

Q- **Define case control studies.**

ANSWER- Case control studies refer to select two groups (taking them as cases), one with an effect (study group) and the other without effect (control group).

Q-define preserve the unitary character of the object under study.

ANSWER-

Preserve the unitary character of the object under study refers to study the case as a whole by collecting the breadth of data about the totality of the unit.

Q-define Case study with example.

ANSWER-Case study is a comprehensive description and analysis of a single situation or a number of specific situations i.e. cases. It is an intensive description and analysis of a case. The examples could be a case study of a highly successful organization, a project (Kala Bagh Dam), a group, a couple, a teacher, and a patient.

Q-Define limitation in FGD.

ANSWER-Results of the focus group discussions cannot usually be generalized beyond the population from where the participants in FGD came. The moderator may influence focus group discussion and may bias the information.

Lesson No.44

Q-Diff b/w Proportionate & Disproportionate Stratified Sampling

ANSWER-

Stratified sampling is a two stage step. In first step heterogeneous groups are divided into homogeneous groups. These strata are mutually exclusive and collectively exhaustive. From each stratum an item is selected by using simple random sampling.

An item by using simple random sampling can be done in two ways:

Proportionate sampling: sample size is proportionate to the size of population in each stratum.

Disproportionate sampling: sample size is not proportionate to the size of population in each stratum.

In the following example total population is 1000. A total sample of 200 is to be extracted.

For proportionate sampling sample will be extracted from each stratum according to size of strata, for this 1000 will be divided by 200 i.e. 1000/200=5. Each stratum will then be divided by factor "5"

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Disproportionate sampling:

Same example is taken for disproportionate sampling, but sample will not be taken proportionately.

A	B	C	D	E	TOTAL
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50	30	45	25	50	200

The choice of selecting proportionate vs. disproportionate sampling depends on the population. If a stratum is very small and proportional taking of items from stratum will not make sample representative then it's better to take disproportionate sampling.

Q-define Appendix

ANSWER-Appendix is a supplementary material at the end of a book, article, document, or other text, usually consisting of an explanatory, statistical, or bibliographic material.

Q-define results, conclusion and recommendations.

ANSWER- Results

This is mostly the longest part of the research report. All the major findings are discussed in it which is obtained from tables and graphs. The results should be organized in coherent and logical way.

- Conclusions and Recommendations

In this part, the researcher links the findings with the management problem. In conclusion sort of solution is given to clients along with the recommendations for the organization.

Q- Define introduction background and significance and objective.

ANSWER-Introduction refers to overview of research. It includes background, significance and objective of study. Background could help in looking at the magnitude of the problem and significance refers to importance of research.

Q- Explain letter of transmutation and letter of authorization.

ANSWER-Letter of Transmittal is written to release the report to the recipient. It also serves to establish some rapport between the reader and the writer Letter of Authorization is a letter to the researcher approving the project, detailing who has responsibility for the project and indicating what resources are available to support it.

Lesson No.45

Q-describe the Deception in easy words?

ANSWER-

Deception includes several types of communications or omissions that serve to distort or omit the complete truth. While exploring your area of interest may require misleading or not completely informing your subjects about the true nature of your research, as a general rule, serious deception should be avoided whenever possible, since it jeopardizes the integrity of informed consent.

Deception includes, but is not limited to:

- intentionally misleading participants about their status
- Giving false information about the investigators or the research purpose
- Omitting information about the real purpose of the research

Lesson 23:	TOOLS FOR DATA COLLECTION

TOOLS FOR DATA COLLECTION: Broadly there are three tools of data collection as part of communication surveys. These are:

1. **Interview schedule**
2. **Questionnaire**
3. **Interview Guide**

Interview schedule and questionnaires both are predesigned list of questions used for communication with the respondents. In the case of *interview schedule*, the list of questions remains in the hands of the interviewer who asks questions from the respondent, gets his/her response, and records the responses. *Questionnaire* is also a list of questions, which is handed over to the respondent, who reads the questions and records the answers himself. For purposes of convenience *questionnaire* will refer to both interview schedule as well as questionnaire.

Interview guide is list of topics that are to be covered during the course of interview. Interview guide is used for purposes of an in-depth interviewing. Questions on the topics are formulated on the spot. Most of the questions are open ended. The interviewer may not use the same wording for each respondent; the number of questions may be different; the sequence of questions may also be different.

GUIDELINES FOR QUESTIONNAIRE DESIGN: A survey is only as good as the questions it asks. Questionnaire design is one of the most critical stages in the survey research process. While common sense and good grammar are important in question writing, more is required in the art of questionnaire design. To assume that people will understand the questions is common error. People may not simply know what is being asked. They may be unaware of topic of interest, they may confuse the subject with something else or the question may not mean the same thing to every respondent. Respondents may simply refuse to answer personal questions. Further, properly wording the questionnaire is crucial, as some problems may be minimized or avoided altogether if a skilled researcher composes the questions.

A good questionnaire forms an integrated whole. The researcher weaves لے کر چلنا questions together so they flow smoothly. He or she includes introductory remarks and instructions for clarification and measures each variable with one or more survey questions.

WHAT SHOULD BE ASKED?

The problem definition will indicate which type of information must be collected to answer the research question; different types of questions may be better at obtaining certain type of information than others.

1. **Questionnaire Relevancy**
2. **Questionnaire Accuracy**
3. **Avoid Ambiguity, Confusion, and Vagueness.**
4. **Avoid Double-Barreled Questions**
5. **Avoid Leading Questions**
6. **Avoid Loaded Questions**
7. **Avoid Burdensome Questions that may Tax the Respondent's Memory**
8. **Arrange Questions in a Proper Sequence**
9. **Use Filter Question, if Needed**
10. **Layout of the questionnaire**

1. Questionnaire Relevancy: A questionnaire is relevant if no unnecessary information is collected and if the information that is needed to solve the problem is obtained. Asking the wrong or an irrelevant question is a pitfall غلطی to be avoided. If the task is to pinpoint compensation problems, for example, questions asking for

general information about morale may be inappropriate. To ensure information relevancy, the researcher must be specific about data needs and there should be a rationale for each item of information.

2. Questionnaire Accuracy: Once the researcher has decided what should be asked, the criterion of accuracy becomes of primary concern. Accuracy means that the information is reliable and valid. While experienced researchers believe that one should use simple, understandable, unbiased, unambiguous and nonirritating words. Obtaining accurate answer from respondents is strongly influenced by the researcher's ability to design a questionnaire that facilitates recall and that will motivate the respondent to cooperate. Therefore avoid jargon, slang, and abbreviations. The respondents may not understand some basic terminology. Words used in the questionnaire should be readily understandable to all respondents.

3. Avoid Ambiguity, Confusion, and Vagueness: Ambiguity and vagueness ابہام plague مصیبت میں ڈالنا most question writers. A researcher might make implicit assumptions without thinking of respondents' perspectives. For example, the question, "what is your income?" could mean weekly, monthly, or annual: family or personal; before taxes or after taxes; for this year or last year; from salary or from all sources. The confusion causes inconsistencies in how different respondents assign meaning to and answer the question. Another source of ambiguity is the use indefinite words or response categories. Consider the words such as *often, occasionally, usually, regularly, frequently, many, good, fair,* and *poor*. Each of these words has many meanings. For one person frequent reading of *Time* magazine may be reading six or seven issues a year; for another it may be two issues a year. The word *fair* has great variety of meanings; the same is true for many indefinite words.

4. Avoid Double-Barreled Questions: Make each question about one and only one. A double barreled question consists of two or more questions joined together. It makes the respondent's answer ambiguous. For example, if asked, "Does this company have pension and health insurance benefits?" a respondent at the company with health insurance benefits only might answer either yes or no. The response has an ambiguous meaning and the researcher cannot be certain of the respondent's intentions. When multiple questions are asked in one question, the results may be exceedingly difficult to interpret.

5. Avoid Leading Questions: Make respondents feel that all responses are legitimate باقاعدہ/صحیح. Do not let them aware of an answer that the researcher wants. A leading question is the one that leads the respondent to choose one response over another by its wording. For example, the question, "you don't smoke, do you?" leads respondents to state that they do not smoke. "Don't you think that women should be empowered?" In most the cases the respondent is likely to agree with the statement.

6. Avoid Loaded Questions: Loaded questions suggest a socially desirable answer or are emotionally charged. "Should the city government repair all the broken streets?" Most of the people are going to agree with this question simply because this is highly socially desirable. A question which may be challenging the traditionally set patterns of behavior may be considered as emotionally charged i.e. it is loaded with such material which may hit the emotions of the people. Look at some behaviors associated with masculinity in Pakistani society. Let us ask a husband "Have you ever been beaten up by your wife?" Straight away this question may be considered to be a challenge to the masculinity of the person. Hence it may be embarrassing for the person to admit such an experience. Therefore, even if the husband was beaten up by his wife, he might give a socially desirable answer.

7. Avoid Burdensome Questions that may Tax the Respondent's Memory: A simple fact of human life is that people forget. Researchers writing questions about past behavior or events should recognize that certain questions may make serious demand on the respondent's memory. "How did you feel about your brother when you were 6 years old?" It may very difficult to recall something from the childhood.

8. Arrange Questions in a Proper Sequence: The order of question or the question sequence may serve several functions for the researcher. If the opening questions are interesting, simple to comprehend and easy to answer then respondent's cooperation and involvement can be maintained throughout the questionnaire. If respondent's curiosity is not aroused at the outset, they can become disinterested and terminate the interview. Sequencing specific questions before asking about broader issues is a common cause of question order bias. In some situations it may be advisable to ask general question before specific question to obtain the freest opinion of the respondent. This procedure is known as **funnel technique.**

9. Use Filter Question, if Needed: Asking a question that doesn't apply to the respondent or that the respondent is not qualified to answer may be irritating or may cause a biased response. Including filter question minimizes the chance of asking questions that are inapplicable. **Filter question** is that question which screens out respondents not qualified to answer a second question. For example the researcher wants to know about the bringing up of one's children. "How much time do you spend playing games with your oldest child?" What if the respondent is unmarried? Even if the respondent is married but does not have the child. In both these situations the question is inapplicable to him/her. Before this question the person may put a filter question whether or not the respondent is married.

10. Layout of the questionnaire: There are **two format or layout** issues: **the overall physical layout of the questionnaire** and **the format of questions and responses**. Good lay out and physical attractiveness is crucial in mail, Internet, and other self-administered questionnaires. For different reason it is also important to have a good layout in questionnaires designed for personal and telephone interviews. Give each question a number and put identifying information on questionnaire. Never cramp questions together or create a confusing appearance. Make a cover sheet or face sheet for each, for administrative use. Put the time and date of the interview, the interviewer, the respondent identification number, and interviewer's comments and observations on it. Give interviewers and respondents instructions on the questionnaire. Print instructions in a different style from question to distinguish them. Lay out is important for mail questionnaires because there is no friendly interviewer to interact with the respondent. Instead the questionnaire's appearance persuades the respondents. In mail surveys, include a polite, professional cover letter on letterhead stationery, identifying the researcher and offering a telephone number for any questions. Always end with "Thank you for your participation."

Lesson 24:	**PILOT TESTING OF THE QUESTIONNAIRE**

PILOT TESTING OF THE QUESTIONNAIRE: Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire. Conventional wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for extra training the research team.

Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. The main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

Administering a questionnaire exactly as planned in the actual study often is not possible.

WHAT ASPECTS TO BE EVALUATED DURING PILOT TESTING?

1. **Reactions of Respondents:**
2. **Discovering errors in the instrument:**
3. **Sampling procedure can be checked:**
4. **Staffing and activities of research team can be checked:**
5. **Procedure for data processing and analysis can be evaluated:**

1. Reactions of Respondents:

The reactions of the respondents can be looked at from different angles. The researcher may be familiar with the local culture; still getting the first hand experience is always useful. Going to the field, contacting the people, and their reactions to the different aspects of research may be a learning experience.

• **Availability of study population timing.** In case we are doing interviewing then pre-testing might help to find out the most appropriate time when the respondent shall be available. The researcher can plan the interviewing accordingly.

• **Acceptability of the questions asked.** An important purpose of pre-testing is to discover participants' reaction to the questions.

• **Pre-testing will also provide the opportunity to see the acceptability of the wording of the questions in the local cultural context**.

• **Willingness of the respondents to co-operate.** Field testing of the questionnaire will give the idea about the level of cooperation the research team is likely to get from the respondents, particularly if they have to interview them.

2. Discovering errors in the instrument:

• **Do the tools provide you the information? Reliability and Suitability for analysis.** Tabulation of the results /of a pretest helps determine whether the questionnaire will meet the objectives of the research.

• **Time taken/needed to interview/conduct the observation.** Pre-testing can indicate the time taken for interview or to conduct the observation. Too long questionnaires may not be recommended and, therefore, need modification. It can also help in estimating average time being taken to collect information form a respondent. Such an exercise can help in budget estimations.

• **If there is any need to revise the format of the tool.** Question arrangement can play a significant role in the success of the instrument. May be we should start with stimulating questions and place sensitive questions last. Such a situation might be handled through pretesting. Therefore, pre-testing may help in putting questions in proper sequence, using acceptable wording, doing appropriate translation, question spacing, structuring of answers, coding system, and needing instructions for interviewers (probing).

3. Sampling procedure can be checked:

• **The extent to which instructions given are followed.** Field functionaries are given the instructions for following a sampling procedure. Depending upon the type of sampling to be followed, the field worker must follow the guidelines otherwise the quality of the study will be hampered. During the pre-testing one could see not only the extent to which the instructions are being followed but also locate the problems in carrying out those instructions. Also what could be the solutions to those problems?

• **How much time is needed to locate the respondents?** By following the instructions how easy it is to locate the respondents, and how much time is needed to do that activity. It could help in calculating the overall time for data collection, having relevancy for budgeting the resources.

4. Staffing and activities of research team can be checked:

• **How successful the training has been?** Pre-testing can be seen as a period of extra training. The pre-testing exercise can provide a good opportunity to make an evaluation of the achievement of the objectives of training. For any deficiencies additional training may be provided.

• **What is the work output of each member?** The researcher can calculate the average output of each fieldworker and accordingly calculate the number of workers needed to finish the work on time. It can also help in making the budget estimates.

• **How well the research team works together?** It is a good opportunity to observe the kind of coordination the research team has. The integrated work is likely affect the efficiency of the team. Any shortcomings could be looked after.

• **Is the logistical support adequate?** Of course we are leaving the field functionaries in isolation. They shall be in need of other logistical support like the transportation, boarding, lodging, guidance and supervision. Some of these aspects could also be appraised during the pre-testing

5. Procedure for data processing and analysis can be evaluated:

• **Make dummy tables.** See how can we tabulate the data and use the appropriate statistics for purposes of interpretations

Lesson 25:	**INTERVIEWING**

INTERVIEWING: A personal interviewer administering a questionnaire door to door, a telephone interviewer calling from a central location, an observer counting pedestrians in a shopping mall and others involved in the collection of data and the supervision of that process are all **fieldworkers.** The activities they perform vary substantially. The supervision of data collection for a mail survey differs from the data collection in an observation study. Nevertheless there are some basic issues in all kinds of fieldwork. Just for convenience, in this session we shall focus on the interviewing process conducted by personal interviewers. However, many of the issues apply to all fieldworkers, no matter what their specific setting.

WHO CONDUCTS THE FIELDWORK?

Data collection in a sponsored study is rarely carried out by the person who designs the research project. For a student, depending upon the sample size, data collection is usually done by the student himself/herself. However, the data collection stage is crucial, because the research project is no better than the data collected in the field. Therefore, it is important that the research administrator selects capable people who may be entrusted to collect the data.

There are Field Interviewing Services, who specialize in data gathering. These agencies perform door-to-door surveys, central location telephone interviewing, and other forms of fieldwork for fee. These agencies typically employ field supervisors who oversee and train interviewers, edit questionnaires completed in the field, and confirm that the interviews have been conducted.

Whether the research administrator hires in-house interviewers or selects a field interviewing service, it is desirable to have fieldworkers meet certain job requirements. Although the job requirements for different types

of surveys vary, normally interviewers should be healthy, outgoing, honest, accurate, responsible, motivated, and of pleasing appearance – well groomed and properly dressed. An essential part of the interviewing process is establishing rapport with the respondent.

In-House Training: After personnel are selected, they must be trained. The training that the interviewer will receive after being selected by a company may vary from virtually no training to one week program. Almost always there will be a **briefing session** on the particular project.

The **objective of training** is to ensure that the data collection instrument is administered uniformly by all fieldworkers. The **goal of training session** is to ensure that each respondent is provided with common information. If the data are collected in a uniform manner from all respondents, the training session will have been success. More extensive training programs are likely to cover the following topics:

1. How to make initial contact with the respondent and secure the interview?

2. How to ask survey questions?

3. How to probe?

4. How to record responses? How to terminate the interview?

The Role of the Interviewer: Survey research interviewing is a specialized kind of interviewing. As with most interviewing, its goal is to obtain accurate information from another person.

The survey interview is a social relationship. Like other social relationships, it involves social roles, norms, and expectations. The interview is a short-term, secondary social interaction between two strangers with the explicit purpose of one person's obtaining specific information from the other. The social roles are those of the interviewer and the interviewee or respondent. Information is obtained in a structured conversation in which the interviewer asks prearranged questions and records answers, and the respondent answers.

The role of interviewer is difficult. They obtain cooperation and build rapport, yet remain neutral and objective. They encroach on respondents' time and privacy for information that may not benefit the respondents. They try to reduce embarrassment, fear, and suspicion so that respondents feel comfortable revealing information. They explain the nature of the survey research or give hints about social roles in an interview. Good interviewers monitor the pace and direction of the social interaction as well as content of the answers and the behavior of the respondents.

Survey interviewers are nonjudgmental and do not reveal their opinions, verbally or nonverbally. If the respondent asks for an interviewer's opinion, he or she politely redirects the respondent and indicate that such questions are inappropriate.

STAGES OF AN INTERVIEW

- ***Making Initial Contact and Securing the Interview***
- ***Asking the Questions***
- ***Probing ---- تفشیش کرنا***
- ***Recording the Responses***
- ***Terminating the Interview***

Making Initial Contact and Securing the Interview

The interview proceeds through stages, beginning with introduction and entry. Interviewers are trained to make appropriate opening remarks that will convince the person that his or her cooperation is important.

Asslaam-o-Alaykum, my name is _____ and I am working for a National Survey Company. We are conducting a survey concerning "women empowerment." I would like to get a few of your ideas.

For the initial contact in a telephone interview, the introduction might be:

Asslaam-o-Alaykum, my name is _____. I am calling from Department of Social Research, Virtual University.

By indicating that telephone call is a long distance, interviewers attempt to capitalize on the fact that most people feel a long distance call is something special, unusual, or important. Giving one's personal name personalizes the call.

Personal interviewers may carry a letter of identification that will indicate that the study is bona fide research project and not a salesman's call. The name of the research agency is used to assure the respondent that the caller is trustworthy.

Asking the Questions

The purpose of the interview is, of course, to have the interviewer ask questions and record the respondent's answers. Training in the art of stating questions can be extremely beneficial, because interviewer bias can be a source of considerable error in survey research.

There are five major principles for asking questions:

- Ask the questions exactly as they are worded in the questionnaire.
- Read each question very slowly.
- Ask the question in the order in which they are presented in the questionnaire.
- Repeat questions those are misunderstood or misinterpreted.
- Do not take shortcuts when the task becomes monotonous.

Although interviewers are generally trained in these procedures, when working in the field many interviewers do not follow them exactly. Interviewers may shorten questions or rephrase unconsciously when they rely on their memory of the question rather than reading the question as it is worded. If the respondents do not understand a question, they will usually ask for some clarification. The recommended procedure is to repeat the question, or if the respondent does not understand a word, the interviewer should respond with "just whatever it means to you.

Often the respondents volunteer information relevant to a question that is supposed to be asked at a later point in the research. In this situation the response should be recorded under the question that deals specifically with that subject. Then rather than skip the question that was answered out of sequence, the interviewers should be trained to say something like "We have briefly discussed this, but let me ask you…." By asking every question, the interviewer can be sure that complete answers are recorded.

Probing: Probing means the verbal prompts made by field worker when the respondent must be motivated to communicate his or her answer or to enlarge on, clarify or explain an answer. **Probing may be needed for two types of situations. First, it is necessary** when the respondent must be motivated to enlarge on, clarify, or explain his or her answer. The interviewer must encourage the respondent to clarify or expand on answers by providing a stimulus that will not suggest the interviewer's own ideas. The ability to probe with neutral stimuli is the mark of an experienced interviewer. **Second, probing may be necessary** in situations in which he

respondent begins to ramble or lose track of the question. In such cases the respondent must be led to focus on specific content of the interview and to avoid irrelevant and unnecessary information. Probing is also needed when the interviewer recognizes an irrelevant or inaccurate answer.

He interviewer has several possible probing tactics to choose from, depending on the situation:

• **Repetition of the question.** The respondent who remains completely silent may not have understood the question or may not have decided how to answer it. Mere repetition may encourage the respondent to answer in such cases. For example, if the question is "What is there that you do not like about your supervisor?" and the respondent does not answer, the interviewer may probe: "just to check, is there anything you do not like about your

supervisor?"

• **An expectant pause.** If the interviewer believes the respondent has more to say, the "silent probe," accompanied by an expectant look may motivate the respondent to gather his/her thoughts and give a complete response.

• **Repetition of the respondent's reply**. As the interviewer records the response, he or she may repeat the respondent's reply verbatim. This may stimulate the respondent to expand on the answer.

• **Neutral questions or comments**. Asking neutral question may indicate the type of information that the interviewer is seeking. For example, if the interviewer believes that the respondent's motives should be clarified, he or she might ask, "Why do you feel that way?" If the interviewer feels that there is a need to clarify a word or phrase, then he/she might ask, "What do you mean by _____?"

Recording the Responses: The rules for recording responses to closed ended questions vary with the specific question. The general rule, however, is to place a check in the box that correctly reflects the respondent's answer.

The general instructions for recording answers to open-ended response questions is to record the answer verbatim, a task that is difficult for most people. Some of the suggestions are:

• Record responses during the interview.

• Use the respondent's own words.

• Do not summarize or paraphrase the respondent's answer.

• Include everything that pertains to the question objectives.

• Include all your probes.

Terminating the Interview: Fieldworkers should not close the interview before all the information has been secured. The interviewer whose departure is hasty will not be able to record those spontaneous comments respondents sometimes offer after all formal questions have been asked. Avoiding hasty departures is also a matter

of courtesy. Fieldworkers should also answer to the best of their ability any questions the respondent has concerning the nature and purpose of the study. Always leave by observing the local cultural customs. "Don't burn

your bridges." Because the fieldworker may be required to re-interview the respondent at some future time, he or she should leave the respondent with positive feeling about having cooperated in a worthwhile undertaking. It

is extremely important t thank the respondent for his or her cooperation. The interviewer then goes to a quite and private place to edit the questionnaire and record other details such as the date, time, and place of interview; a thumbnail sketch of the respondent and interview situation, the respondent's attitude; and any unusual circumstances. The interviewer also records personal feelings and anything that was suspected.

PRINCIPLES OF INTERVIEWING (The Basics)

- *Have integrity and be honest.* This is the cornerstone of all professional inquiry, regardless of its purpose.
- *Have patience and tact.* Interviewers ask for information from people they do not know. Thus all the rules of human relations that apply to inquiry situations – patience, tact, courtesy – apply "in spades" to interviewing.
- *Have attention to accuracy and detail.* Among the greatest interviewing "sins" are inaccuracy and superficiality, for the professional analyst can misunderstand, and in turn mislead, a client. Do not record the answer unless you fully understand it yourself. Probe for clarification and detailed full answers.
- *Exhibit a real interest in the inquiry at hand, but keep your opinions to yourself.* Impartiality is imperative.
- *Be a good listener.* Some interviewers talk too much, wasting time when respondents could be supplying more pertinent facts or opinions on the topic.
- *Keep the inquiry and respondents' responses confidential.* Do not discuss the studies you are doing with relatives, friends, or associates. Never quote one respondent's opinion to another.
- *Respect others' rights.* Survey research depends on the goodwill of others to provide information. There should be no coercion زبردستی. Impress on prospective respondents that their cooperation is important and valuable.

INTERVIEW BIAS -----میلان/طرف داری/رجحان

• Information obtained during interview should be as free as possible of bias.

• Bias could be introduced by the interviewer, interviewee, or the situation.

Interviewer Bias :Interviewer bias falls into six categories:

1. Interviewer could bias the data if proper rapport is not established Errors by the respondent – forgetting, embarrassment, misunderstanding, or lying because of the presence of others.

2. Unintentional errors or interviewer sloppiness – contacting the wrong person, misreading a question, omitting questions, reading questions in the wrong order, recording wrong answer, or misunderstanding the respondent.

3. Intentional subversion by the interviewer – purposeful alteration of answers, omission or rewording of questions, or choice of an alternative respondent.

4. Influence due to the interviewer's expectations about a respondent's appearance, living situation, or other answers.

5. Failure of an interviewer to probe or to probe properly.

6. Influence on the answers due to the interviewer's appearance, tone, attitude, reactions to answers, or comments made outside of the interview schedule.

Interviewee Bias: Errors made by the respondent can be

1. Interviewees can bias the data when they do not come out with their true opinion but provide information that they think what the interviewer expects of them or would like to hear.

2. They do not understand the question, they may feel difficult or hesitant to clarify.

3. Some interviewees may be turned off because of the personal liking, or the dress of the interviewer, or the manner in which questions are put. So they may not provide truthful answers.

4. Some may provide socially undesirable answers.

Situational Bias: Situational biases in terms of:

1. Non-participants – Unwillingness or inability to participate. Bias the sample.

2. Trust levels and rapport established by different interviewers. Elicit answers of different degrees of openness.

3. The physical setting of the interview. Respondent may not feel comfortable to be interviewed at work.

SOME TIPS FOR INTERVIEWING

- Know the culture of the people in advance.
- Appearance – wear acceptable dress.
- Pleasantness and flexibility.
- Carry the letter of authority.
- Establish credibility and rapport. Motivating individuals to respond.
- Familiarity with the questionnaire.
- Following the question wording/ question order
- Recording responses exactly.
- Probing for responses.
- Closing the interview. No false promises. Also don't burn your bridges.
- Edit the questionnaire in the first available opportunity.

Lesson 26:	**SAMPLE AND SAMPLING TERMINOLOGY**

SAMPLE: **A sample is a subset, or some part, of a larger whole**. **A larger whole** could be anything out which sample is taken. That 'whole' could be a bucket of water, a bag of sugar, a group of organizations, a group of students, a group of customers, or a group mid-level managers in an organization. A complete group of entities sharing some common set of characteristics is population. In other words, the totality out of which sample is drawn is referred to as **population.**

WHY SAMPLE?

1. ***Saves Cost, Labor, and Time***
2. ***Quality Management/supervision***
3. ***Accurate and Reliable Results***
4. ***Sampling may be the Only Way***
5. ***Determine the Period of Study***

1. Saves Cost, Labor, and Time

Applied research projects usually have budget and time constraints. Since sample can save financial cost as well as time, therefore, to go for sample study is pragmatic. Of course, a researcher investigating a population with an extremely small number of population elements may elect to conduct a study on the total population rather than a sample because cost, labor, and time constraints are relatively insignificant. Although sample study cuts costs, reduces labor requirements, and gathers vital information quickly, yet there could be other reasons.

2. Quality Management/supervision

Professional fieldworkers are a scarce commodity. In a large study rather than employing less qualified staff it may be advisable to do a sample study and employ highly qualified professional fieldworkers. It can certainly affect the quality of the study. At the same time it may be easier to manage a small group and produce quality information. Supervision, record keeping, training, and so forth would all be more difficult in a very large study.

3. Accurate and Reliable Results

Another major reason for sampling is that samples, if properly selected, are sufficiently accurate in most of the cases. If the elements of a population are quite similar, only a small sample is necessary to accurately portray

the characteristics of interest. Most of us have had blood samples taken from the finger, the arm, or another part of body. The assumption is that blood is sufficiently similar through out the body, the characteristics of the blood can be determined on the basis of sample.

When the elements of population are highly homogenous, samples are highly representative of the population. Under these circumstances almost any sample is as good as another. Samples may be more accurate than census. In a census study of large population there is a greater likelihood of non-sampling errors. In a survey mistakes may occur that are unrelated to the selection of people in the study. For example, a response may be coded incorrectly, or the keyboard operator might make data entry error. Interviewer mistakes, tabulation errors, and other non-sampling errors may increase during census because of the increased volume of work. In sample increased accuracy is possible because the fieldwork and tabulation of the data can be closely supervised than would be possible in a census. In field survey, a small, well trained, closely supervised group may do a more careful and accurate job of collecting information than a large group of nonprofessional interviewers trying to contact everyone.

4. Sampling may be the Only Way

Many research projects, especially those in quality control testing, require the destruction of the items being tested. If the manufacturer of firecrackers wished to find out whether each product met a specific production standard, there would be no product left after testing. Similarly, consider the case of electric bulbs. In testing the life of bulbs, if we were to burn every bulb produced, there would be none left to sell. This is destructive sampling.

5. Determine the Period of Study

Interviewing every element of a large population without sampling requires lot of time, may be a year or more. In such a long period study, even the seasonal variation may influence the response pattern of the respondents. For example, if the study was aimed at measuring the level of unemployment in a given large city, the unemployment rate produced by the survey data would not refer to the city as of the beginning of interviewing or as of the end. Researcher may be forced to attribute the unemployment to some hypothetical date, representing to the midpoint of the study period. Hence it will be difficult to determine the exact timing to which the data of the study pertains.

SAMPLING TERMINOLOGY

There are a number of technical terms used in books on research and statistics which need explanation. Some of the important terms are:

Element: An element is that unit about which information is collected and which provides the basis of analysis. Typically, in survey research, elements are people or certain types of people. It is that unit about which information is collected and that provides the basis of analysis. It can be a person, groups, families, organizations, corporations, communities, and so forth.

Population: A population is the theoretically specified aggregation of study elements. It is translating the abstract concept into workable concept. For example, let us look at the study of "college students." Theoretically who are the college students? They might include students registered in government colleges and/or private colleges, students of intermediate classes and/or graduate classes, students of professional

colleges and/or non-professional colleges, and many other variations. In this way the pool of all available elements is population.

Target Population: Out of the conceptual variations what exactly the researcher wants to focus on. This may also be called a target population. Target population is the complete group of specific population elements relevant to the research project. Target population may also be called *survey population* i.e. that aggregation of elements from which the survey sample is actually selected. At the outset of the sampling process, it is vitally important to carefully define the target population so the proper source from which the data are to collected can be identified. In our example of 'college students" finally we may decide to study the college students from government institutions located in Lahore, who are studying social sciences, who are aged 19 years of age, and hailing from rural areas.

Sampling: The process of using a small number of items or parts of a larger population to make conclusions about

the whole population. It enables the researchers to estimate unknown characteristics of the population.

Sampling Frame: In actual practice the sample will be drawn from a list of population elements that is often different from the target population that has been defined. A sampling frame is the list of elements from which the sample may be drawn. A simple example could be listing of all college students meeting the criteria of target population and who are enrolled on the specified date. A sampling frame is also called the *working population* because it provides the list that can be worked with operationally. In our example, such a list could be prepared with help of the staff of the selected colleges.

Sampling Frame Error: A sampling frame error occurs when certain sample elements are excluded or when the entire population is not accurately represented in the sampling frame. The error that occurs when certain sample elements are not listed or available and are not represented in the sampling frame.

Sampling Unit: A sampling unit is that element or set of elements considered for selection in some stage of sampling. Sampling may be done in single stage or in multiple stages. In a simple, single-stage sample, the sampling units are the same as the elements. In more complex samples, however, different levels of sampling units may be employed. For example, a researcher may select a sample of *Mohallahs* in a city, and then select a sample of households from the selected *Mohallahs*, and finally may select a sample of adults from the selected households. The sampling units of these three stages of sampling are respectively *Mohallah*, households, and adults, of which the last of these are the elements. More specifically, the terms "primary sampling units," "secondary sampling units," and "final sampling units" would be used to designate the successive stages.

Observation Unit: An observation unit, or unit of data collection, is an element or aggregation of elements from which the information is collected. Often the unit of analysis and unit of observation are the same – the individual

person – but this need not be the case. Thus the researcher may interview heads of household (the observation units) to collect information about every member of the household (the unit of analysis).

Parameter: A parameter is the summary description of a given variable in a population. The mean income of all families in a city and the age distribution of the city's population are parameters. An important part portion of survey research involves the estimation of population parameters on the basis of sample observation.

Statistic: A statistic is the summary description of a given variable in a survey sample. Thus the mean income computed from the survey sample and the age distribution of that sample are statistics. Sample statistics are used to make estimates of the population parameters.

Sampling Error: Probability sampling methods seldom, if ever, provide statistics exactly equal to the parameters that they are used to estimate. Probability theory, however, permits us to estimate the error to be expected for a given sample (more information to be sought from professional in Statistics).

Lesson 27:	**PROBABILITY AND NON-PROBABILITY SAMPLING**

PROBABILITY AND NON-PROBABILITY SAMPLING: There are several alternative ways of taking a sample. The major alternative sampling plans may be grouped into probability techniques and non-probability techniques.

In **probability sampling** every element in the population has a *known nonzero probability* of selection. The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizeability.

When time or other factors, rather than generalizeability become critical then non-probability sampling is generally used. In **non-probability sampling** the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate. Nevertheless, there are occasions when non-probability samples are best suited for the researcher's purpose.

Types of non-probability sampling: In non-probability sampling designs, the elements in the population do not have any probabilities attached to their being chosen as sample subjects. This means that the findings from the study of the sample cannot be confidently generalized to the population. Sometimes non-probability could be the only way to collect the data.

1. **Convenience Sampling**
2. **Purposive Sampling**
3. **Quota Sampling**
4. **Snowball Sampling**
5. **Sequential Sampling**
6. **Theoretical Sampling**

1. ***Convenience Sampling:*** Convenience sampling (also called *haphazard or accidental sampling*) refers to sampling by obtaining units or people who are most conveniently available. For example, it may be convenient and economical to sample employees in companies in a nearby area. The person-on-the street interview conducted by TV programs is another example. Another example of haphazard sample is that of a newspaper that asks the readers to clip a questionnaire from the paper and mail it in but the sample cannot be used to generalize accurately to the population.

Convenience samples are least reliable but normally the cheapest and easiest to conduct. Convenience sampling is most often used during the exploratory phase of a research project.

2. ***Purposive Sampling:*** Depending upon the type of topic, the researcher lays down the criteria for the subjects to be included in the sample. Whoever meets that criteria could be selected in the sample. The

researcher might select such cases or might provide the criteria to somebody else and leave it to his/her judgment for the actual selection of the subjects. That is why such a sample is also called as **judgmental or expert opinion sample.**

3. ***Quota Sampling:*** A sampling procedure that ensures that certain characteristics of a population sample will be represented to the exact extent that the researcher desires. In this case the researcher first identifies relevant categories of people (e.g. male and female; or under age 30, ages 30 to 60, over 60, etc) then decides how many to get in each category. Thus the number of people in various categories of sample is fixed. This is quota sampling. Once the quota has been fixed then the researcher may use convenience sampling. The convenience sampling may introduce **bias**. Speed of data collection, lower costs, and convenience are the major advantages of quota sampling compared to probability sampling. Quota sampling becomes necessary when a subset of a population is underrepresented, and may not get any representation if equal opportunity is provided to each.

4. ***Snowball Sampling:*** Snowball sampling (also called *network, chain referral, or reputational sampling*) is a method for identifying and sampling (or selecting) cases in the network. It is based on an analogy to a snowball, which begins small but becomes larger as it is rolled on wet snow and picks up additional snow. It begins with one or a few people or cases and spreads out on the basis of links to the initial cases. This design has been found quite useful where respondents are difficult to identify and are best located through referral networks. In the initial stage of snowball sampling, individuals are discovered and may or may not be selected through probability methods. This group is then used to locate others who possess similar characteristics and who, in turn, identify others. The "snowball" gather subjects as it rolls along. For example, a researcher examines friendship networks among teenagers in a community.

5. ***Sequential Sampling:*** Sequential sampling is similar to purposive sampling with one difference. In purposive sampling, the researcher tries to find as many relevant cases as possible, until time, financial resources, or his or her energy is exhausted. The principle is to get every possible case. For example, a researcher locates and plans in-depth interviews with 60 widows over 70 years old who have been living without a spouse for 10 or more years. Depending on the researcher's purposes, getting an additional 20 widows whose life experiences, social background, and worldview differ little from the first 60 may be unnecessary.

6. ***Theoretical Sampling:*** In theoretical sampling, what the researcher is sampling (e.g. people, situation, events, time periods, etc.) is carefully selected, as the researcher develops grounded theory. A growing theoretical interest guides the selection of sample cases. The researcher selects cases based on new insights they may provide. For example, a field researcher may be observing a site and a group of people during week days. Theoretically, the researcher may question whether the people act the same at other times or when other aspects of site change. He or she could then sample other time periods (e.g. nights and weekends) to get more full picture and learn whether important conditions are the same.

Lesson 28:	**TYPES OF PROBABILITY SAMPLING**

TYPES OF PROBABILITY SAMPLING: Probability samples that rely on random processes require more work than nonrandom ones. A researcher must identify specific sampling elements (e.g. persons) to include in the sample. For example, if conducting a telephone survey, the researcher needs to try to reach the specific sampled person, by calling back several times, to get an accurate sample.

Random samples are most likely to yield a sample that truly represents the population. In addition, random sampling lets a researcher statistically calculate the relationship between the sample and the population – that is the **size of *sampling error*. A non-statistical definition of the sampling error is the deviation between sample result and a population parameter due to random process.**

1. **Simple Random Sample:**
2. **Systematic Random Sample:**
3. **Stratified Random Sample:**
4. **Multistage Sampling:**
5. **Cluster Sampling:**
6. **Double Sampling:**

1. *Simple Random Sample:*

- Assumption –homogeneous population.
- Develop an accurate sampling frame.
- Locate the exact element/s to be selected i.e. decide the sample size.
- Number all the elements in the sampling frame.
- Use the list/table of random numbers.

2. Systematic Random Sample:

- Short cut for simple random sampling.
- Number each element in the sampling frame.
- Instead of using the table of random numbers, researcher develops a system for the selection of elements.
- Calculate the sampling interval. Basis of random selection
- Sample size/Population size X 100 i.e. sample as % of Pop.
- Begin with a random start.
- Draw sample by choosing every Nth case.
- If elements are organized in some pattern in the sampling frame then problem. Possible periodicity in population.

3. Stratified Random Sample:

- In case population is heterogeneous, then stratify. Each stratum in itself becomes homogeneous.
- Researcher's decision about homogeneity of the population.
- Draw the sample from each stratum by using simple random sampling procedure.

Reasons for stratification: There are three reasons why a researcher chooses a stratified random sample:

- To increase a sample's statistical efficiency. Each stratum gets represented. Reduces random sampling error makes the sample representative.
- Provides adequate data for analyzing the various subpopulations.
- Enables to use different research procedures in different strata.

***Stratification Process*:** The ideal stratification would be based on the primary variable (the dependent variable) under study. The criterion is identified as an efficient basis for stratification. The criterion for stratification is that it is a characteristic of the population elements known to be related to the dependent variable or other variables of interest. The variable chosen should increase homogeneity within each stratum and increase heterogeneity between strata. Next, for each separate subgroup or stratum, a list of population

elements must be obtained. Serially number the elements within each stratum. Using a table of random numbers or some other device, a separate simple random sample is taken within each stratum. Of course the researcher must determine how large a sample must be drawn from each stratum.

4. ***Cluster Sampling:***

- Purpose: sample economically and retaining probability sampling.
- Heterogeneity within clusters but homogeneity between clusters.
- Random selection of clusters.
- Random selection of elements within the selected clusters.
- Cluster sampling addresses two problems
- Researchers lack good sampling frame for dispersed population
- Reaching each sampled element is costly.
- The unit costs of cluster sampling are much lower than those of other probability sampling designs.
- However, cluster sampling exposes itself to greater biases at each stage of sampling.

5. ***Multistage Sampling***

- Researcher draws several samples in stages. For example:
- Cluster sampling is usually multistage
  Stage 1– Random sampling of big clusters
  Stage 2– Random sampling of small clusters within the big ones
  Stage 3– Random sampling of elements within the selected small clusters
- City blocks – households – individuals

6. ***Double Sampling:***

- When further information is needed from a subset of the group from which some information has already been collected for the same study.
- Want to examine the matter in more detail.
- Sub-sample of the primary sample

What is the Appropriate Sample Design? The most common criteria for appropriate sample design is:

1. **Degree of Accuracy**
2. **Resources**
3. **Advance Knowledge of the Population**
4. **National versus Local Project (Geographic proximity of population)**
5. **Need for Statistical Analysis**

1.*Degree of Accuracy*: Selecting a representative sample is important to all researchers.

2.*Resources:* The costs associated with different sampling techniques vary tremendously. If the researcher's financial and human resources are restricted then this limitation of resources will eliminate certain methods.

3.*Advance Knowledge of the Population:* Advance knowledge of population characteristics is an important criterion. A lack of adequate list may automatically rule out any type of probability sampling.

4.*National versus Local Project:* Geographic proximity of population elements will influence sample design. When population elements are unequally distributed geographically a cluster sampling may become more attractive.

5.*Need for Statistical Analysis:* The need for statistical projections based on the sample is often a criterion.

Lesson 29:	**DATA ANALYSIS**

DATA ANALYSIS:

- Data processing and analysis is part of research design –decisions already made.
- During analysis several interrelated procedures are performed to summarize and rearrange data.
- Goal of most research is to provide information.
- **Information**: a body of facts that are in a format suitable for decision making.

- **Data**: simply recorded measures of certain phenomenon.

STAGES OF DATA ANALYSIS

1. **Editing**
2. **Coding**
3. **Data Entry**
4. **Data Cleaning**

Transformation of raw data into information

- It is a lengthy process. Starts with: Editing, coding, and data storage.
- For storage and processing, many advantages to use computer.
- For computer storage follow the coding procedure.

Errors

- Possibility of mistakes by fieldworker's e.g. Recording improbable answer, Interviewing ineligible respondent, Contradictory answers
- Editing should take care of such mistakes prior to coding

1. *Editing*

- Process of checking and adjusting the data for omissions, legibility and consistency.
- Purpose is to ensure the completeness, consistency and readability of data to be transferred storage.
- Editor may have to reconstruct some data (weekly income converted into monthly income)
- Bring to light the hidden values without bias.
- Editing is not coding because Coding is assignment of numbers of symbols to previously edited data.
- Editing can be field editing or In-house Editing.
- ***Field Editing***
    - Fieldworkers and then supervisors to do the field edits.
    - Do the editing the same day.
    - Catch technical omissions (blank page)
    - Check legibility of hand writing.
    - Clarify inconsistent responses.
    - Supervisor can question the interviewers.
    - Callback if needed.
    - Do the interviewers need additional training?
- ***In-House Editing***
    - Rigorously investigates the results of data collection.
    - For mailed questionnaire in-house editing is the only option.
- ***Editing for consistency:*** Adjust the inconsistent and contradictory responses and Editor should adjust for consistency.
- ***Editing for completeness:*** Editor to complete the missing answer.
- Editing questions answered out of order.
- Tabulation of "don't know"

2. *Coding*

- Coding involves assigning numbers or other symbols –responses can be grouped into classes or categories
- Helps in efficient analysis of data.
- Keep the data in raw form so far it is possible. Computer can help in grouping.

- Codes allow data processing in a computer.
- Organize data into field, record and file.
    - **A field**: collection of characters (single No., letter, symbol) representing single type of data.
    - **A record**: collection of related fields.
    - **A file**: collection of related records.
- File, records, and fields are stored on magnetic tapes, floppy disks, or hard drives.
- Researchers use a coding procedure and codebook.
- A **coding procedure** is a set of rules stating that certain numbers are assigned to variable attributes. For example, a researchers codes males as 1 and females as 2.
- A **codebook** is a document (i.e. one or more pages) describing the coding procedure and the location of data for variables in a format that computers can use.
- A survey researcher pre-codes a questionnaire before collecting the data. Pre-coding means placing the code categories (e.g. 1 for male, 2 for female) on the questionnaire. Next, the researcher transfers the information from each questionnaire into a format that computers can read.
- ***Code Construction:*** When the question has a fixed-alternative (closed ended) format, the number of categories requiring codes is determined during the questionnaire design stage. The codes 8 and 9 are conventionally given to "don't know" (DK) and "no answer" (NA) respectively.
- ***There are two basic rules for code construction***.
    - Coding categories should be exhaustive i.e. coding category provided for every possible response.
    - Coding categories should be mutually exclusive and independent.
- A major objective in code-building process is to accurately transfer the meaning from written answers to numeric codes.
- ***Code Book:*** A book identifying each variable in a study and its position in the data matrix. The book is used to identify a variable's description, code name, and field.
- ***Production Coding:*** Transferring the data from the questionnaire or data collection form after the data have been collected is called production coding. Depending upon the nature of the data collection form, codes may be written directly on the instrument or on a special coding sheet.

3. Data Entries

- The process of transforming data from the research project to computers.
- Optical scanning systems may be used to read material directly to the computer's memory into the computer's memory. Optical scanners process the marked-sensed questionnaires and store the answers in a file.

4. Cleaning Data

- The final stage in the coding process is the error checking and verification, or "data cleaning" stage.
- Error checking and verification –make sure that all codes are legitimate.
- No errors in coding process, as well as entering data into a computer.

Lesson 30:	DATA TRANFORMATION

DATA TRANSFORMATION:

Data transformation is the process of changing data from their original form to a format that is more suitable to perform a data analysis that will achieve the research objectives.

- Researchers often modify the values of a scalar data or create new variables.
- Collapsing or combining categories of a variable is a common data transformation that reduces the number of categories. For example five categories of Likert scale response categories to a question may be combined like: the "strongly agree" and the "agree" response categories are combined. The "strongly disagree" and the "disagree" response categories are combined into a single category. The result is the collapsing of the five-category scale down to three. We had constructed number of statements on each element with 5 response categories using Likert scale i.e. strongly agree, agree, undecided, disagree, and strongly disagree.
- Creating new variables by re-specifying the data numeric or logical transformations is another important data transformation.
- The researchers have created numerous different scales and indexes to measure social phenomenon. For example scales and indexes have been developed to measure the degree of formalization in bureaucratic organization, the prestige of occupations, the adjustment of people in marriage, the intensity of group interaction, the level of social activity in a community and the level of socio-economic development of a nation.

Keep it in mind that

- Very social phenomenon can be measured. Some constructs can be measured directly and produce precise numerical values (e.g. family income). Other constructs require the use of surrogates or proxies that indirectly measure a variable (e.g. job satisfaction).
- A lot can be learned from measures used by other researchers. We are fortunate to have the work of thousands of researchers to draw on. It is not always necessary to start from a scratch. We can use a past scale or index or we can modify it for our own purposes. The process of creating measures for a construct evolves over time.
- Measurement is an ongoing process with constant change; new concepts are developed, theoretical definitions are refined, and scales or indexes that measure old or new constructs are improved.

INDEXES AND SCALES

- Scales and indexes are often used interchangeably. One researcher's scale is another's index.
- Both produce ordinal or interval level measures of variable. To add to the confusion, scale and index techniques can be combined in one measure.
- Scales and indexes give a researcher more information about variables and make it possible to assess the quality of measurement.
- Scales and indexes increase reliability and validity.
- They aid in **data reduction;** that is condense and simplify the information that is collected.

A scale **is a measure in which the researcher captures the intensity, direction, level, or potency of a variable construct.** It arranges responses or observation on a continuum. A scale can use single indicator or multiple indicators. Most are at the ordinal level of measurement.

An index **is a measure in which a researcher adds or combines several distinct indicators of a construct into a single score.** This composite score is often a simple sum of multiple indicators. It is used for content or convergent validity. Indexes are often measured at the interval or ratio level.

Researchers sometimes combine the features of scales and indexes in a single measure. This is common when a researcher has several indicators that are scales. He or she then adds these indicators together to yield a single score, thereby an index.

UNIDIMENSIONALITY: It means that all the items in a scale or index fit together or measure a single construct. It says if you combine several specific pieces of information into a single score or measure, have all the pieces measure the same thing (Each sub dimension is part of the construct's overall content). For example, we define the construct "feminist ideology" as a general ideology about gender. Feminist ideology is a highly abstract and general construct. It includes a specific beliefs and attitudes towards

social, economic, political, family, sexual relations. The ideology's five belief areas parts of a single general construct. The parts are mutually reinforcing and together form a system of beliefs about dignity, strength, and power of women.

INDEX CONSTRUCTION: **An index is combination of items into a single numerical score.** Various components or subgroups of a construct are each measured and then combined into one measure.

There are many types of indexes. For example, if you take an exam with 25 questions, the total number of questions correct is a kind of index. It is a composite measure in which each question measures a small piece of knowledge, and all the questions scored correct or incorrect are totaled to produce a single measure. One way to demonstrate that indexes are not a very complicated is to use one. Answer yes or no to the seven questions that follow on the characteristics of an occupation. Base your answers on your thoughts regarding the following four occupations: long-distance truck driver, medical doctor, accountant, telephone operator. Score each answer 1 for yes and 0 for no.

1. Does it pay good salary?
2. Is the job secure from layoffs or unemployment?
3. Is the work interesting and challenging?
4. Are its working conditions (e.g. hours, safety, time on the road) good?
5. Are there opportunities for career advancement and promotion?
6. Is it prestigious or looked up to by others?
7. Does it permit self-direction and the freedom to make decisions?

Total the seven answers for each of the four occupations. Which had the highest and which had the lowest score? The seven questions are our operational definition of the construct good occupation.

Each question represents a subpart of our theoretical definition. Creating indexes is so easy that it is important to be careful that every item in the index has face validity. Items without face validity should be excluded. Each part of the construct should be measured with at least one indicator. Of course, it is better to measure the parts of a construct with multiple indicators.

WEIGHTING: An important issue in index construction is whether to weight items. Unless it is otherwise stated, assume that an index is un-weighted. Likewise, unless we have a good reason for assigning different weights, use equal weights. A weighted index gives each item equal weight. It involves adding up the items without modification, as if each were multiplied by 1 (or – 1 for negative items that are negative).

Lesson 31:	**DATA PRESENTATION**

Tables and graphs (pictorial presentation of data) may simplify and clarify the research data. Tabular and graphic representation of data may take a number of forms, ranging from computer printouts to elaborate pictographs. The purpose of each table or graph, however, is to facilitate the summarization and communication of the meaning of the data. Although there are a number of standardized forms for presenting data in table or graphs, the creative researcher can increase the effectiveness of particular presentation. Bar charts, pie charts, curve diagrams, pictograms, and other graphic forms of presentation create a strong visual impression.

Commercial packages eliminate the need to write a new program every time you want to tabulate and analyze data with a computer. SAS, Statistical Package for the Social Sciences (SPSS), SYSTAT, Epi. Info. And MINITAB is commonly used statistical packages. These user friendly packages emphasize statistical calculations and hypothesis testing for varied types of data. They also provide programs for entering and editing data. Most of these packages contain sizeable arrays of programs for descriptive analysis and univariate, bivariate, and multivariate statistical analysis.

RESULTS WITH ONE VARIABLE (FREQUENCY DISTRIBUTION): Several useful techniques for displaying data are in use. The easiest way to describe the numerical data of one variable is with a frequency distribution. It can be used with nominal-, ordinal-, interval-, or ratio-level data and takes many forms. For example we have data of 400 students. We can summarize the data on the gender of the students at a glance with raw count or a frequency distribution. We can present the same information in a

graphic form. Some common types of graphic presentations are the histograms, bar chart, and pie chart. Bar charts or graphs are used for discrete variables. They can have vertical or horizontal orientation with small space between the bars. The terminology is not exact, but histograms are usually upright bar graphs for interval or ratio data.

Table 1: Frequency distribution of students

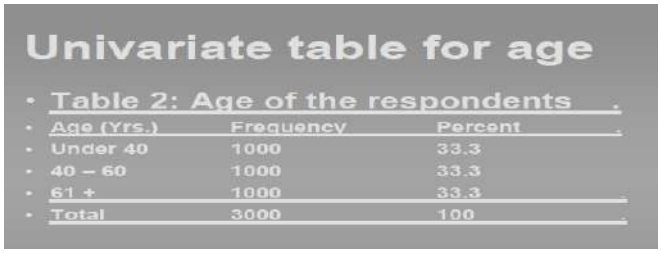
Gender	Frequency	Percent
Male	300	75
Female	100	25
Total	**400**	**100**

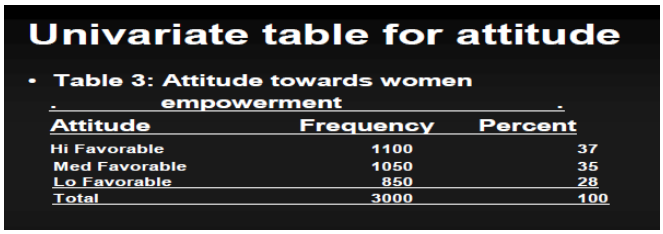
The presented data has to be interpreted. The purpose of interpretation is to explain the meanings of the data so that we can make inferences and formulate conclusions.

In order for interpretation, the data have to be meaningfully analyzed. For purposes of analysis the researchers use statistics. The word statistics has several meanings. It can mean a set of collected numbers (e.g. numbers telling how many people living in a city) as well as a branch of applied mathematics used to manipulate and summarize the features of numbers.

Descriptive statistics describe numerical data. They can be categorized by the number of variables involved: univariate, bivariate, or multivariate (for one, two, and three or more variables). Univariate statistics describe one variable.

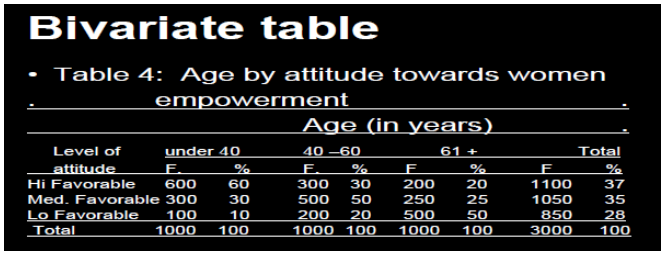
Researchers often want to summarize the information about one variable into a single number. They use three measures of central tendency, or measures of the center of the frequency distribution: mean, median and mode, which are often called averages (a less precise and less clear way to say the same thing). The **mode** is simply the most common or frequently occurring number. The **median** is the middle point. The **mean** also called the arithmetic average, is the most widely used measure of central tendency. A particular central tendency is used depending upon the nature of the data.





BIVARIATE TABLES: The bivariate contingency table is widely used. The table is based on cross-tabulation (cross classification); that is the cases are organized in the table on the basis of two variables at the same time.A contingency table is formed by cross-tabulating the two or more variables.

CONSTRUCTING PERCENTAGE TABLES: It is to construct a percentage table, but there are ways to make it look professional. Let us take two variables like the age of the respondents and their attitude towards "women empowerment." Assuming that age affects the attitude towards women empowerment let us hypothesize: the lower the age, the higher the favorable attitude towards "women empowerment." The age range of the respondents is 25 to 70, and the attitude index has three categories of "highly favorable," "medium favorable," and "low favorable." The age variable has so many categories that making a table with that number becomes unwieldy and meaningless. Therefore, we regroup (recode) the age categories into three i.e. under 40 years, 40 – 60 years, and 61 + years.



Lesson 32:	**THE PARTS OF THE TABLE**

Each table has following seven parts

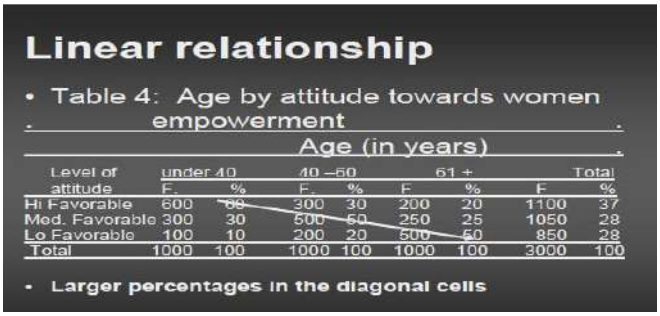
1. Give each table a number.
2. Give each table a title.
3. Label the row and columns variables and give name to each of the variable categories.
4. Include the totals of the columns and rows. These are called *marginals.*
5. Each place that corresponds to the intersection of a category for each variable is a *cell of a table.*
6. The numbers with the labeled variable categories and the totals are called the *body of the table.*
7. If there is missing information, report the number of missing cases near the table.

Researchers convert raw count tables into percentages to see bi-variate relationship. There are three ways to percentage a table: by row, by column, and for the total.

***Reading a percentage Table*:** Once we understand how table is made, reading it and figuring out what it says are much easier. To read a table, first look at the title, the variable labels, and any background information. Next, look at the direction in which percentages have been computed – in rows or columns. Researchers read percentaged tables to make comparisons. Comparisons are made in the opposite direction from that in which percentages are computed. A rule of thumb is to compare across rows if the table is percentaged down (i.e. by column) and to compare up and down in columns if the table is percentaged across (i.e. by row).

It takes practice to see a relationship in a perentaged table.

- If there is no relationship in a table, the cell percentages look approximately equal across rows or columns.
- A linear relationship looks like larger percentages in the diagonal cells.
- If there is curvilinear relationship, the largest percentages form a pattern across cells. For example, the largest cells might be the upper right, the bottom middle, and the upper left. It is easiest to see a relationship in a moderate-sized table (9 to 16 cells) where most cells have some cases (at least five cases are recommended) and the relationship is strong and precise.



Statistical Control: Showing an association or relationship between two variables is not sufficient to say that an independent variable *causes* a dependent variable. In addition to temporal order and association, a researcher must eliminate **alternative explanations** (Explanations that can make the hypothetical relationship spurious). In experimental research, researchers do this by choosing a research design that physically controls potential alternative explanations for results (i.e. that threaten internal validity). In non-experimental research, a researcher controls for alternative explanations with statistics. He or she measures possible alternative explanations with *control variables*, and then examines the control variables with multivariate tables and statistics that help him or her to decide whether a bivariate relationship is spurious.

A researcher controls for alternative explanation in multivariate (more than two variables) analysis by introducing a third (sometimes fourth, or fifth) variable.

Statistical control is a key idea in advanced statistical techniques. A measure of association like the correlation co-efficient only suggests a relationship. Until a researcher considers control variables, the bivariate relationship could be spurious. Researchers are cautious in interpreting bivariate relationships until they have considered control variables.

After they introduce control variables, researchers talk about the *net effect* of an independent variable. There are two ways to introduce control variables: trivariate percentaged tables and multiple regression analysis.

1. Constructing Trivariate Tables:
2. Multiple Regression Analysis:

1. ***Constructing Trivariate Tables:***

Trivariate tables differ slightly from bivariate tables; they consist of multiple bivariate tables. A trivariate table has a bivariate table of the independent and dependent variable for each category of the control variable. These new tables are called *partials*. The number of partials depends on the number of categories in control variable. Partial tables look like bivariate tables but they use a subset of the cases.

Trivariate tables have three limitations.

1. First, they are difficult to interpret if a control variable has more that four categories.
2. Second, control variables can be at any level of measurement, but interval or ratio control variables must be grouped (i.e. converted to an ordinal level), and how cases are grouped can affect the interpretation of effects.
3. Finally, the total number of cases is a limiting factor because the cases are divided among cells in partials. The number of cells in the partials equals the number of cells in the bivariate relationship multiplied by the number of categories in the control variables.

   An example of a trivariate table with "gender" as control variable for the bivariate table is shown here:

Partial table for males

	Age (in years)							
Level of	Under 40		40—60		61+		Total	
Attitude	F	%	F	%	F.	%	F.	%.
High	300	60	200	33	30	6	530	33
Medium	140	28	270	45	120	24	530	33
Low	60	12	130	22	350	70	540	34
Total	500	100	600	100	500	100	1600	100

Partial table for females

	Age (in years)							
Level of	Under 40		40—60		61+		Total	
Attitude	F	%	F	%	F.	%	F.	% .
High	350	70	200	50	20	4	570	41
Medium	150	30	150	38	220	44	520	37
Low	-	-	50	12	260	52	310	22
Total	500	100	400	100	500	100	1400	100

The replication pattern is the easiest to understand. It is when the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable. It means that the control variable has no effect.

The specification pattern is the next easiest pattern. It occurs when one partial replicate the initial bivariate relationship but other partials do not.

The interpretation pattern describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

The suppressor variable pattern occurs when the bivariate tables suggest independence but relationship appears in one or both of the partials.

Multiple Regression Analysis: Multiple regression controls for many alternative explanations of variables simultaneously (it is rarely possible to use more than one control variable using percentaged tables). Multiple regression is a technique whose calculation you may have learnt in the course on statistics.

Lesson 33:	**EXPERIMENTAL RESEARCH**

EXPERIMENTAL RESEARCH

- Experimental research **builds on the principles of positivist approach** more directly than do the other research techniques.
- The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated.
- The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.
- Experimentation is a research design that allows evaluation of causal relationship among variables.

To establish that variable X cause's variable Y, *all three* of the following conditions should be met:

1. Both X and Y should co-vary (i.e. when one goes up, the other should also simultaneously go up (or go down).

2. X (the presumed causal factor) should precede Y. In other words, there must be a time sequence in which the two occur.

3. No other factor should possibly cause the change in the dependent variable Y.

The Language of Experiments: Experimental research has its own language or set of terms and concepts. One important term frequently used is *subjects* or *test units*. In experimental research, the cases or people used in research projects and on whom variables are measured are called the *subjects* or *test units*. In other words these are those entities whose responses to the experimental treatment are measured or observed. Individuals, organizational units, sales territories, or other entities may be the test units. Similar terminology is used on different component parts of the experiments.

Parts of Experiments: We can divide the experiments into seven parts and for each part there is a term. Not all experiments have all these parts, and some have all seven parts plus others. The following seven usually make up a true experiment.

1. Treatment or independent variable.
2. Dependent variable.
3. Pretest.
4. Posttest.
5. Experimental group.
6. Control group.
7. Assignment of subjects.

1. ***Treatment or independent variable:*** The experimenter has some degree of control over the independent variable. The variable is independent because its value can be manipulated by the experimenter to whatever he or she wishes it to be. Its value may be changed or altered independently of any other variable.
2. ***Dependent Variable***: The criterion or standard by which the results are judged. It is assumed that the changes in the dependent variable are consequence of changes in the independent variable. For example, measures of turnover, absenteeism, or morale might be alternative choices for the dependent variable, depending on the purpose of the training.

The outcomes in the experimental research are the physical conditions, social behaviors, attitudes, feelings, or beliefs of subjects that change, in response to treatment. Dependent variables can be measured by paper-and-pencil indicators, observations, interviews, or physiological responses (e.g. heartbeat, or sweating palms). Selection of dependent variable is crucial decision in the design of an experiment.

3. ***Pretests:*** The *pretest* is the measurement of the dependent variable prior to the introduction of the treatment.
4. ***Posttests:*** The posttest is the measurement of the dependent variable after the treatment has been introduced into the experimental situation.
5. ***Experimental Group:*** Experimental researchers often divide subjects into two or more groups for purposes of compassion. A simple experiment has only two groups, only one of which receives the treatment. The experimental group is the group that receives the treatment or in which the treatment is present.
6. ***Control Groups:*** The group that does not receive the treatment is called the "control group." When the independent variable takes on many different values, more than one experimental group is used. In the simplest type of experiment, only two values of the independent variable are manipulated.

Assignment of Subjects/Test Units: *Random assignment* (Randomization) is a method for assigning the cases (e.g. individuals organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way. It is a mechanical method; the assignment is automatic, and the researcher cannot make assignments on the basis of personal preference or the features of specific cases. Random assignment is random in statistical/mathematical sense, not in everyday sense. **In everyday speech**, random means unplanned, haphazard, or accidental, but it has a special meaning in mathematics. **In probability theory**, random describes a process in which each case has a known chance of being selected.

Random assignment or randomization is unbiased because a researcher's desire to confirm a hypothesis

Three Types of Controls:

1. **Manipulation of the Independent Variable:**
2. **Holding Conditions Constant**
3. **Control over the Composition of Groups:**

1. ***Manipulation of the Independent Variable:*** In order to examine the causal effects of an independent variable on a dependent variable, certain manipulations need to be tried. Manipulation simply means control over the stimulus that is we create different levels of the independent variable to assess the impact on the dependent variable
2. ***Holding Conditions Constant:*** When we postulate cause-and-effect relationships between two variables X and Y, it is possible that some other factor, say A, might also influence the dependent variable Y. In such a case, it will not be possible to determine the extent to which Y occurred only because of X, since we do not know how much of the total variation of was caused by the presence of the other factor A. If the true effect of the X is to be assessed, then the effect of A has to be controlled. This is also called as controlling the effect of contaminating factors or confounding factors.
3. ***Control over the Composition of Groups:*** If the experimental and control groups have such characteristics that could contaminate the results then the researcher may have to take note of such factors, if there are any. The group differences should not confound the effect of X variable that happens to be under study. The experimental and control groups need to be balanced. For this purpose the researcher may use **random selection** of the subjects and allocating to different groups. Finally the experimental and control groups should also be selected randomly. Another way to have identical groups is by following the procedure of **matching**.

Random Assignment: *Random assignment* is a method for assignment cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way. It is
mechanical method; the assignment is automatic, and the researcher cannot make assignments on the basis of personal preference or the features of specific cases.
Random assignment is random in a statistical or mathematical sense, not in an everyday sense. In everyday speech, random means unplanned, haphazard, or accidental, but it has a specialized meaning in mathematics. In probability theory, *random* describes a process in which each case has a known chance of being selected. Random assignment lets a researcher calculate the odds that a specific case will be sorted into one group over another.
Random assignment or randomization is unbiased because a researcher's desire to confirm a hypothesis or a research subject's personal interest does not enter into selection process.
Matching: It implies to match the characteristics (such as age, sex) of the cases in each group. Matching is an
alternative to random assignment, but it is an infrequently used one.

Lesson 34:	EXPERIMENTAL RESEARCH (CONT.)

Steps in Conducting an Experiment
Following the basic steps of the research process, experimenters decide on a topic, narrow it into a testable research problem or question, and then develop a hypothesis with variables. Once a researcher has the hypothesis, the steps of experimental research are clear. Broadly there are about 12 steps in conducting an experiment, which are as below:

1. Begin with a straightforward hypothesis that is appropriate for experimental research.
2. Decide on an experimental design that will test the hypothesis within practical limitations.
3. Decide how to introduce the treatment or the independent variable.
4. Develop a valid and reliable measure of the dependent variable.
5. Set up an experimental setting and conduct a pilot test of the variable measures.
6. Locate appropriate subjects or cases.
7. Randomly assign subjects to groups.
8. Gather data for the pretest measure of the dependent variable for all groups.
9. Introduce the treatment to the experimental group only and monitor all groups.
10. Gather data for posttest measure of the dependent variable.
11. Debrief the subjects by informing them of the true purpose and reasons for the experiment.
12. Examine data collected and make comparisons between different groups.

Types of Designs:
Researchers combine parts of experiment (e.g. pretests, control groups, etc.) together into an experimental design. For example some designs lack pretests, some do not have control groups, and others have many experimental groups. Certain widely used standard designs have names.
Classical Experimental Design: All designs are variations of the classical experimental design, which has random assignment of subjects, a pretest and a posttest, an experimental group, and a control group.

1. **Quasi Experimental Designs**
2. **True Experimental Designs**
3. ***Solomon's Four Group Design***

1. Quasi-Experimental Designs:

1.1. ***One-shot Case Study Design:*** Also called the one-group posttest-only design.
The one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment.

1.2. ***One Group Pretest-posttest Design*:** This design has one group, a pretest, a treatment, and a posttest. It lacks a control group and random assignment.

1.3. ***Two Groups Posttest-only Design:*** It has two groups, a random assignment of subjects, a posttest, and a treatment. It has all parts of the classical design except a pretest.

2. **True Experimental Designs:** Experimental designs, which have at least two groups, a random assignment of subjects to experimental and control groups, only experimental group is exposed to treatment, both groups record information before and after the treatment, are known as ex-post facto experimental designs.

   2.1. ***Pretest and Posttest Experimental and Control Group Design:*** Two groups, one control group and the other experimental group, are formed randomly. Both the groups are exposed to pretest and posttest. The experimental group is exposed to treatment while the control group is not. Measuring the difference between the differences in the post- and pretests of the two groups would give the net effects of the treatment.

3. ***Solomon's Four Group Design:*** To gain more confidence in internal validity in experimental designs, it is advisable to set up two experimental groups and two control groups. One experimental group and one control group can be given the both pretest and the posttest. The other two groups will be given only the posttest.

Lesson 35:	**EXPERIMENTAL RESEARCH (CONT.)**

Validity in Experiments: Experiments are judged by two measures; internal validity and external validity.

The first, internal validity indicates whether the independent variable was the sole cause of the change in the dependent variable. It implies the researcher's ability to eliminate alternative explanations of the dependent variable. Variables, other than the treatment, that affect the dependent variable are threats to internal validity. They threaten the researcher's ability to say that the treatment was the true causal factor producing change in the dependent variable.

The second measure, external validity, indicates the extent to which the results of the experiment are applicable in the real world.

Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalisability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situations.

Factors Affecting Internal Validity: There are eight major types of extraneous variables that may jeopardize internal validity:

1. **History effect**
2. **Maturation effect**
3. **Testing effect**
4. **Instrumentation effect**
5. **Selection bias effect**
6. **Selection bias effect**
7. **Statistical regression**
8. **Mortality**
9. **Mechanical loss**

1. History Effect: A specific event in the external environment occurring between the first and second measurement that is beyond the control of the experimenter and that affects the validity of an experiment. The researcher does not have control on such happenings which have an impact on the X and Y relationship.

2. Maturation Effect: **C**ause and effect relationship can also be contaminated by the effects of the passage of time – another uncontrollable variable. Such contamination is called **maturation effect.** Examples of maturation processes could include *growing older, getting tired, feeling hungry, and getting bored*.

3. Testing Effects: Frequently, to test the effects of treatment, subjects are given what is called a ***pretest*** That is, a measure of the dependent variable is taken (pretest), then the treatment given, and after that a

second test, called ***posttest***, administered. The difference between the posttest and the pretest scores is then attributed to the treatment. It is also called **sensitization** through previous testing.

4. Instrumentation Effects: Instrumentation effects are yet another source of threat to internal validity. These might arise because of a change in the measuring instrument between pretest and posttest, and not because of the instrument's differential impact at the end.

5. Selection Bias Effect: Selection bias is the threat that subjects will not form equivalent groups. It is a problem in design without random assignment, hence differential selection of the subjects for the comparison groups. It occurs when subjects in one experimental group have a characteristic that affects the dependent variable.

6. Statistical Regression: Statistical regression is not easy to grasp intuitively. It is a problem of extreme values or a tendency for random error to move group results towards the average. If extremes are taken then they tend to regress toward the mean. Those who are on either end of the extreme would not truly reflect the cause and effect relationship.

One situation arises when subjects are unusual with regard to dependent variable. Because they begin as unusual or extreme, subjects are likely to respond further in the same direction.

If participants chosen for experimental group have extreme scores on the dependent variable to begin with then the laws of probability say that those with very low scores on a variable have a greater probability to improve and scoring closer to mean on the posttest after treatment. This phenomenon of low scorers tending to score closer to the mean is known as "regressing toward the mean."

7. Mortality: Mortality, or attrition, arises when some subjects do not continue throughout the experiment. Although the word mortality means death, it does not necessarily mean that subjects have died.

8. Mechanical Loss: A problem may be experienced due to equipment failure.

9. Experimenter Expectancy: In addition to the usually listed eight factors affecting the internal validity sometimes **experimenter expectancy** may threaten the causal logic of the relationship between the variables. A researcher may threaten internal validity, not purposefully unethical behavior but by indirectly communicating experimenter expectancy to the subjects. Researchers may highly committed to the hypothesis and indirectly communicate desired findings to subjects. The ***double-blind experiment*** is designed to control experimenter expectancy.

External Validity: Even if the researcher eliminates all concerns for internal validity, external validity remains a potential problem. External validity is the ability to generalize experimental findings to real life situations. Without external validity, the findings are of little use for both basic and applied research i.e. we shall not be able to develop any theories that could be applicable to similar other situations.

Reactivity: A Threat to External Validity: Subjects may react differently in an experiment than they would in real life; because they know they are in a study. ***The Hawthorn Effect,*** a specific kind of reactivity to the experimental situation is a good example in this respect. The experiment was conducted in the Hawthorn Electric Company where the performance of the participants was supposed to change due to the change in the environmental conditions i.e. improvement on the environmental conditions will have a positive effect on the performance. Here the workers did not respond to treatment (modification of working conditions) but to the additional attention they received (being in the experiment ad being the focus of attention).

Demand characteristic is another type of reactivity. Here the participants change their behavior as a reaction to the demands of the experimenter who may have inadvertently told the subjects about the expected outcome of the treatment.

Ethical Issues in Lab Experiments: The following actions may be unethical:

- Putting pressure on individuals to participate in experiments through coercion or apply social pressure.
- Asking demeaning questions from the subjects that hurt their self respect.
- Deceiving subjects by deliberately misleading them as to the true purpose of research.
- Exposing participants to physical or mental stress.

• Not allowing subjects to withdraw from the experiment when they want to.
• Using research results to disadvantage the participants, or for purposes not to their liking.
• Not explaining the procedures to be followed in the experiment.
• Exposing subjects to hazardous and unsafe environments.
• Not debriefing the participants fully and accurately after the experiment is over.
• Not preserving the privacy and confidentiality of the information given by the participants.
• Withholding benefits from the control group.

Human Subjects Committee: In order to protect the rights of participating subjects the research institutions have usually set up Ethics Committees. Sometime project specific ethics committees are also formed. Such committees try to look after the rights of the subjects participating in the experiments, as well as in other research techniques.

Lesson 36:	NON-REACTIVE RESEARCH

NON-REACTIVE RESEARCH: Experiments and surveys research are both ***reactive***; that is, the people being studied are aware of the fact that they are being studied. In ***non-reactive*** research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

The Logic of Non-Reactive Research

The critical thing about non-reactive or *unobtrusive measures* (i.e. the measures that are not obtrusive or intrusive) is that the people being studied are not aware of it but leave evidence of their social behavior or actions 'naturally." The researcher infers from the evidence to behavior or attitudes without disrupting those being studied. Unnoticed observation is also a type of non-reactive measure. For example, a researcher may be observing the behavior of drivers from a distance whether drivers stopped at red sign of the traffic lights. The observations can be made both at the day time and at night. It could also be noted whether the driver was a male or a female; whether the driver was also or with passengers; whether other traffic was present; and whether the car came to a complete stop, a slow stop, or no stop.

Varieties of Non-Reactive Observations

Non-reactive measures are varied and researchers have been creative in inventing indirect ways to measures behaviors. Because the measures have little in common except being non-reactive, they are best learned through examples like:

Physical Traces:

• ***Erosion*:** Wear and tear suggests a greater use. For example, a researcher examines children's toys at a children's play centre that were purchased at the same time. Worn out toys suggest greater interest of children in them.

• ***Accretion:*** Accumulation of physical evidence suggests behavior. A researcher examines the soft drink cans or bottles in the garbage collection. That might indicate the brands and types of soft drinks that are very popular.

Archives:

• ***Running Records:*** Regularly produced public records may reveal lot of information. For example, a researcher may examine marriage records for brides' and grooms' recorded ages. The differences might indicate that males marrying younger females are greater than the other way around.

• ***Other Records:*** Irregular or private records can reveal a lot. For example, a researcher may look into the number of reams of paper purchased by a college principal's office for the last 10 years and compare it with students' enrollment.

Observations:

• ***External Appearance:*** How people appear may indicate social factors. For example, a researcher watches students to see whether they are more likely to wear their college's colors and symbols after the college team won or lost.

• ***Count Behaviors:*** Counting how many people do something can be informative. For example a researcher may count the number of men and women who come to a full stop and those who come to a rolling stop at a traffic stop sign. This suggests gender difference in driving behavior.

• ***Time Duration:*** How long people take to do things may indicate their intention. For example a researcher may measure how long men and women pause in front of a particular painting. Time taken may indicate their interest in the painting.

Recording and Documentation: Creating non-reactive measures follows the logic of quantitative measurement, although qualitative researchers also use non-reactive observations. A researcher first conceptualizes a construct, and then links the construct to non-reactive empirical evidence, which is its measure. The operational definition of the variable includes how the researcher systematically notes and records observations.

Content Analysis: Content analysis is a technique for gathering and analyzing the content of a text. The *content* refers to words, meanings, pictures, symbols, ideas, themes, or any message that can be communicated. The *text* is anything written, visual, or spoken that serves as a medium of communication. ***Possible artifacts for study*** could be books, newspaper or magazine articles, advertisements, poems, letters, laws, constitutions, dramas, speeches, official documents, films or videotapes, musical lyrics, photographs, articles of clothing, or works of arts. All these works may be called as documents. The documents can be:

• Personal – letters, diary, autobiography.

• Non-personal – interoffice memos, official documents, proceedings of a meeting.

• Mass media – newspapers, magazines, fiction, films, songs, poems, works of arts.

Content analysis goes back nearly a century and is used in many fields – literature, history, journalism, political science, education, psychology, sociology, and so on. It is also called a study of communication, which means who says what, to whom, why, how, and with what effect. In content analysis, the researcher uses objective and systematic counting and recording procedures to produce a quantitative description of the symbolic content in a text. It may also be called **"textual coding."** There are qualitative versions of content analysis. The emphasis here is quantitative data about a text's content.

Content Analysis is Non-Reactive: It is non-reactive because the placing of words, messages, or symbols in a text to communicate to the reader or receiver occurs without influence from the researcher who analyzes its contents. There is no interaction between the researcher and the creator of the text under analysis.

Measurement and Coding: Careful measurement is crucial in content analysis because a researcher takes different and murky symbolic communication and turns it into precise, objective, quantitative data. He or she carefully designs and documents the procedures for coding to make replication possible.

Constructs in content analysis are operationalizing with a ***coding system,*** a set of instructions or rules on how to systematically observe and record content from text.

Observations can be structured: Measurement in content analysis uses *structured observation* i.e. systematic, careful observation based on written rules. The rules explain how to categorize and classify observations in terms of frequency, direction, intensity and space.

• ***Frequency***: Frequency simply means counting whether or not something occurs and how often (how many times).

• ***Direction****:* Direction is noting the direction of messages in the content along some continuum (e.g., positive or negative, supporting or opposed).

• ***Intensity****:* Intensity is the strength or power of a message in a direction.

• ***Space:*** A researcher can record the size of the text message or the amount of space or volume allocated to it. Space in written text is measured by counting words, sentences, paragraphs or space on a page (e.g. square inches) for video or audio text, space can be measured by the amount of time allocated. For example, a TV character may be present for a few seconds or continuously in every seen of a two hour program.

The unit analysis can vary a great deal in content analysis. It can be a word, a phrase, a theme, a plot,, a news paper article, a character, and so forth.

Coding: The process of identifying and classifying each item and giving labels to each category. Later on each category may be assigned a numerical value for its entry into the computer. In content analysis one can look at the manifest coding and latent coding.

i. **Manifest Coding:** Coding the visible, surface content in a text is called manifest coding. For example, a researcher counts the number of times a phrase or word (e.g. red) appears in the written text, or whether a specific action (e.g. shaking hands) appears in a photograph or video scene. The coding system lists terms or actions or characters that are then located in text. A researcher can use a computer program to search for words or phrases in the text and have a computer do the counting work.
Manifest coding is highly reliable because the phrase or the word either is or is not present.

ii. **Latent Coding:** A researcher using *latent coding* (**also called *semantic analysis***) looks for the underlying meaning in the content of a text. For example, the researcher reads the entire paragraph and decides whether it contains vulgar themes or a romantic mood. His or her coding system has general rules to guide his or her interpretation of the text and for determining whether particular themes or mood are present.
Latent coding tends to be less reliable than the manifest coding. It depends on a coder's knowledge of language and its social meaning. Training, practice, and written rules improve reliability, but still it is difficult to consistently identify themes, moods, and the like.

How to Conduct Content Analysis Research

1. **Question Formulation**
2. **Unit of Analysis**
3. **Sampling**
4. **Variables and Constructing Coding Categories**
5. **Inferences**

1. **Question Formulation:** As in most research, content analysis researchers begin with a research question. When the question involves variables that are messages or symbols, content analysis may be appropriate.
2. **Unit of Analysis:** A researcher decides on the unit of analysis (i.e. the amount of text that is assigned a code). In the previous example each advertisement may be a unit of analysis.
3. **Sampling:** Researchers often use random sampling in content analysis. First, they define the population and the sampling element. For example, the population might be all words, all sentences, all paragraphs or all articles in certain type of documents over a period of specified period.
4. **Variables and Constructing Coding Categories:** Say a researcher is interested in women's portrayal in significant leadership roles. Define "significant leadership role" in operational terms and put it as written rules for classifying people named in the articles. Say the researcher is further interested in positive leadership roles, so the measure will indicate whether the role was positive or negative. In addition to written rules for coding decisions, a content analysis researcher creates a recording sheet (also called a *coding form* or *tally sheet*) on which to record the information. Each unit should have a separate recording sheet.
5. **Inferences:** The inference a researcher can or cannot make on the basis of results is critical in content analysis. Content analysis describes what is in the text. It cannot reveal the intentions of those who created the text or the effects that messages in the text have on those who receive them.

Lesson 37:	**USE OF SECONDARY DATA**

ADVANTAGES AND DISADVANTAGES OF THE USE OF CONTENT ANALYSIS:

Prior to the discussion of secondary data, let us look at the advantages and disadvantages of the use of content analysis that was covered in the last lecture. In a way content analysis is also the study of

documents through which the writers try to communicate, though some of the documents (like population census) may simply contain figures.

Advantages of Content Analysis:

1. ***Access to inaccessible subjects:*** One of the basic advantages of content analysis is that it allows research on subjects to which the researcher does not have physical access.
2. ***Non-reactivity:*** Document study shares with certain types of observations (e.g., indirect observation or non participant observation through one-way mirror) the advantage of little or no reactivity.
3. ***Can do longitudinal analysis:*** Like observation and unlike experiments and survey, document study is especially well suited to study over a long period of time. Many times the objective of the research could be to determine a trend.
4. ***Use Sampling:*** The researcher can use random sampling. One could decide on the population, develop sampling frame and draw sample random sample by following the appropriate procedure.
5. ***Can use large sample size:*** Larger the sample closer the results to the population. In experimentation as well as in survey research there could be limitations due to the availability of the subjects or of the resources but in document analysis the researcher could increase the sample and can have more confidence in generalization.
6. ***Spontaneity:*** The spontaneous actions or feelings can be recorded when they occurred rather than at a time specified by the researcher. If the respondent was keeping a diary, he or she may have been recording spontaneous feelings about a subject whenever he or she was inspired to do so. The contents of such personal recording could be analyzed later on.
7. ***Confessions:*** A person may be more likely to confess in a document, particularly one to be read only after his or her death, than in an interview or mailed questionnaire study
8. ***Relatively low cost:*** Although the cost of documentary analysis can vary widely depending on the type of document analyzed, how widely documents are dispersed, and how far one must travel to gain access to them, documentary analysis can be inexpensive compared to large-scale surveys.
9. ***High quality:*** Although documents vary tremendously in quality, many documents, such as news paper columns, are written by skilled commentators and may be more valuable than, for example, poorly written responses to mailed questionnaires.

Disadvantages of Content Analysis:

1. ***Bias:*** Many documents used in research were not originally intended for research purposes. The various goals and purposes for which documents are written can bias them in various ways.
2. ***Selective survival:*** Since documents are usually written on paper, they do not withstand the elements well unless care is taken to preserve them. Thus while documents written by famous people are likely to be preserved, day-to-day documents such as letters and diaries written by common people tend either to be destroyed or to be placed in storage and thus become inaccessible.
3. ***Incompleteness:*** Many documents provide incomplete account to the researcher who has had no prior experience with or knowledge of the events or behavior discussed.
4. ***Lack of availability of documents:*** In addition to the bias, incompleteness, and selective survival of documents, there are many areas of study for which no documents are available. In many cases information simply was never recorded. In other cases it was recorded, but the documents remain secret or classified, or have been destroyed.
5. ***Sampling bias:*** One of the problems of bias occurs because persons of lower educational or income levels are less likely to be represented in the sampling frames
6. ***Limited to verbal behavior:*** By definition, documents provide information only about respondent's verbal behavior, and provide no direct information on the respondent's nonverbal behavior, either that of the document's author or other characters in the document.

7. ***Lack of standardized format:*** Documents differ quite widely in regard to their standardization of format.
8. ***Coding difficulties:*** For a number of reasons, including differences in purpose for which the documents were written, differences in content or subject matter, lack of standardization, and differences in length and format, coding is one of the most difficult tasks facing the content analyst.
9. ***Data must be adjusted for comparability over time:*** Although one of the advantages of document study is that comparisons may be made over a long period of time, since external events cause changes so drastic that even if a common unit of measure is used for the entire period, the value of this unit may have changed so much over time that comparisons are misleading unless corrections are made. Look at the change in measuring distance, temperature, currency, and even literacy in Pakistan.

Use of Secondary Data or Existing Statistics/Documents

Secondary Data: Secondary data refer to information gathered by someone other than the researcher conducting the present study. Secondary data are usually historical, already assembled and do not require access to respondents or subjects.

Secondary data may be collected by large bureaucratic organization like the Bureau of Statistics or other government or private agencies. These data may have been collected for policy decisions or as part of public service.

The data may be a time bound collection of information (population census) as well as spread over long periods of time (unemployment trends, crime rate). Secondary data are used for making comparisons over time in the country (population trends in the country) as well as across the countries (world population trends).

Selecting Topic for Secondary Analysis

- Search through the collections of information with research question and variables in mind, and then reassemble the information in new ways to address the research question.
- It is difficult to specify topics that are appropriate for existing statistics research because they are so varied.
- Experiments are best for topics where the researcher controls a situation and manipulates an independent variable.
- Survey research is best for topics where the researcher asks questions and learns about reported attitudes and behavior.
- Content analysis is best for topics that involve the content of messages in cultural communication.
- Existing statistics research is best for topics that involve information collected by large bureaucratic organizations. Thus existing statistics research is appropriate when a researcher wants to test hypotheses involving variables that are also in official reports of social, economic and political conditions. These include descriptions of organizations or people in them. Often, such information is collected over long periods. For example, existing statistics can be used by researcher who wants to see whether unemployment and crime rates are associated in 100 cities across a 20 year period.

  There are so many sources of existing statistics like: UN publications, UNESCO Statistical Yearbook, UN Statistical Yearbook, Demographic Yearbook, Labor Force Survey of Pakistan, and Population Census Data.

Secondary Survey Data: Secondary analysis is a special case of existing statistics; it is reanalysis of previously collected survey or other data that was originally gathered by others. As opposed to primary research (e.g., experiments, surveys, and content analysis), the focus is on analyzing rather than collecting data.

Secondary analysis is increasingly used by researchers. It is relatively inexpensive; it permits comparisons across groups, nations, or time; it facilitates replication; and permits asking about issues not thought by the original researchers.

Large-scale data collection is expensive and difficult. The cost and time required for major national surveys that uses rigorous techniques are prohibitive for most researchers. Fortunately, the organization, preservation, and dissemination of major survey data sets have improved. Today, there are archives of past surveys open to researchers (e.g., data on Population Census of Pakistan, Demographic Survey of Pakistan).

Reliability and Validity: Existing statistics and secondary data are not trouble free just because a government agency or other source gathered the original data. Researchers must be concerned with validity and reliability, as well as with some problems unique to this research technique.

Validity: Validity problems occur when the researcher's theoretical definition does not match that of the government agency or organization that collected the information. Official policies and procedures specify definitions for official statistics. For example, a researcher defines a *work injury* as including minor cuts, bruises, and sprains that occur on the job, but the official definition in government reports only includes injuries that require a visit to a physician or hospital. Many work injuries as defined by the researcher will not be in the official statistics.

Another validity problem arises when official statistics are a proxy for a construct in which the researcher is really interested. This is necessary because the researcher cannot collect original data. For example, the researcher wants to know how many people have been robbed, so he or she uses police statistics on robbery arrests as a proxy. But the measure is not entirely valid because many robberies are not reported to the police, and reported robberies do not always result in an arrest.

Another validity problem arises because the researcher lacks control over how information is collected. All information, even that in official government reports, is originally gathered by people in bureaucracies as part of their job. A researcher depends on them for collecting organizing, reporting, and publishing data accurately. Systematic errors in collecting the initial information (e.g., census people who avoid poor neighborhoods and make-up information, or people who put a false age on their ID card); errors in organizing and reporting information (e.g., police department that is sloppy about filing crime reports and loses some); errors in publishing information (e.g., a typographical error in a table) all reduce measurement validity.

Reliability: Stability reliability problems develop when official definition or the method of collecting information changes over time. Official definitions of work injury, disability, unemployment, literacy, poverty, and the like change periodically. Even if the researcher learns of such changes, consistent measurement over time is impossible.

Equivalence reliability can also be a problem. For example, studies of police department suggest that political pressures to increase arrests are closely related to the number of arrests. It could be seen when political pressures in one city may increase arrests (e.g., a crackdown on crime), whereas pressures in another city may decrease arrests (e.g., to show drop in crime shortly before an election in order to make officials look better).

Researchers often use official statistics for international comparisons but national governments collect data differently and the quality of data collection varies.